Let the Machine Reveal Itself

Machine formation and equilibrium

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We begin with a physical event: two light nuclei fuse and release energy. Each step between that event and a working magnetic confinement system introduces a new demand. Following those demands will give us the tokamak's components, the quantities that describe them, and the equations that tie them together.

1. The first requirement: make nuclei approach

The fuel considered here is an equal mixture of deuterium and tritium, the two heavy isotopes of hydrogen. Their reaction is

$$^2\mathrm{H} + {}^3\mathrm{H} \longrightarrow {}^4\mathrm{He}\ (3.5\,\mathrm{MeV}) + n\ (14.1\,\mathrm{MeV}), \qquad Q = 17.6\,\mathrm{MeV}. \tag{1}$$

The energy comes from the decrease in rest mass. In the centre-of-mass frame, the two products carry equal and opposite momenta, so their kinetic energies are inversely proportional to their masses. The neutron takes about four fifths of the released energy; the helium nucleus, or alpha particle, takes the rest. That division will later determine how much energy heats the fuel and how much must be collected outside it.

Both reacting nuclei carry charge $+e$. Their electrostatic repulsion is

$$U(r) = \frac{e^2}{4\pi\epsilon_0 r}. \tag{2}$$

At a separation of 2 fm this is about 0.72 MeV. Quantum tunnelling allows fusion at much lower collision energies. At temperatures of order 10–20 keV, collisions in the energetic part of the ion distribution have a useful probability of penetrating the barrier. We use temperature in energy units throughout: T denotes $k_B T_K$, so 10 keV corresponds to 1.16×10^8 K.

Fuel at this temperature must be kept away from material surfaces. Contact would remove its energy, erode the surface and introduce impurities into the fuel. A *vacuum vessel* provides an evacuated chamber that limits contact with cold gas; the hot fuel inside it must be confined by electromagnetic fields. Reaction energies and the reactor-design framework used here follow Freidberg, Mangiarotti and Minervini.[1]

> **Design consequence.** An evacuated vessel limits cold-gas contact with the hot fuel. Keeping the plasma clear of its wall requires magnetic confinement.

[1] J. P. Freidberg, F. J. Mangiarotti and J. Minervini, "Designing a Tokamak Fusion Reactor—How Does Plasma Physics Fit In?" (2015), open manuscript. This is the reference for the reduced reactor model developed later.

2. Charged particles give us a handle

At these temperatures the fuel is ionized: electrons and nuclei form a plasma. On lengths large compared with its electrostatic screening length, the positive and negative charge densities nearly cancel. This property is called *quasineutrality*. Each constituent still responds to the Lorentz force,

$$m\frac{d\mathbf{v}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \tag{3}$$

Set $\mathbf{E} = 0$ and $\mathbf{B} = B\hat{\mathbf{z}}$. Since $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0$, the magnetic force does no work: it preserves the particle's kinetic energy while turning its velocity. With $w = v_x + iv_y$, the perpendicular motion satisfies $\dot{w} = -i(qB/m)w$. Thus $w = w(0)e^{-iqBt/m}$, and integration gives

$$\omega_c = \frac{|q|B}{m}, \qquad r_L = \frac{v_\perp}{\omega_c} = \frac{mv_\perp}{|q|B}. \tag{4}$$

Here ω_c is the gyrofrequency and r_L the gyroradius. Parallel motion remains free, so the orbit is a helix around a magnetic field line. A 10 keV deuteron with $v_\perp = \sqrt{2T/m_D} = 9.8 \times 10^5\,\mathrm{m\,s^{-1}}$ has $r_L = 6.8\,\mathrm{mm}$ in a 3 T field. A metre-scale chamber can therefore contain small transverse orbits. Its next problem is escape along the field.

> **Design consequence.** A magnetic field bounds transverse motion while leaving particles free to stream along it. Confinement must therefore address parallel escape.

3. A straight magnetic corridor

A coil is a conductor wound into turns so that their magnetic fields add. For a long solenoid, the turns surround a cylindrical volume. Ampère's law gives

$$\oint_C \mathbf{B} \cdot d\boldsymbol{\ell} = \mu_0 I_{\mathrm{linked}}. \tag{5}$$

Choose a rectangular path with length L inside the solenoid and its return far outside. *Model used here.* For a solenoid long compared with its radius, the exterior field is small. With n_{turn} turns per unit length, each carrying current I_c,

$$BL = \mu_0(n_{\mathrm{turn}}L)I_c, \qquad B \simeq \mu_0 n_{\mathrm{turn}} I_c. \tag{6}$$

This field guides particles down the tube. At $v_\parallel \sim 10^6\,\mathrm{m\,s^{-1}}$, a particle traverses a 10 m device in about $10\,\mu$s. Increasing B reduces the gyroradius but leaves this parallel transit time unchanged. Magnetic mirrors can reflect a range of particle pitch angles; here we pursue another way to remove direct end loss: join the ends of the corridor.

> **Design consequence.** The straight solenoid leaves a direct escape path at each end. Closing that path requires a different geometry.

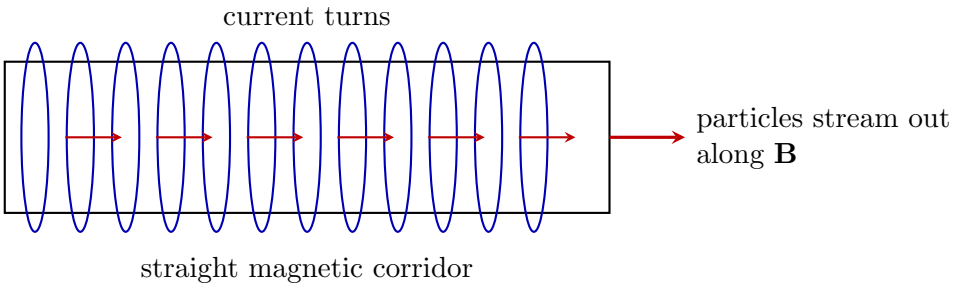


Figure 1. The axial field guides particles through the open ends.

4. Closing the corridor creates the torus

The resulting tube is a torus. Its long-way direction is *toroidal*; the short way around a tube cross-section is *poloidal*. The major radius R_0 measures from the symmetry axis to the cross-section's centre. The minor radius a measures the horizontal extent of the plasma from that centre. Their ratio is

$$A \equiv \frac{R_0}{a} \qquad \text{(aspect ratio)}. \tag{7}$$

Toroidal-field (TF) coils wrap around the tube at successive positions around the ring. Approximating them by a continuous, axisymmetric winding, a circular Ampère path of radius R links N turns carrying I_c:

$$2\pi R B_\varphi(R) = \mu_0 N I_c, \qquad B_\varphi(R) = \frac{\mu_0 N I_c}{2\pi R}. \tag{8}$$

Model used here. The continuous-winding approximation is axisymmetric. Discrete TF coils introduce a periodic field variation called *ripple*. The field is stronger on the *inboard* side, nearest the symmetry axis, than on the *outboard* side:

$$\frac{B_{\text{in}}}{B_{\text{out}}} = \frac{R_0 + a}{R_0 - a} = \frac{A+1}{A-1}. \tag{9}$$

The closed path removes the ends, but its curvature and field gradient change the particle orbits. We must calculate that change before accepting the magnetic bottle.

> **Design consequence.** Closing the corridor creates field curvature and a gradient, $B_\varphi \propto 1/R$. Their effect on particle orbits must now be checked.

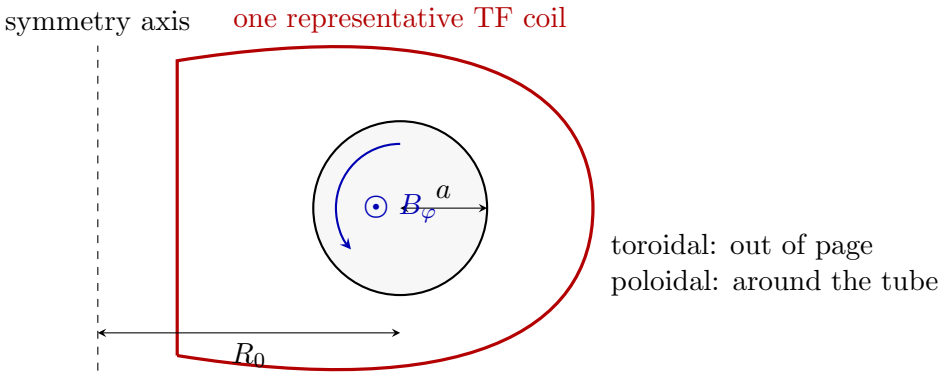


Figure 2. Torus geometry in a poloidal section: the TF winding surrounds the plasma tube; R_0 locates its centre and a measures its minor radius. The blue arrow marks the poloidal direction.

5. Why the toroidal bottle still leaks

When $r_L/R_0 \ll 1$, we can average over gyromotion and follow the orbit's centre, called the *guiding centre*. For a slowly varying perpendicular force $\mathbf{F}$, seek a constant drift velocity satisfying $q\mathbf{v}_F \times \mathbf{B} + \mathbf{F} = 0$. Crossing with $\mathbf{B}$ yields

$$\mathbf{v}_F = \frac{\mathbf{F} \times \mathbf{B}}{qB^2}. \tag{10}$$

Two effective forces arise in a toroidal field. The magnetic-moment magnitude of a gyro-orbit follows from its circulating charge:

$$\mu = \frac{|q|\omega_c}{2\pi}\,\pi r_L^2 = \frac{mv_\perp^2}{2B}.$$

For fields varying slowly over a gyroperiod and gyroradius, μ is an adiabatic invariant. The perpendicular energy μB then gives a gradient force $-\mu\nabla B$. Following a curved field line also gives the effective outward curvature force $(mv_\parallel^2/R)\hat{\mathbf{R}}$. The guiding-centre ordering underlying these expressions is developed systematically by Brizard and Hahm.[2]

Because $B \propto 1/R$, we have $\nabla B = -(B/R)\hat{\mathbf{R}}$. Both effective forces point outward. With $\mathbf{B} = B\hat{\boldsymbol{\varphi}}$ and $\hat{\mathbf{R}} \times \hat{\boldsymbol{\varphi}} = \hat{\mathbf{Z}}$,

$$\mathbf{v}_{\nabla B} = \frac{mv_\perp^2}{2qBR}\hat{\mathbf{Z}}, \qquad \mathbf{v}_c = \frac{mv_\parallel^2}{qBR}\hat{\mathbf{Z}}, \qquad \mathbf{v}_d = \frac{m}{qBR}\left(v_\parallel^2 + \frac{v_\perp^2}{2}\right)\hat{\mathbf{Z}}. \tag{11}$$

Ions and electrons drift vertically in opposite directions. The resulting charge separation creates a vertical electric field. Setting $\mathbf{F} = q\mathbf{E}$ in Eq. (10) gives

$$\mathbf{v}_E = \frac{\mathbf{E} \times \mathbf{B}}{B^2}. \tag{12}$$

This drift is common to both species and carries the plasma outward toward the wall. A purely toroidal field therefore leaves a systematic escape mechanism.

The remedy is to make field lines visit the top and bottom of the tube. Then a vertical drift alternates between outward and inward motion relative to the local magnetic surface. Adding a poloidal field provides this rotation and allows bounded orbit excursions in the axisymmetric confinement picture.

> **Design consequence.** Toroidal drifts separate charge and drive outward motion. A poloidal field is needed to carry particles around the cross-section.

6. Plasma current supplies the second field

External helical windings can produce the required twist; that is the stellarator route. The tokamak uses a toroidal current in the plasma to generate the principal poloidal field. For a circular cross-section at large aspect ratio, Ampère's law gives

$$B_\theta(r) \simeq \frac{\mu_0 I_{\mathrm{enc}}(r)}{2\pi r}, \tag{13}$$

where $I_{\mathrm{enc}}(r)$ is the plasma current inside radius r. The combination $\mathbf{B} = B_\varphi\hat{\boldsymbol{\varphi}} + B_\theta\hat{\boldsymbol{\theta}}$ produces helical field lines.

[2] A. J. Brizard and T. S. Hahm, "Foundations of nonlinear gyrokinetic theory," *Reviews of Modern Physics* **79**, 421–468 (2007), doi:10.1103/RevModPhys.79.421.

To initiate that current, place a solenoid through the central hole and change its magnetic flux. Faraday's law,

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi}{dt}, \tag{14}$$

creates a toroidal electric field. The central solenoid is the primary winding of a transformer; the plasma ring acts as its one-turn secondary. This gives the *plasma current* I_p a physical source. The TF coils provide the long-way field, and the induced plasma current provides the short-way field.

Design consequence. Changing central-solenoid flux drives plasma current and its poloidal field. The resulting helical field needs a measure of its pitch.

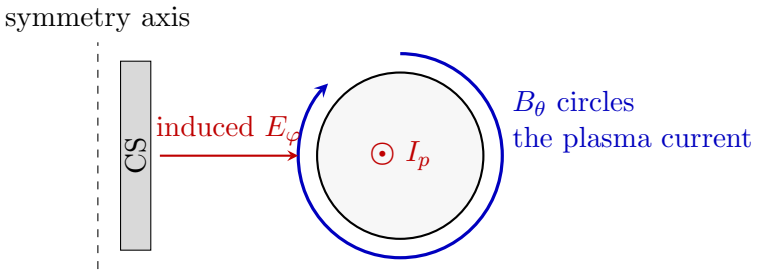


Figure 3. Induced toroidal electric field drives current; its poloidal field circles the plasma.

7. The helix has a pitch: the safety factor

A field-line displacement is parallel to $\mathbf{B}$. On a circular surface its toroidal and poloidal lengths therefore satisfy

$$\frac{R\,d\varphi}{r\,d\theta} = \frac{B_\varphi}{B_\theta}. \tag{15}$$

At large aspect ratio take $R \simeq R_0$ and $B_\varphi \simeq B_0$. During one poloidal circuit the field line advances toroidally through $2\pi q$, where

$$q(r) \equiv \frac{\Delta\varphi}{2\pi} \simeq \frac{rB_0}{R_0 B_\theta(r)}. \tag{16}$$

The *safety factor* q counts toroidal turns per poloidal turn. Its historical name reflects its use in stability analysis. Substituting Eq. (13) gives

$$q(r) \simeq \frac{2\pi r^2 B_0}{\mu_0 R_0 I_{\mathrm{enc}}(r)}, \qquad q_a \simeq \frac{2\pi a^2 B_0}{\mu_0 R_0 I_p}. \tag{17}$$

Thus an imposed lower bound on the edge safety factor becomes a ceiling on current:

$$I_p \simeq \frac{2\pi a^2 B_0}{\mu_0 R_0 q_a}. \tag{18}$$

Increasing plasma current strengthens the poloidal field and reduces the number of toroidal turns per poloidal circuit. Size, current, field and pitch are now coupled. For $q = m/n$ in lowest terms, the ideal field line closes after n poloidal turns and m toroidal turns. An irrational q produces a line that densely samples its surface.

Design consequence. An adopted minimum safety factor bounds plasma current at fixed size and toroidal field. Force balance must also be satisfied.

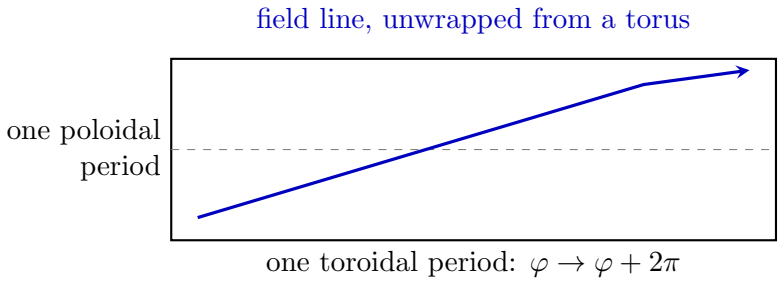


Figure 4. Unwrapping the magnetic surface shows the field-line pitch.

8. The plasma pushes back: pressure and beta

Electrons and ions both contribute to pressure. With singly charged fuel, $n_e \simeq n_i \equiv n$, and equal temperatures,

$$p = n_e T_e + n_i T_i \simeq 2nT. \tag{19}$$

At $n = 10^{20}\,\mathrm{m}^{-3}$ and $T = 10\,\mathrm{keV}$ this gives $p \simeq 3.2 \times 10^5\,\mathrm{Pa}$. The pressure gradient drives expansion. On scales large compared with particle orbits, a single-fluid model with scalar pressure gives

$$\rho_m \frac{d\mathbf{u}}{dt} = -\nabla p + \mathbf{j} \times \mathbf{B}. \tag{20}$$

Here ρ_m is mass density, $\mathbf{u}$ bulk velocity and $\mathbf{j}$ current per unit area. We use static magnetohydrodynamic (MHD) equilibrium: negligible bulk acceleration, isotropic pressure and negligible viscous force. Then

$$\mathbf{j} \times \mathbf{B} = \nabla p. \tag{21}$$

Dotting with $\mathbf{B}$ immediately gives

$$\mathbf{B} \cdot \nabla p = 0. \tag{22}$$

Pressure is constant along each field line, and on a nested surface densely sampled by those lines. Magnetic geometry must support this pressure distribution.

A natural pressure scale follows from the work needed to establish a field. For a long solenoid of N turns, length L and area S, its linked flux is NBS and $B = \mu_0 N I_c / L$. Integrating $dW_B = I_c\, d(NBS)$ gives

$$W_B = \frac{B^2}{2\mu_0} SL. \tag{23}$$

The energy density $B^2/(2\mu_0)$ has pressure units. Its ratio to plasma pressure defines the dimensionless parameter

$$\beta \equiv \frac{p}{B^2/(2\mu_0)}. \tag{24}$$

Beta measures how much plasma pressure is supported relative to the magnetic energy density. Later global estimates use representative or volume-averaged pressure and B_0, with that convention stated in the calculation. The allowable beta requires a stability model; the definition itself sets no universal limit.

Design consequence. Plasma pressure and current produce forces that require external equilibrium fields. The next task is to identify the field that balances radial expansion.

9. External coils hold the plasma in position

The plasma ring's pressure and self-generated magnetic field create an outward expansion tendency. A vertical field supplies an adjustable radial force on its toroidal current. For a ring element,

$$d\mathbf{F} = I_p R_0 \, d\varphi \, \hat{\boldsymbol{\varphi}} \times B_v \hat{\mathbf{Z}} = I_p R_0 B_v \, d\varphi \, \hat{\mathbf{R}}. \tag{25}$$

Changing the sign of B_v reverses the local force. The circumferentially integrated *radial load* is $2\pi R_0 I_p B_v$; vector forces around a complete symmetric ring cancel in translation while still expanding or compressing the ring.

External *poloidal-field coils*, or PF coils, are circular windings around the symmetry axis, placed at selected radii and heights. Their fields lie in the R–Z plane. By changing their currents we can supply the vertical field that balances radial expansion, and later adjust plasma height and shape. A power supply is the circuit that controls each coil's current; the coil is the magnetic actuator. The necessary field magnitude depends on the plasma pressure and current profiles, which enter the equilibrium equation derived below.

The coil roles are now distinct. TF coils generate B_φ. The central solenoid induces I_p, whose field supplies the main poloidal twist. The external PF system positions and shapes the resulting plasma.

Design consequence. A controlled vertical field supplies radial force on the plasma ring. PF coils must control its cross-sectional boundary as well as its position.

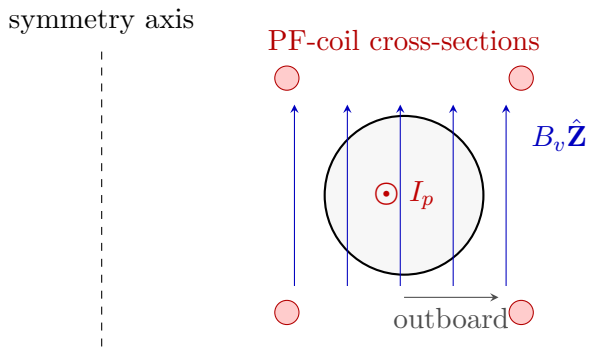


Figure 5. A vertical field acts radially on the toroidal plasma current.

10. Flux surfaces define the magnetic boundary

In cylindrical coordinates (R, φ, Z), axisymmetry means $\partial/\partial\varphi = 0$. The divergence-free condition becomes

$$\partial_R(RB_R) + \partial_Z(RB_Z) = 0. \tag{26}$$

Introduce a flux function $\psi(R, Z)$ through

$$B_R = -\frac{1}{R}\partial_Z\psi, \qquad B_Z = \frac{1}{R}\partial_R\psi, \qquad \mathbf{B}_p = \frac{\nabla\psi \times \hat{\boldsymbol{\varphi}}}{R}. \tag{27}$$

The mixed derivatives cancel in Eq. (26). Also,

$$\mathbf{B} \cdot \nabla\psi = 0, \tag{28}$$

so field lines lie on surfaces of constant ψ. With this convention, 2π times a difference in ψ is a poloidal magnetic-flux difference. Rotating a closed contour of ψ about the symmetry axis constructs a magnetic surface.

The outermost closed surface defines the confined plasma boundary. The vacuum vessel surrounds this magnetic configuration. PF-coil currents and plasma current together set its flux contours; controlling these currents lets us change the boundary's position and shape. The point inside the nested contours about which they contract is the *magnetic axis*.

This flux description is the starting point for the Grad–Shafranov equilibrium calculation in §§21–24. An analytic treatment is given by Cerfon and Freidberg.[3]

> **Design consequence.** The magnetic boundary is a contour of ψ, controlled by PF currents. The vessel must leave room around that confined plasma.

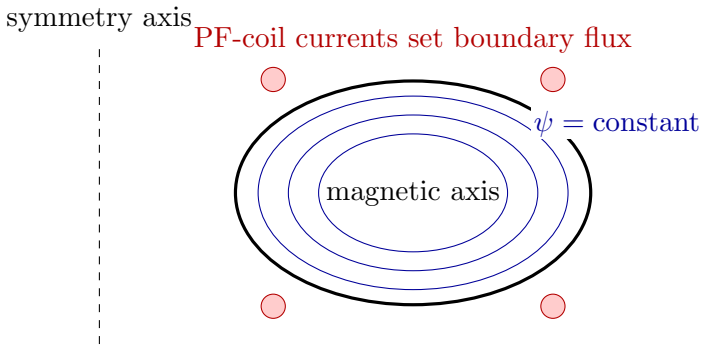


Figure 6. PF currents set the flux contours that form the plasma boundary.

11. The transformer has a finite flux budget

Faraday's law around the plasma ring defines the loop voltage:

$$V_{\text{loop}} \equiv \oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi_{\text{link}}}{dt}. \tag{29}$$

Over a discharge,

$$\int V_{\text{loop}}\, dt = -\Delta\Phi_{\text{link}}. \tag{30}$$

One volt-second equals one weber of linked flux. The central solenoid can swing only through a finite flux range before reaching its current, field or mechanical limits. The flux budget pays for initiating and ramping the plasma current and for maintaining it against resistance. Sustained transformer-driven operation therefore has a finite pulse length. Longer operation needs another source of current or a new pulse after resetting the solenoid.

Electrical resistance also provides *ohmic heating*. In a resistive fluid model,

$$P_{\text{ohm}} = \int \eta j^2\, dV \simeq I_p^2 R_p, \tag{31}$$

where η is resistivity and R_p an effective plasma resistance. During a current ramp, part of the electrical input changes magnetic energy. For a simple circuit with fixed inductance L_p, $V_{\text{loop}} = L_p \dot{I}_p + R_p I_p$; only when the inductive term is small does $I_p V_{\text{loop}} \simeq P_{\text{ohm}}$.

[3] A. J. Cerfon and J. P. Freidberg, "One size fits all" analytic solutions to the Grad–Shafranov equation, *Physics of Plasmas* **17**, 032502 (2010), doi:10.1063/1.3328818.

In a fully ionized collisional plasma, classical Spitzer resistivity scales as

$$\eta \propto T_e^{-3/2} \tag{32}$$

for fixed composition and approximately fixed Coulomb logarithm.[4] At comparable current density, heating falls as the electrons get hotter. Resistance is useful during startup, but reaching fusion temperature calls for auxiliary heating.

Design consequence. The transformer has a finite volt-second budget, while ohmic heating weakens as temperature rises. Additional heating must sustain the hot plasma.

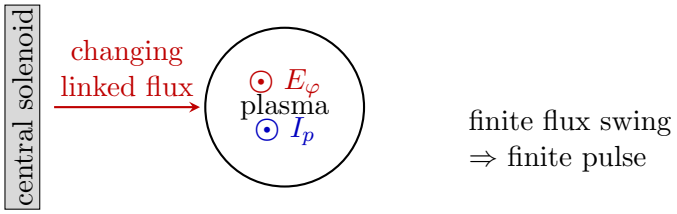


Figure 7. The transformer couples changing linked flux to the plasma ring.

12. Heating must balance energy loss

A Maxwellian species has thermal energy density $3n_sT_s/2$. For approximately uniform fuel at equal ion and electron temperatures,

$$W \simeq 3nTV, \qquad P_{\text{loss}} \equiv W/\tau_E. \tag{33}$$

Here τ_E is the energy-confinement time. In this section P_{loss} includes transport and radiation losses together; each loss channel is counted once. The thermal-energy balance is

$$\frac{dW}{dt} = P_{\text{ohm}} + P_{\text{aux}} + P_\alpha - P_{\text{loss}}. \tag{34}$$

The powers on the right are deposited in the plasma. Auxiliary heating must raise W against losses while fusion power is small. Later, confined alpha particles from Eq. (1) transfer their energy to the fuel, providing P_α. Neutrons carry their energy out of the magnetically confined plasma.

A *neutral-beam injector* accelerates ions and neutralizes them before injection. The fast neutral atoms cross the confining magnetic field, become ionized inside the plasma, and transfer energy and momentum through collisions. A *radio-frequency* (RF) system instead sends electromagnetic waves through an antenna or launcher; suitable frequencies and wave patterns transfer energy to electrons or ions. These systems require access openings, or ports, through the vessel. With suitable injection or wave conditions, they can also drive plasma current without transformer induction.

We now have the equipment that produces a hot current-carrying plasma. Its useful volume and stable pressure depend on the shape that the PF system can create and hold.

Design consequence. Beam injectors or RF launchers supply auxiliary heat through vessel ports. Their deposited power must cover losses and any increase in stored energy.

[4] Naval Research Laboratory, *NRL Plasma Formulary* (2019), collisional transport and resistivity relations, open formulary.

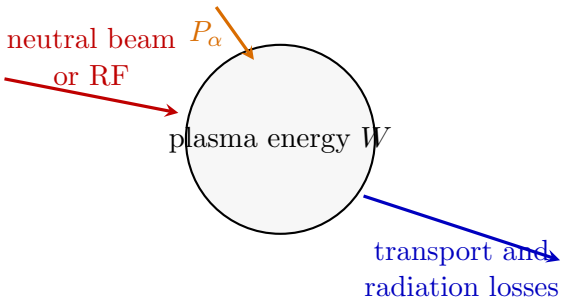


Figure 8. Deposited heating competes with energy carried out of the plasma.

13. Elongation buys volume

An elliptical cross-section has horizontal semi-axis a and vertical semi-axis κa, where

$$\kappa \equiv \frac{\text{vertical semi-axis}}{\text{horizontal semi-axis}} \tag{35}$$

is the *elongation*. A convenient boundary parameterization is $R = R_0 + a\cos\theta$, $Z = \kappa a \sin\theta$. Its area is $\pi\kappa a^2$. Rotating the ellipse about the symmetry axis gives

$$V = 2\pi R_0 \, \pi\kappa a^2 = 2\pi^2 R_0 a^2 \kappa. \tag{36}$$

Scope. The volume formula is exact for the specified ellipse with $R_0 > a$. At fixed R_0 and a, increasing κ increases volume in direct proportion. At a fixed plasma state, that volume can contain more fuel and produce more fusion power.

Creating the ellipse requires a spatially varying PF field. Coil locations and currents are selected so that the combined plasma and external fields support a tall boundary. Shape is therefore an additional task for the same PF system that holds the plasma's radial position. Elongation also changes its response to displacement.

> **Design consequence.** Elongation increases volume at fixed R_0 and a. Its benefit must be assessed together with the response to vertical displacement.

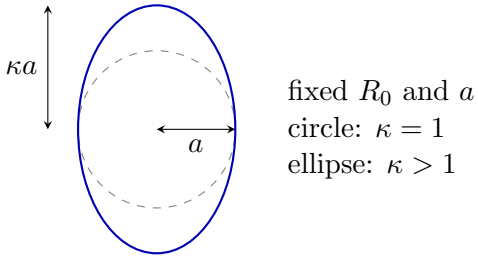


Figure 9. Elongation increases cross-sectional area from πa^2 to $\pi\kappa a^2$. The added height also changes the vertical-stability problem.

14. A taller plasma needs position feedback

To expose the stability question, represent a small vertical displacement by z. Expand its force about equilibrium:

$$F_Z(z) = F_Z(0) + K_z z + \cdots, \qquad K_z = \left.\frac{\partial F_Z}{\partial z}\right|_0. \tag{37}$$

Since $F_Z(0) = 0$, a one-coordinate inertial model gives

$$M\ddot{z} = K_z z. \tag{38}$$

For $K_z < 0$ a displacement produces a restoring force. For $K_z > 0$ the growing solution is $z \propto e^{\sqrt{K_z/M}t}$. This model explains the sign of the feedback needed. *Scope.* A real vertical mode also couples to induced currents in nearby conducting structures and to the coil circuits; its growth rate requires those dynamics.

Elongation generally increases the difficulty of vertical stabilization.[5] Magnetic sensors measure signals from which plasma position is estimated. A controller compares that position with the requested one and commands power supplies to change control-coil currents. The resulting field must oppose the displacement quickly enough to keep the plasma clear of the vessel. Conducting structures can slow the motion and give the active system time to respond.

Design consequence. An unstable vertical displacement requires active correction. Sensors, a controller and powered coils must close the restoring feedback loop.

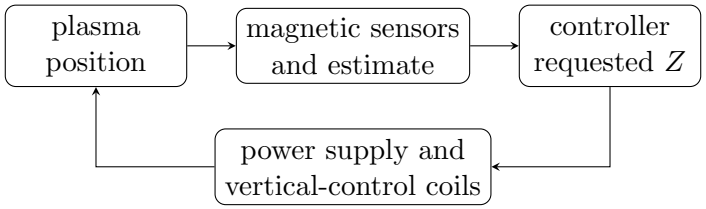


Figure 10. The feedback loop turns a measured displacement into a corrective magnetic force. Sensor, circuit and controller response times enter the achievable elongation.

15. Shape also changes the current allowance

The safety-factor relation from §7 can be generalized to the elongated geometry. The reduced reactor model of Freidberg, Mangiarotti and Minervini adopts

$$q_* = \frac{2\pi a^2 B_0}{\mu_0 R_0 I_p} \frac{1+\kappa^2}{2}. \tag{39}$$

The first factor is the circular edge estimate. The second accounts approximately for elongation in this large-aspect-ratio description. We take this shaped relation from the model, which uses it as a global measure of field-line pitch.

A *kink* is a collective displacement that bends the current-carrying plasma. The model represents its global current limit by requiring $q_* > q_K$, with an adopted threshold $q_K \simeq 2$. Substitution gives

$$I_p < \frac{2\pi a^2 B_0}{\mu_0 R_0 q_K} \frac{1+\kappa^2}{2}. \tag{40}$$

The shaped estimate and threshold are the closures used in Eqs. (30) and (39) of the 2015 reference. *Model used here.* Profiles, conducting-wall response and operating regime affect the physical stability boundary. The current ceiling here uses the stated large-aspect-ratio closure and $q_K \simeq 2$.

[5] J. P. Freidberg, A. Cerfon and J. P. Lee, "Tokamak elongation—how much is too much? Part 1. Theory," *Journal of Plasma Physics* **81**, 515810607 (2015), open manuscript.

Within that model, stronger toroidal field, greater minor radius and greater elongation permit more current. Increasing R_0 while holding the other quantities fixed reduces the allowance. This matters because the current that creates the poloidal confinement field also enters stability limits. Elongation offers both more volume and a larger current allowance, while demanding stronger position control. These consequences must be considered together.

> **Design consequence.** Elongation raises the current allowance in the adopted model. The chosen shape must also remain within the vertical-control capability.

16. The current profile determines the interior pitch

Total current is an area integral. In the circular approximation,

$$I_{\mathrm{enc}}(r) = 2\pi \int_0^r j_\varphi(r')r'\,dr', \qquad q(r) \simeq \frac{2\pi r^2 B_0}{\mu_0 R_0 I_{\mathrm{enc}}(r)}. \tag{41}$$

Two plasmas with the same $I_p = I_{\mathrm{enc}}(a)$ can therefore have different interior $q(r)$ if their current density is distributed differently. A centrally concentrated current places more enclosed current inside a small radius and lowers q there relative to a broader distribution with the same total current.

For example, a uniform $j_\varphi = j_0$ gives $I_{\mathrm{enc}} = \pi j_0 r^2$ and a constant $q \simeq 2B_0/(\mu_0 R_0 j_0)$ in this approximation. A varying profile changes that pitch from surface to surface. The radial variation of pitch is called *magnetic shear*. Both pitch and shear influence how disturbances couple across magnetic surfaces. Rational surfaces, where $q = m/n$, allow perturbations with matching periodicity to remain aligned with the field.

The transformer and auxiliary current-drive systems influence where current flows, as well as how much flows. Current control must account for $j_\varphi(r)$ and $q(r)$ alongside I_p.

> **Design consequence.** Total current alone does not fix $q(r)$. Current control must account for the distribution $j_\varphi(r)$ as well as I_p.

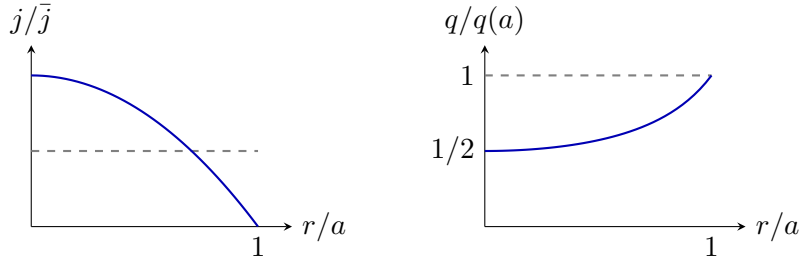


Figure 11. Illustrative circular, large-aspect-ratio profiles with the same total current: uniform $j = \bar{j}$ (dashed) and peaked $j = 2\bar{j}[1 - (r/a)^2]$ (blue), where $\bar{j} = I_p/(\pi a^2)$. The corresponding normalized pitches are 1 and $[2 - (r/a)^2]^{-1}$; the peaked profile lowers the interior pitch while preserving $q(a)$.

17. Why field and volume dominate a first power estimate

For an equal D–T mixture with total fuel-ion density n, $n_D = n_T = n/2$. In a uniform plasma the reaction rate per unit volume is $n_D n_T \langle\sigma v\rangle = n^2\langle\sigma v\rangle/4$. The *reactivity* $\langle\sigma v\rangle$ averages the cross-section

times relative speed over the ion distributions. At fixed temperature and fuel mixture,

$$P_{\text{fus}} = \frac{n^2}{4}\langle\sigma v\rangle Q V \propto n^2 V. \tag{42}$$

Using $p = 2nT$ and $p = \beta B_0^2/(2\mu_0)$ gives $n = \beta B_0^2/(4\mu_0 T)$, and hence

$$P_{\text{fus}} \propto \beta^2 B_0^4 V \propto \beta^2 B_0^4 R_0 a^2 \kappa. \tag{43}$$

Scope. This scaling holds temperature, composition and profile factors fixed, and uses B_0 as the reference magnetic field. For example, a 10% field increase gives a factor $1.1^4 \simeq 1.46$ in fusion power if beta and volume can be maintained. The fourth power explains the attraction of high field. Maintaining those other quantities requires compatible confinement, current and stability; producing the field also imposes loads on the magnet. The scaling identifies the incentive that later engineering calculations must price.

> **Design consequence.** The conditional $\beta^2 B_0^4 V$ scaling rewards field, pressure and volume. These quantities must be achievable together in one operating state.

18. The machine and its coupled variables

A component enters the ledger only after a section has forced it. We can now identify each component by the physical task that introduced it:

Component	Purpose
Vacuum vessel	Maintains the evacuated environment around the magnetically confined fuel.
TF coils	Produce the toroidal field that guides particle motion around the ring.
Central solenoid	Changes linked flux to initiate and ramp plasma current.
Plasma current	Generates the principal poloidal field and sets field-line pitch.
PF and control coils	Position and shape the plasma; apply corrective forces.
Sensors, controller, power supplies	Measure the state and command the coil response.
Beam or RF systems	Deposit auxiliary heat and, under suitable conditions, drive current.

The associated variables form a coupled set. Geometry (R_0, a, κ) determines volume; B_0 and I_p determine pitch through Eq. (39); the current profile sets interior $q(r)$. Density and temperature set pressure and reaction rate, while beta compares pressure with magnetic energy density. Finally, τ_E connects stored energy to the heating requirement.

The remaining task in this block is to put shape and force balance into one equation. We first add the D-shaped boundary parameter, then derive the equation that determines which boundary, pressure profile and current distribution can coexist.

> **Design consequence.** The components now have defined physical tasks. Their settings must support a common plasma state, requiring a shared description of shape and force balance.

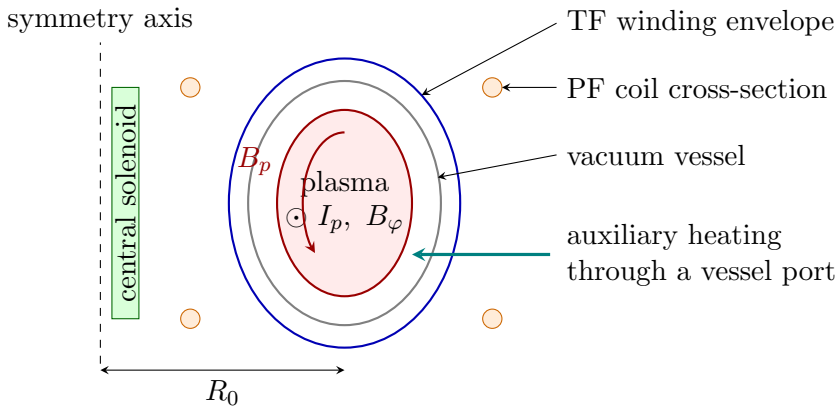


Figure 12. Component map in a poloidal cross-section. The TF winding wraps around the vessel; separate PF rings appear as small coil cross-sections. The central solenoid changes flux through the plasma ring. The winding and vessel outlines are schematic; a heating port is shown only by its access direction.

19. Triangularity adds a D-shaped boundary

Elongation specifies height relative to width. A second parameter measures the inward shift of the upper and lower tips of the cross-section. A useful boundary family is

$$\begin{aligned} R(\theta) &= R_0 + a\cos(\theta + x\sin\theta), \qquad x = \arcsin\delta, \\ Z(\theta) &= \kappa a \sin\theta. \end{aligned} \tag{44}$$

At the top, $\theta = \pi/2$, so $R_{\text{top}} = R_0 - a\sin x = R_0 - \delta a$. Thus

$$\delta = \frac{R_0 - R_{\text{top}}}{a} \tag{45}$$

is the *triangularity*. Positive δ pulls the top and bottom inward, producing a D-like cross-section. At $\delta = 0$ the boundary is the ellipse of §13. *Model used here.* The Miller parameterization specifies a boundary family. The small-δ harmonic expansion below retains terms through first order.

For small δ, expand $\cos(\theta + x\sin\theta)$ and use $\sin^2\theta = (1 - \cos 2\theta)/2$:

$$R(\theta) = R_0 - \frac{a\delta}{2} + a\cos\theta + \frac{a\delta}{2}\cos 2\theta + O(a\delta^2). \tag{46}$$

The added second angular harmonic and centring shift describe a shape change independent of elongation. The PF system must supply another adjustable spatial pattern to control it.

> **Design consequence.** Triangularity adds an independent boundary parameter. The PF system needs enough adjustable field patterns to control it alongside position and elongation.

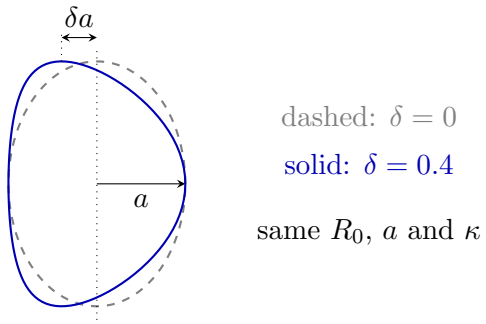


Figure 13. Triangularity shifts the upper and lower tips toward the central hole. Elongation and horizontal half-width are held fixed.

20. Coil placement controls the available shape patterns

In a current-free vacuum region away from the coils, $\nabla \times \mathbf{B} = 0$ and $\nabla \cdot \mathbf{B} = 0$. Locally we can write $\mathbf{B} = -\nabla \Phi_m$, so the magnetic scalar potential satisfies $\nabla^2 \Phi_m = 0$. A two-dimensional local example organizes the first patterns as

$$\Phi_1 = C_1 r \sin\theta, \qquad \Phi_2 = C_2 r^2 \sin 2\theta, \qquad \Phi_3 = C_3 r^3 \sin 3\theta.$$

Each is harmonic. With local Cartesian coordinates $x = r\cos\theta$, $z = r\sin\theta$, the first two become $C_1 z$ and $2C_2 xz$. Their fields are, respectively, uniform and linear in position. The third potential, $C_3(3x^2 z - z^3)$, gives a quadratic field.

A uniform vertical field changes the radial force on the plasma. The linear, *quadrupole* pattern contributes to elliptical shaping. The next, *hexapole* pattern supplies a further shaping degree of freedom useful for triangularity. These names describe successive spatial patterns of the field. The actual response of the boundary follows from force balance in toroidal geometry.

Each coil influences several patterns. Its position fixes that influence, while its power supply sets the current. Independent control of position and shape therefore requires enough independently adjustable coil circuits, with suitable placement and current range. More detailed shaping makes this an increasingly coupled control problem.

Design consequence. Each coil contributes to several field patterns. Coil placement and independently powered circuits must provide the required shaping freedom.

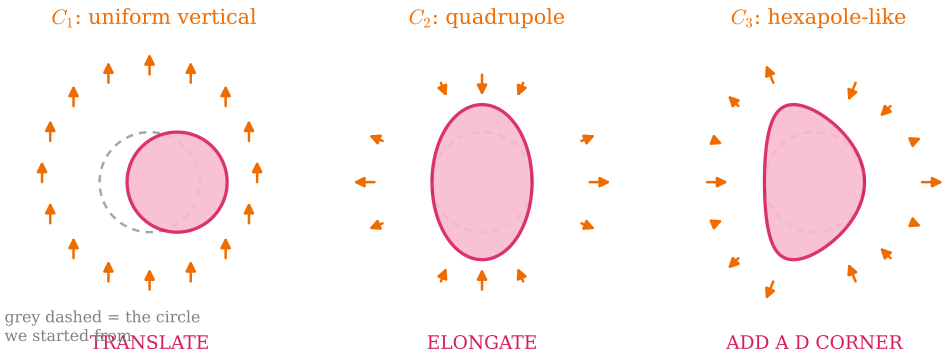


Figure 14. Successive field patterns supply additional shaping freedom.

21. Equilibrium joins pressure to magnetic surfaces

The static fluid balance derived in §8 is $\nabla p = \mathbf{j} \times \mathbf{B}$. Its projection along the field gave $\mathbf{B} \cdot \nabla p = 0$. In the nested-surface equilibrium considered here, we therefore write $p = p(\psi)$: one pressure value for each flux surface. The pressure profile specifies how that value changes from the hot interior to the edge.

This is the point where particle confinement becomes a collective design problem. Pressure, plasma current and the field from the coils must satisfy force balance at every point. *Model used here.* We retain the static, axisymmetric, isotropic-pressure model introduced in §8. Substantial flow or pressure anisotropy requires a generalized equilibrium. Each surface is labelled by its poloidal flux ψ, so $p = p(\psi)$ gives both a pressure value and, through $dp/d\psi$, the pressure change from one surface to the next. The spatial gradient is $\nabla p = (dp/d\psi)\nabla\psi$: its direction is normal to the surface and its magnitude depends on the spacing of the contours. This links a profile specified as a function of flux to the physical force density that the magnetic field must balance. Within the stated model, the geometry can therefore be reduced to a scalar differential equation.

Design consequence. Force balance makes pressure constant on connected magnetic surfaces. Describing those surfaces requires a common flux coordinate.

22. Two scalar functions describe the field

Combine the poloidal flux representation of §10 with a toroidal component written as $B_\varphi = F(R, Z)/R$:

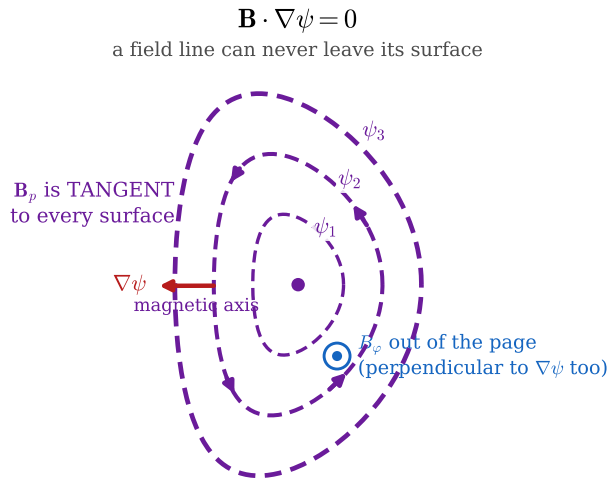
$$\mathbf{B} = \frac{1}{R}\nabla\psi \times \hat{\boldsymbol{\varphi}} + \frac{F}{R}\hat{\boldsymbol{\varphi}}. \tag{47}$$

Axisymmetry and $\nabla \cdot \mathbf{B} = 0$ permit this representation away from the coordinate axis. The contours of ψ locate the magnetic surfaces, and the function F sets the toroidal field strength. The factor $1/R$ retains the toroidal geometry.

We have already established $p = p(\psi)$. The toroidal component of force balance will now establish $F = F(\psi)$ as well. The two functions carry different information. Derivatives of $\psi(R, Z)$ give B_R and B_Z, so closely spaced contours mean a stronger poloidal field at the same R. Its contour shapes locate the magnetic surfaces and boundary. Meanwhile $F = RB_\varphi$ fixes the toroidal field; it has units of tesla-metres, whereas ψ has flux units. In a current-free toroidal-field region F is constant, recovering $B_\varphi \propto 1/R$. In the plasma, $F(\psi)$ can vary across surfaces and contributes to the current and force balance.

Specifying $p(\psi)$ and $F(\psi)$ thus supplies two surface profiles. It still leaves $\psi(R, Z)$ to be found: the map from those surface labels to real space. The Grad–Shafranov equation determines that map with the imposed boundary or external-coil conditions. Its solution simultaneously supplies the poloidal field, flux-surface geometry and toroidal current density.

Design consequence. The flux function specifies the surfaces and poloidal field. The next step is to solve for it using pressure and toroidal-field profiles.



Explanatory sketch. No hardware is drawn: this is the mathematical map that makes flux surfaces intelligible, not a picture of a machine.

Figure 15. Flux contours and their normal identify the poloidal field geometry.

23. Deriving the Grad–Shafranov equation

Insert Eq. (47) into $\mu_0\mathbf{j} = \nabla\times\mathbf{B}$. The toroidal current density comes from the curl of the poloidal field:

$$j_\varphi = -\frac{1}{\mu_0 R}\Delta^*\psi, \qquad \Delta^*\psi \equiv R\frac{\partial}{\partial R}\left(\frac{1}{R}\frac{\partial\psi}{\partial R}\right) + \frac{\partial^2\psi}{\partial Z^2}. \tag{48}$$

To check the sign, use $(\nabla\times\mathbf{B})_\varphi = \partial_Z B_R - \partial_R B_Z$ and the components in Eq. (27). The remaining components give

$$\mathbf{j}_p = \frac{1}{\mu_0 R}\nabla F\times\hat{\boldsymbol{\varphi}}.$$

Since pressure is axisymmetric, the toroidal component of $\mathbf{j}\times\mathbf{B}$ must vanish. Its value is proportional to $\mathbf{B}_p\cdot\nabla F$, so F is constant along the poloidal field. On the nested surfaces we can write $F = F(\psi)$ and hence $\mathbf{j}_p = F'(\psi)\mathbf{B}_p/\mu_0$.

Only two cross products remain: $j_\varphi\hat{\boldsymbol{\varphi}}\times\mathbf{B}_p$ and $\mathbf{j}_p\times B_\varphi\hat{\boldsymbol{\varphi}}$. Using $\hat{\boldsymbol{\varphi}}\times\mathbf{B}_p = \nabla\psi/R$, they give

$$\mathbf{j}\times\mathbf{B} = -\frac{\Delta^*\psi + FF'}{\mu_0 R^2}\nabla\psi, \qquad \nabla p = p'\nabla\psi.$$

Equating coefficients normal to a regular flux surface yields

$$\boxed{\Delta^*\psi = -\mu_0 R^2\frac{dp}{d\psi} - F\frac{dF}{d\psi}.} \tag{49}$$

This is the *Grad–Shafranov equation*. The pressure gradient and toroidal-field profile supply its source terms; the solution supplies the flux surfaces.

Scope. At a smooth magnetic axis or null, the equation is interpreted through the regular solution. The intermediate cancellation of $\nabla\psi$ applies where that gradient is nonzero.

The derivation reveals why geometry alone is insufficient to specify a tokamak. Changing pressure or current changes the differential equation that its magnetic surfaces must satisfy. Coil settings then have to support the resulting equilibrium.

24. From an equilibrium equation to an operating machine

There are two useful ways to pose the calculation. In a *fixed-boundary* calculation we prescribe the plasma boundary and the profiles $p(\psi)$ and $F(\psi)$, then solve for the interior flux surfaces and current. In a *free-boundary* calculation we include the external-coil fields and surrounding vacuum; the plasma boundary is part of the solution. A desired shape can thus be tested first, then translated into requirements on coil locations and currents. The analytic equilibria cited in §10 illustrate how profiles and boundary geometry enter this problem.

The magnetic axis generally shifts outward relative to the geometric centre of the boundary. This *Shafranov shift* reflects the pressure and current contributions to toroidal force balance. Its size depends on the profiles and geometry. When those profiles change during heating or a current ramp, the equilibrium changes, and the PF currents needed to maintain the requested boundary change with it.

A D-shaped plasma and a D-shaped TF coil have separate design purposes. Triangularity specifies a plasma boundary chosen through equilibrium and stability considerations. A D-like TF-coil shape can instead be chosen to carry magnetic loads mainly in tension and reduce conductor bending. The engineering design must accommodate both the required magnetic surfaces and the forces on the conductors that produce their field.

We have reached a working description of machine formation. Hot charged particles require a vacuum environment and magnetic guidance; closing the field introduces curvature drifts; plasma current supplies the poloidal twist; pressure demands external equilibrium fields; transformer limits demand auxiliary heating and current drive; useful shaping demands feedback. The Grad–Shafranov equation joins these choices into a calculable equilibrium. The next block asks what fuel state and confinement this machine must sustain to produce useful fusion power.

Design consequence. Pressure and current evolution change the equilibrium. Each operating state needs compatible profiles, a magnetic boundary and PF-coil settings.

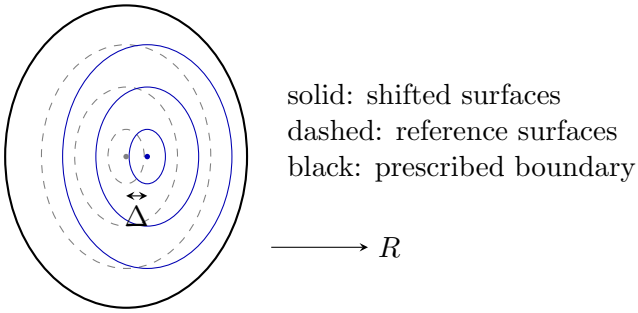


Figure 16. Schematic outward displacement of the magnetic axis relative to a fixed boundary. The interior contours illustrate the shift only; an equilibrium calculation determines their deformation and spacing for the specified profiles.

Let the Machine Reveal Itself

Fusion power, operating limits and radial build

Jony (TUNEM)

Compression review v2 — Sections 25–50 — October 2026

This block continues the revised opening. Section and equation numbers retain their thesis sequence; references to §§1–24 refer to the approved 18-page opening.

25. From individual collisions to a fusion-power density

Consider a small volume dV containing $n_D dV$ deuterons and $n_T dV$ tritons. Choose one deuteron. During a short time dt, tritons with relative speed v sweep through the effective collision tube of area $\sigma(v)$ and length $v dt$. The expected number of its reactions is

$$n_T\, \sigma(v)\, v\, dt.$$

Particles in a thermal plasma do not all have the same relative speed. Average this expression over the normalized relative-speed distribution and multiply by the number $n_D dV$ of deuterons. Dividing by $dV\, dt$ gives the reaction rate per unit volume:

$$\mathcal{R}_{DT} = n_D n_T \langle \sigma v \rangle, \qquad \langle \sigma v \rangle = \int_0^\infty \sigma(v) v\, f_{\mathrm{rel}}(v)\, dv. \tag{53}$$

The measured cross-section $\sigma(v)$ supplies the nuclear reaction probability; $f_{\mathrm{rel}}(v)$ supplies the thermal statistics.

If each reaction releases total energy $E_f = 17.6\,\mathrm{MeV}$, then

$$p_{\mathrm{fus}} = E_f \mathcal{R}_{DT} = E_f n_D n_T \langle \sigma v \rangle, \qquad P_{\mathrm{fus}} = p_{\mathrm{fus}} V. \tag{54}$$

This gives the factors behind the scaling in §17: density, volume and the temperature-dependent reactivity jointly determine fusion output.

Model used here. Equation (53) assumes independent binary collisions and a thermal distribution. At reactor density, D–T fusion remains dilute enough for that description; collective plasma physics controls confinement, not the short-range nuclear reaction once two ions meet.

> **Design consequence.** Fusion output depends on the product $n_D n_T$. At fixed fuel density, the mixture must maximize that product.

26. Why the fuel mixture is nearly fifty–fifty

Let the total fuel-ion density be fixed:

$$n_i = n_D + n_T.$$

Write $n_D = x n_i$. Then $n_T = (1-x)n_i$, and the density product in Eq. (54) is

$$n_D n_T = x(1-x)n_i^2. \tag{55}$$

Complete the square:

$$x(1-x) = x - x^2 = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2.$$

The squared term is nonnegative, so the largest possible value is $1/4$, achieved at $x = 1/2$. Thus

$$n_D = n_T = \frac{n_i}{2} \quad \Longrightarrow \quad p_{\text{fus}} = \frac{1}{4} n_i^2 \langle \sigma v \rangle E_f. \tag{56}$$

Equal fuel fractions maximize the reaction rate at fixed total fuel-ion density and temperature.

The pressure used in magnetic confinement also includes electrons. A quasi-neutral 50–50 D–T mixture has electron density $n_e = n_i$. We write the primary fuel density as $n \equiv n_i$. If the ions and electrons share a temperature T measured in energy units, their ideal-gas pressures add:

$$p = (n_i + n_e)T = 2 n_i T. \tag{57}$$

Combining Eqs. (56) and (57) explains the scaling chain on §17, but with its factors visible: at fixed T, $n_i = p/(2T)$ and hence $p_{\text{fus}} \propto p^2$.

> **Design consequence.** The density needed for fusion also creates pressure. The same fuel state must satisfy the magnetic-pressure and density constraints.

27. Where the fusion energy goes decides what the machine must add

The D–T products are an alpha particle and a neutron. In the center-of-mass frame they leave with equal and opposite momentum, so their momentum magnitudes are equal: $m_\alpha v_\alpha = m_n v_n$. Their kinetic energies are $E_\alpha = p^2/(2m_\alpha)$ and $E_n = p^2/(2m_n)$. Therefore

$$\frac{E_n}{E_\alpha} = \frac{m_\alpha}{m_n} \simeq 4, \qquad E_\alpha = \frac{1}{5} E_f \simeq 3.5\,\text{MeV}, \quad E_n = \frac{4}{5} E_f \simeq 14.1\,\text{MeV}.$$

The charged alpha particle can remain confined long enough to slow down by collisions and heat the plasma. The neutron has no electric charge, so magnetic fields do not bend it; it escapes the plasma and deposits its energy only after meeting material outside the vacuum region.

With complete alpha-energy deposition, $P_\alpha = (E_\alpha/E_f)P_{\text{fus}} \simeq P_{\text{fus}}/5$. Neutron power is $P_n \simeq 4P_{\text{fus}}/5$. The first term enters the plasma self-heating balance; the second must be captured outside the confined plasma. *Model used here.* The power balances below assume thermal fuel, common ion and electron temperature, and complete alpha-energy deposition. Alpha escape reduces the deposited fraction.

> **Design consequence.** Alpha energy can sustain the plasma; neutron energy must be collected in surrounding material. The two fractions enter different power balances.

28. Stored energy and the meaning of confinement time

Before asking whether alpha heating is enough, we need a careful measure of loss. For an ideal gas with three translational degrees of freedom, the internal energy density is $3nT/2$. In the equal-temperature D–T plasma there are n_i ions and $n_e = n_i$ electrons, so the total thermal energy is

$$W = \left(\frac{3}{2}n_iT + \frac{3}{2}n_eT\right)V = 3n_iTV. \tag{58}$$

The energy-confinement time is defined, rather than guessed, by

$$\tau_E \equiv \frac{W}{P_{\text{loss}}}, \qquad P_{\text{loss}} = \frac{W}{\tau_E}. \tag{59}$$

It answers a plain question: if all heating stopped and the loss rate stayed at its instantaneous value, how long would the present thermal energy take to leave?

Here P_{loss} includes transport and radiation, each counted once. The instantaneous ratio W/P_{loss} remains useful even when the energy decay is not exponential.

Using Eq. (57), Eq. (58) can also be written $W = (3/2)pV$. Thus a high-pressure plasma stores more energy at a given volume, but it also demands proportionally more heating power if τ_E is fixed.

> **Design consequence.** The loss power is W/τ_E. Ignition requires alpha heating to cover it at the proposed density and temperature.

29. The Lawson condition follows from one inequality

Ignition means that alpha heating alone can replace the plasma losses:

$$P_\alpha \geq P_{\text{loss}}. \tag{60}$$

Use the equal-mixture fusion result, Eq. (56), but retain only the alpha energy E_α:

$$P_\alpha = \frac{1}{4}n_i^2\langle\sigma v\rangle E_\alpha V.$$

Use Eq. (58) and Eq. (59) for the loss:

$$P_{\text{loss}} = \frac{3n_iTV}{\tau_E}.$$

Substitution into Eq. (60) gives

$$\frac{1}{4}n_i^2\langle\sigma v\rangle E_\alpha V \geq \frac{3n_iTV}{\tau_E}.$$

Cancel the positive common factor n_iV, then multiply by $4\tau_E$:

$$\boxed{n_i\tau_E \geq \frac{12T}{\langle\sigma v\rangle E_\alpha}.} \tag{61}$$

Every factor has a visible origin: $1/4$ from the equal fuel mix, 3 from ions plus electrons and three translational degrees of freedom, and E_α rather than E_f because neutrons do not heat the confined plasma.

Multiplying Eq. (61) by T gives the triple product $n_iT\tau_E$. Its threshold depends on the chosen temperature and D–T reactivity.

Design consequence. Ignition requires a compatible combination of density, temperature and confinement. External heating supplies any remaining power deficit.

30. Gain is a ratio, not a synonym for ignition

A plasma can make useful fusion power without being ignited if external heating supplies the missing energy. Define the plasma gain

$$Q_{\text{plasma}} \equiv \frac{P_{\text{fus}}}{P_{\text{aux}}}, \tag{62}$$

where P_{aux} is the heating delivered to the plasma by neutral beams, radio frequency systems, or other actuators. In steady state, a stripped-down energy balance is

$$P_\alpha + P_{\text{aux}} = P_{\text{loss}}. \tag{63}$$

As P_{aux} tends to zero while the equality remains possible, Q_{plasma} tends to infinity: that limiting case is ignition. At finite Q_{plasma}, the plant must supply auxiliary power and also pay the inefficiency of producing and delivering it.

Net electric power additionally depends on neutron-energy capture, thermal conversion and recirculating loads such as refrigeration, pumps and controls. Q_{plasma} measures the plasma contribution to that wider power balance.

Design consequence. Finite plasma gain requires auxiliary power. Its required level depends on transport through τ_E.

31. Why transport is the unresolved clock

A particle in a uniform magnetic field circles with a characteristic gyroradius ρ. A collision changes the direction of its velocity; the center of the next gyro-orbit is displaced by an amount of order ρ. If collisions occur at rate ν, then in time t there are approximately $N = \nu t$ independent steps. Random-walk addition gives

$$\langle \Delta r^2 \rangle \sim N\rho^2 = \nu t \rho^2.$$

By the definition $\langle \Delta r^2 \rangle \sim 2Dt$ in one radial direction, the classical cross-field diffusion estimate is

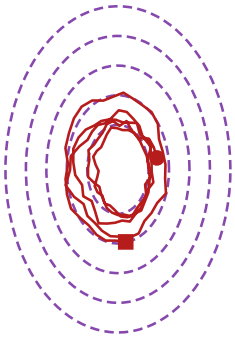
$$D_{\text{cl}} \sim \frac{\rho^2 \nu}{2}. \tag{64}$$

A container of minor radius a would then have a diffusion time of order

$$\tau_{\text{diff}} \sim \frac{a^2}{D_{\text{cl}}}. \tag{65}$$

This is a transparent reason strong field helps: $\rho \propto 1/B$, so the classical diffusion coefficient falls roughly as $1/B^2$ if the collision rate is held fixed.

Measured energy transport often exceeds this classical estimate. Fluctuations and turbulence correlate motion across flux surfaces. The random walk supplies a baseline; reactor prediction uses a confinement model calibrated to experiments.



one guiding centre, 150 collisions:
each kick moves it about one gyro-radius ρ

the estimate that follows

$N = \nu t$ collisions in time t

$\langle \Delta r^2 \rangle \sim N\rho^2 = \nu t\, \rho^2$

$D_{\mathrm{cl}} \sim \rho^2 \nu / 2$

a diffusion coefficient built from ONE step size and ONE step rate --- this is the CLASSICAL estimate

Real tokamaks lose heat faster than this. Toroidal geometry adds a neoclassical enhancement, and TURBULENCE adds a further channel that this picture does not model at all.

Explanatory sketch; schematic, not to scale.

Figure 17. Independent collisional steps provide a classical transport scale; correlated fluctuations add further transport.

Design consequence. Cross-field transport sets the loss time. Reactor calculations therefore need a confinement model with a stated applicability range.

32. Confinement is calibrated to experiments

For reactor studies, the working confinement time is often represented as

$$\tau_E = H\, \tau_{E,\mathrm{ref}}(I_p, B_0, n, P_{\mathrm{loss}}, R_0, a, \kappa, \ldots). \tag{66}$$

Here $\tau_{E,\mathrm{ref}}$ is a regression fitted to a specified database of experiments, and H is the ratio of the observed confinement to that reference. Its use requires the database, operating regime, units, heating conditions and shape range of the selected regression. Treat the regression as data with a stated applicability range.

The distinction between a derived equation and a fit matters especially here. Equation (64) followed from a stated random-walk model. Equation (66) is a bookkeeping identity once a reference scaling is chosen; its predictive content lives in the data and the applicability range of $\tau_{E,\mathrm{ref}}$.

A practical design calculation must therefore report at least three things: the selected reference law, the assumed H, and a sensitivity range. An assumed H above comparable experimental performance identifies the confinement improvement the design requires.

Scope. Confinement is represented here by an empirical reference law and multiplier H. The selected law, units and applicability range accompany the numerical reconstruction.

Design consequence. Confinement closes the energy balance alongside geometry and pressure. The assumed reference law and H must be tested together.

33. Pressure cannot rise freely: the origin of a beta limit

The local pressure-versus-line-bending balance gives a scaling estimate. Practical stability thresholds require a specified equilibrium and stability model.

The beta in §8 compares plasma pressure with magnetic pressure:

$$\beta = \frac{p}{B_0^2/(2\mu_0)}.$$

Why should there be an upper limit? Imagine a bulge of a pressure-carrying flux tube. The pressure force trying to enlarge it scales as p times its cross-sectional area. Bending the field lines that surround it costs magnetic energy; the restoring scale is set by B_θ^2/μ_0, where B_θ is the poloidal field that ties the plasma current to the geometry.

Ampere's-law scale at the edge is

$$B_\theta \sim \frac{\mu_0 I_p}{2\pi a}.$$

Equating the pressure and field-bending scales gives

$$p_{\max} \sim \frac{B_\theta^2}{\mu_0} \sim \frac{\mu_0 I_p^2}{4\pi^2 a^2}.$$

Divide by $B_0^2/(2\mu_0)$ to obtain the dimensionless structural form

$$\beta_{\max} \propto \left(\frac{I_p}{aB_0}\right)^2 \quad \text{in this crude local balance.} \tag{67}$$

The balance identifies pressure drive and magnetic restoring force. A global ideal-MHD calculation includes geometry and profiles when determining a stability boundary.

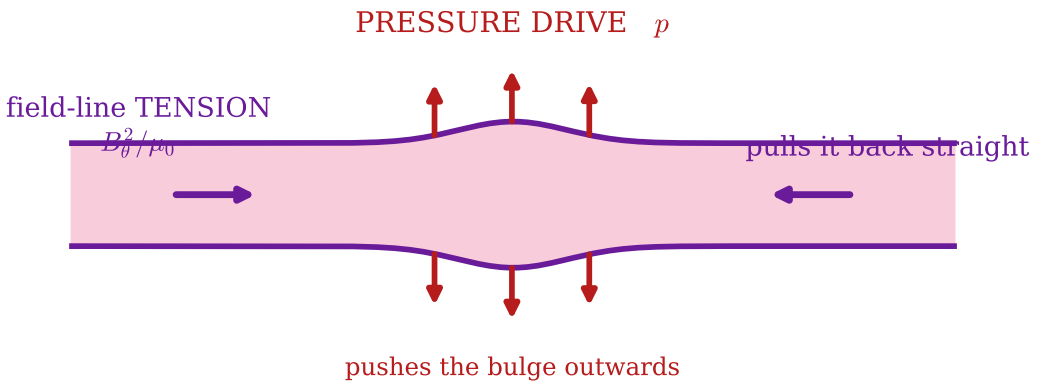


Figure 18. Pressure drives a bulge while magnetic line bending supplies a restoring force. The local balance gives a scaling estimate.

Design consequence. Pressure support depends on current, field and shape. A practical stability constraint must express that coupling.

34. The normalized beta used in reactor design

Detailed ideal-MHD calculations and experiments are commonly summarized by a normalized beta. With β written as a fraction rather than a percentage, define

$$\beta_N \equiv \frac{\beta\, a(\mathrm{m})\, B_0(\mathrm{T})}{I_p(\mathrm{MA})}. \tag{68}$$

This numerical convention uses a in metres, B_0 in tesla and I_p in MA. Using beta in percent instead would give a normalized-beta value 100 times larger. It is useful because comparable tokamak equilibria show an approximate operating boundary of the form

$$\beta \lesssim \beta_{N,\max} \frac{I_p(\mathrm{MA})}{a(\mathrm{m}) B_0(\mathrm{T})}. \tag{69}$$

The coefficient $\beta_{N,\max}$ is empirical/model-dependent; it must be imported from a specified stability study or design paper.

Equation (69) makes the current–pressure coupling explicit: larger allowable current raises allowable pressure, while a larger aB_0 at fixed current lowers this particular normalized beta. Combining it with the current ceiling on §15 makes the coupling visible: the kink limit restricts I_p, which then restricts the pressure that can be claimed for fusion power.

> **Design consequence.** The pressure allowance depends on the current permitted by kink stability. Both constraints must hold at the same operating point.

35. Density has a separate experimental limit

It would be tempting to use Eq. (57) and conclude that a chosen pressure can always be obtained by trading density against temperature. Tokamak operation does not permit an unlimited trade. A widely used experimental reference is the Greenwald density

$$n_G = \frac{I_p(\mathrm{MA})}{\pi a^2(\mathrm{m}^2)} \times 10^{20}\ \mathrm{m}^{-3}. \tag{70}$$

Operational scenarios usually report the fraction $f_G = n_e/n_G$, rather than merely a raw density.

Scope. The Greenwald relation is an empirical operating reference. The chosen scenario must state the density averaging convention and allowable fraction f_G; the numerical comparison uses a line-averaged electron density.

The formula still carries useful transparent geometry. At fixed current, a smaller cross-section raises the reference density as $1/a^2$. At fixed a, more current raises it. But that same current is limited by q_* and the same density contributes to pressure and radiation. The fraction f_G records the resulting density demand relative to the reference.

> **Design consequence.** Density has its own operating constraint. Increasing it must remain compatible with pressure, current and confinement.

36. Toroidal geometry traps some particles

The toroidal field is stronger on the inboard side because approximately $B_\varphi \propto 1/R$. A particle moving into stronger field converts parallel motion into perpendicular motion while conserving, to lowest adiabatic order,

$$\mu = \frac{m v_\perp^2}{2B} \quad \text{and} \quad \frac{1}{2} m (v_\parallel^2 + v_\perp^2).$$

Let the particle start where the field is $B_{\min}$ and let α be its pitch angle there, so $v_\perp = v \sin\alpha$. At the turning point $v_\parallel = 0$. Conservation of μ gives

$$\frac{v^2 \sin^2\alpha}{B_{\min}} = \frac{v^2}{B_{\max}}, \qquad \sin^2\alpha = \frac{B_{\min}}{B_{\max}}.$$

Particles with larger perpendicular fraction reach this condition before the high-field side and bounce back. For isotropically distributed velocity directions evaluated at $B_{\min}$, the local trapped fraction is therefore

$$f_t = \sqrt{1 - \frac{B_{\min}}{B_{\max}}}. \tag{71}$$

The same $1/R$ field that closes the magnetic corridor therefore creates trapped orbits. Neoclassical transport combines these toroidal guiding-centre orbits with Coulomb collisions; pressure gradients also produce a self-generated current.

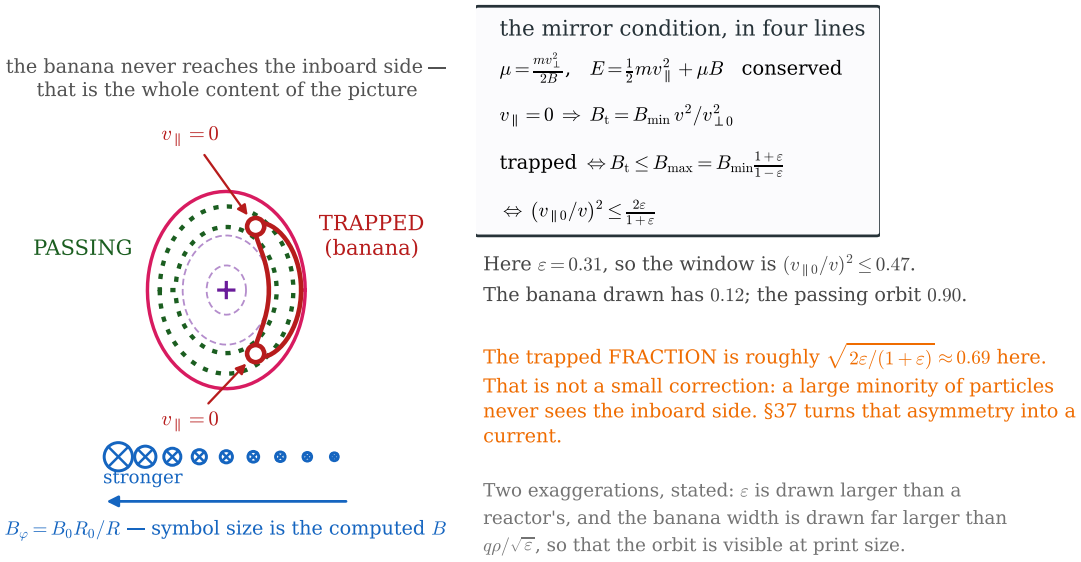


Figure 19. Passing and trapped guiding-centre orbits. Turning points follow from conservation of magnetic moment and energy; orbit widths are schematic.

Design consequence. Toroidal field variation creates trapped particles. Their response to a pressure gradient introduces a source of plasma current.

37. A pressure gradient can generate some of the required current

A trapped particle does not sample the two sides of the torus symmetrically. Its guiding-center drifts and collisions depend on energy and pitch angle. When pressure falls outward, there are more energetic particles on the inner side of a local radial interval than on the outer side. The resulting imbalance in trapped-particle motion is communicated by collisions to passing electrons. The outcome is a net toroidal current called the *bootstrap current*.

The direction and detailed coefficient require kinetic theory, because they depend on collisions, geometry, temperature gradients, and the current profile. What can be stated before importing such a coefficient is the structure:

$$j_{\mathrm{bs}} \propto -\frac{dp}{dr} \quad \text{and} \quad I_{\mathrm{bs}} = \int j_{\mathrm{bs}}\, dA. \tag{72}$$

No pressure gradient means no bootstrap drive in this simplified statement. A steeper gradient can produce more self-driven current, but it also moves the plasma closer to pressure-driven stability limits. Bootstrap current couples the pressure profile to the current needed for magnetic geometry.

Define the bootstrap fraction

$$f_{\mathrm{bs}} \equiv \frac{I_{\mathrm{bs}}}{I_p}. \tag{73}$$

The fraction f_{bs} is the primary parameter f_B. A kinetic model supplies its profile-dependent value, which sets the remaining transformer or external-drive demand.

Model used here. The proportionality describes the pressure-gradient drive in a reduced model. A quantitative prediction requires geometry, collisionality, species gradients and profiles.

> **Design consequence.** The pressure profile supplies part of the current. Its bootstrap contribution must remain compatible with stability and confinement.

38. The full current ledger

The toroidal plasma current can have several sources. Write the accounting identity

$$I_p = I_{\mathrm{ohm}} + I_{\mathrm{CD}} + I_{\mathrm{bs}}. \tag{74}$$

Equation (74) is an accounting identity; performance enters through the three terms. The first term is transformer-induced ohmic current; it is available while the central solenoid changes linked flux. The second is externally driven non-inductive current, for example from waves or beams transferring directed momentum. The third is the pressure-gradient-driven bootstrap current from §37.

For a stationary non-inductive state with negligible transformer contribution, $I_{\mathrm{ohm}} \simeq 0$, and Eq. (74) reduces to

$$I_{\mathrm{CD}} = (1 - f_{\mathrm{bs}}) I_p. \tag{75}$$

The larger the bootstrap fraction, the smaller the externally driven current requirement. But current drive consumes power and its efficiency depends on density, temperature, magnetic field, launch geometry, and the actuator itself. The bootstrap fraction must come from a compatible profile model.

> **Design consequence.** The required current must be supplied by transformer action, bootstrap current and affordable external drive. Their balance depends on operating mode.

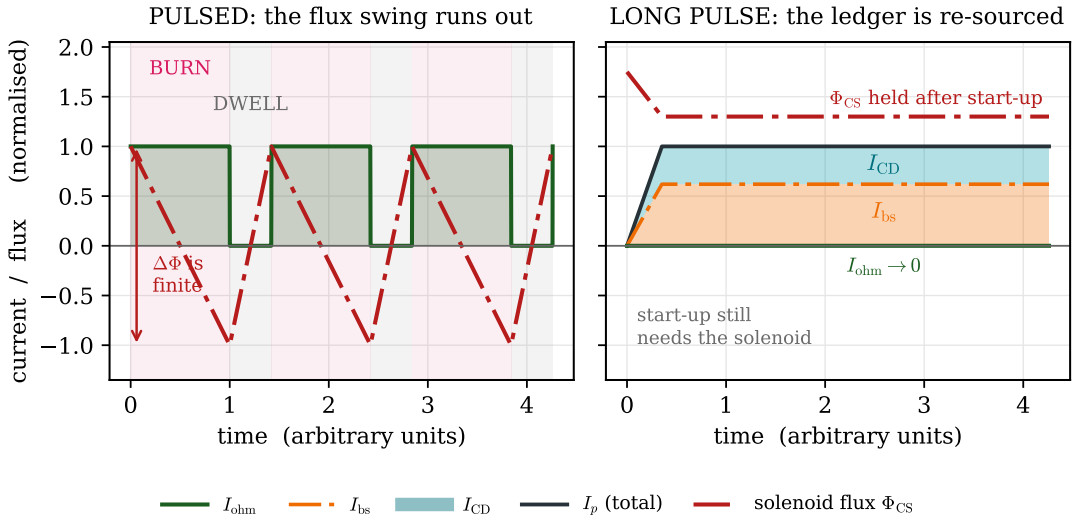
39. Pulsed operation is also a legitimate design choice

The central solenoid is a transformer primary. Faraday's law gives the loop voltage around the torus:

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d\Phi_{\mathrm{link}}}{dt}.$$

For a torus of major radius R_0, a representative toroidal electric field satisfies $2\pi R_0 E_\varphi = -d\Phi_{\mathrm{link}}/dt$. The solenoid has a finite usable flux swing $\Delta\Phi_{\mathrm{link}}$, so it can induce current only for a finite time. This is the physical origin of the transformer clock introduced on §11.

There are two coherent responses. One is a pulsed plant: use transformer current during a burn, then recharge the solenoid between pulses. The other is a long-pulse or steady-state plant: replace the lost ohmic contribution with the current-drive and bootstrap terms in Eq. (74). Neither option is free. Pulsing creates thermal cycling and dwell-time questions; non-inductive operation creates actuator power and accessibility questions. The operating choice must account for both the plant duty cycle and actuator demand.



Left, the only arithmetic on the figure: $V_{\mathrm{loop}} = -d\Phi/dt$ and $I_{\mathrm{ohm}} = V_{\mathrm{loop}}/R_p$ give $t_{\mathrm{burn}} = \Delta\Phi/(I_{\mathrm{ohm}} R_p)$ — a finite swing buys a finite burn. (Internal inductance is neglected; start-up consumes part of $\Delta\Phi$ before the burn begins.) Right, the $I_{\mathrm{bs}}/I_{\mathrm{CD}}$ split drawn is ILLUSTRATIVE: only the sum is constrained here. Neither panel claims its architecture is preferable— each states a different price.

Figure 20. Two current-supply strategies: transformer-supported burns with recharge intervals, or non-inductive current sustaining the burn. The illustrated current shares are schematic.

Design consequence. Transformer flux limits the inductive burn. Extending it requires more usable flux or a larger non-inductive current contribution.

40. The kink limit returns as a constraint on every current source

The current ledger does not allow arbitrary total current. A helical disturbance can be labelled by integers (m, n) through a displacement pattern $\xi \propto \cos(m\theta - n\varphi)$. Along an unperturbed field line, a poloidal turn changes θ by 2π while the toroidal angle changes by $2\pi q$. The phase change is therefore $2\pi(m - nq)$. When $q = m/n$, the disturbance remains in phase after a circuit; field-line bending has the least geometric leverage to average it away.

This is the accessible reason safety factor is a stability variable rather than only a geometric ratio. The full ideal-MHD energy principle determines the precise stability boundary. Reactor studies commonly use a specified global constraint

$$q_* > q_K, \tag{76}$$

with q_K supplied by the chosen stability model. Substituting the shaped safety-factor formula, Eq. (39), reproduces the current ceiling already derived in Eq. (40). Every term in the current ledger must fit under that same ceiling; bootstrap current helps meet a required I_p but does not abolish the stability constraint on I_p itself.

Model used here. The resonance argument motivates the role of field-line pitch. The global threshold q_K is supplied separately by the selected stability model.

> **Design consequence.** Kink stability constrains the total current, regardless of its source. Bootstrap current reduces external drive demand within that same ceiling.

41. The operating constraints must overlap

For an ignited candidate state, the coupled operating requirements include

$$q_* > q_K \qquad \text{current and kink margin,} \tag{77}$$

$$\beta \lesssim \beta_{N,\max}\frac{I_p}{aB_0} \qquad \text{pressure and MHD margin,} \tag{78}$$

$$n_e \lesssim n_G \qquad \text{density reference,} \tag{79}$$

$$n_i\tau_E \geq \frac{12T}{\langle\sigma v\rangle E_\alpha} \qquad \text{self-heating requirement,} \tag{80}$$

$$I_p = I_{\text{ohm}} + I_{\text{CD}} + I_{\text{bs}} \qquad \text{current accounting.} \tag{81}$$

The beta relation uses the metre–tesla–MA convention of §34. The notation $\lesssim$ in the second and third lines is deliberate. Those limits are modelled or empirical margins, not exact equalities. The last line is an identity, but its terms depend on operating mode and profiles. The first and fourth lines are inequalities derived from explicit physical balances, with a stability threshold or reactivity curve still to be specified.

Increasing density might help Eq. (80) but challenge Eq. (79); increasing current can improve one normalized beta bound but challenge Eq. (77); improving confinement changes needed heating and current drive. The correct question is whether the inequalities overlap with usable margin.

> **Design consequence.** An operating point must satisfy the coupled constraints with margin. The remaining geometric input is the radial space required around the plasma.

42. From plasma requirements to the radial build

The plasma-side roles of Freidberg's Table-1 parameters: $a, R_0, \kappa, T, n, p, \tau_E, B_0, \beta, I, q_*, f_B$ have now been introduced. Quantities such as $A, \delta, q(r), \beta_N, H, I_{\text{CD}}, Q_{\text{plasma}}$ remain supporting quantities, not independent primary knobs. The missing hardware parameters are b and c; they must be derived before numerical closure begins. Neutron interactions, magnetic forces and current-carrying area determine their roles.

Design consequence. Neutron capture and magnet forces require radial space. Deriving b and c completes the reactor geometry.

43. A magnetic bottle does not catch the neutron

The D–T reaction in Eq. (1) has two products with very different fates. The charged alpha particle remains magnetically guided and can transfer energy back to the plasma. The neutron has no electric charge. Its Lorentz force is exactly zero:

$$\mathbf{F}_n = 0 \cdot (\mathbf{E} + \mathbf{v}_n \times \mathbf{B}) = 0.$$

It crosses the magnetic surfaces in a nearly straight line until it interacts with surrounding material.

The neutron carries 14.1 MeV, approximately four fifths of the fusion energy. A power-producing machine must therefore do four jobs outside the plasma: survive the first impact, slow and capture neutrons, breed replacement tritium, and protect the magnets and vacuum boundary from the remaining radiation. These jobs force a radial region between plasma and TF coil. We call its total inboard/outboard thickness b.

Design consequence. Neutron energy requires a surrounding nuclear region. Its thickness competes with the radius available to the inboard magnet.

Scope. The following one-dimensional model relates thickness to interaction lengths. Material-specific neutron transport, heat removal and tritium recovery determine the engineered build.

44. Why the nuclear island has a thickness

First consider the slowing-down part of the problem. In a homogeneous material with target number density N and an effective interaction cross section σ_s, the mean distance between relevant collisions is

$$\lambda_s = \frac{1}{N\sigma_s}. \tag{82}$$

This follows directly from probability. In a thin distance dx, a neutron has collision probability $N\sigma_s dx = dx/\lambda_s$. Thus the chance of no collision decreases exponentially with distance.

For a transparent first model, suppose each collision removes a fixed fraction ξ of the neutron energy on average. Over many collisions, the average energy loss per distance can be represented by

$$\frac{dE}{dx} = -\frac{\xi E}{\lambda_s}. \tag{83}$$

Separating variables gives

$$\frac{dE}{E} = -\frac{\xi\, dx}{\lambda_s} \quad \Longrightarrow \quad E(x) = E_0 \exp\!\left(-\frac{\xi x}{\lambda_s}\right).$$

To reduce a 14.1 MeV neutron from E_0 to a chosen capture-friendly energy E_c, the associated slowing-down distance is therefore

$$b_{\text{slow}} = \frac{\lambda_s}{\xi} \ln\!\left(\frac{E_0}{E_c}\right). \tag{84}$$

The logarithm matters: a large energy ratio does not require a proportional thickness, but it does require several interaction lengths. Equation (84) already shows why a thin wall cannot be a reactor solution.

Now consider breeding. One useful reaction is

$$n + {}^{6}\mathrm{Li} \longrightarrow {}^{4}\mathrm{He} + {}^{3}\mathrm{H} + 4.8\,\mathrm{MeV}.$$

The emitted tritium replaces fuel consumed by D–T fusion. If slow neutrons are absorbed with macroscopic probability per length $\Sigma_B = N_6\sigma_B$, their surviving flux Γ obeys

$$\frac{d\Gamma}{dx} = -\Sigma_B\Gamma, \qquad \Gamma(x) = \Gamma(0)e^{-x/\lambda_B}, \qquad \lambda_B = \frac{1}{\Sigma_B}.$$

Capturing a chosen fraction f_{cap} then requires

$$b_{\mathrm{breed}} = -\lambda_B \ln(1 - f_{\mathrm{cap}}). \tag{85}$$

Both λ_s and λ_B depend on material and neutron energy. Thus Eqs. (84)–(85) retain those material and energy dependencies.

45. From microscopic collisions to the reactor dimension b

The nuclear region performs several functions. At the level needed for a first reactor layout, write

$$b = b_{\mathrm{FW}} + b_{\mathrm{mult}} + b_{\mathrm{slow}} + b_{\mathrm{breed}} + b_{\mathrm{shield}} + b_{\mathrm{vac}}. \tag{86}$$

Here b_{FW} is the first-wall allowance; b_{mult} represents any neutron-multiplying region; b_{slow} and b_{breed} are motivated by the preceding transport calculation; b_{shield} protects coils and surrounding structures; and b_{vac} is the vacuum-boundary allowance. Some concepts combine or reorder these functions. The terms in Eq. (86) are non-overlapping radial allowances; combined functions are counted once.

The first wall also sees a power-density demand. In the large-aspect-ratio circular approximation, a torus with major radius R_0 and minor radius a has surface area

$$S_{\mathrm{wall}} \simeq (2\pi R_0)(2\pi a) = 4\pi^2 R_0 a.$$

If P_n is the neutron power and W_n is an allowed average neutron wall loading,

$$\frac{P_n}{4\pi^2 R_0 a} \le W_n \qquad \Longrightarrow \qquad R_0 a \ge \frac{P_n}{4\pi^2 W_n}. \tag{87}$$

Elongation and detailed surface shape modify this estimate; they do not remove its message. At fixed fusion-neutron power, a smaller first-wall area means a harsher wall loading. Nuclear performance therefore couples the two plasma dimensions R_0 and a even before we make a plasma-transport calculation.

The more immediate geometric consequence of b is radial crowding. On the inboard side, the blanket lies between the plasma edge and the TF coil. It displaces the coil toward the symmetry axis, where a toroidal field is stronger. The next section turns that qualitative statement into an equation.

Design consequence. The nuclear build b separates plasma from magnet. That separation changes the attainable field through the inboard geometry.

46. The field gets stronger as the coil moves inward

The continuous-coil model already gave

$$B_\varphi(R) = \frac{\mu_0 N I_c}{2\pi R}.$$

At the plasma centre $R = R_0$, write this as B_0. The inboard edge of the plasma is at $R_0 - a$. After the nuclear island and an inboard TF-coil build have been added, the innermost high-field location lies approximately at

$$R_{\text{in}} = R_0 - a - b - c. \tag{88}$$

The sign is important: moving inward means decreasing R, which increases the toroidal field for fixed coil current.

Using $B_\varphi(R)R = B_0 R_0$, the maximum field in this simplified radial build is

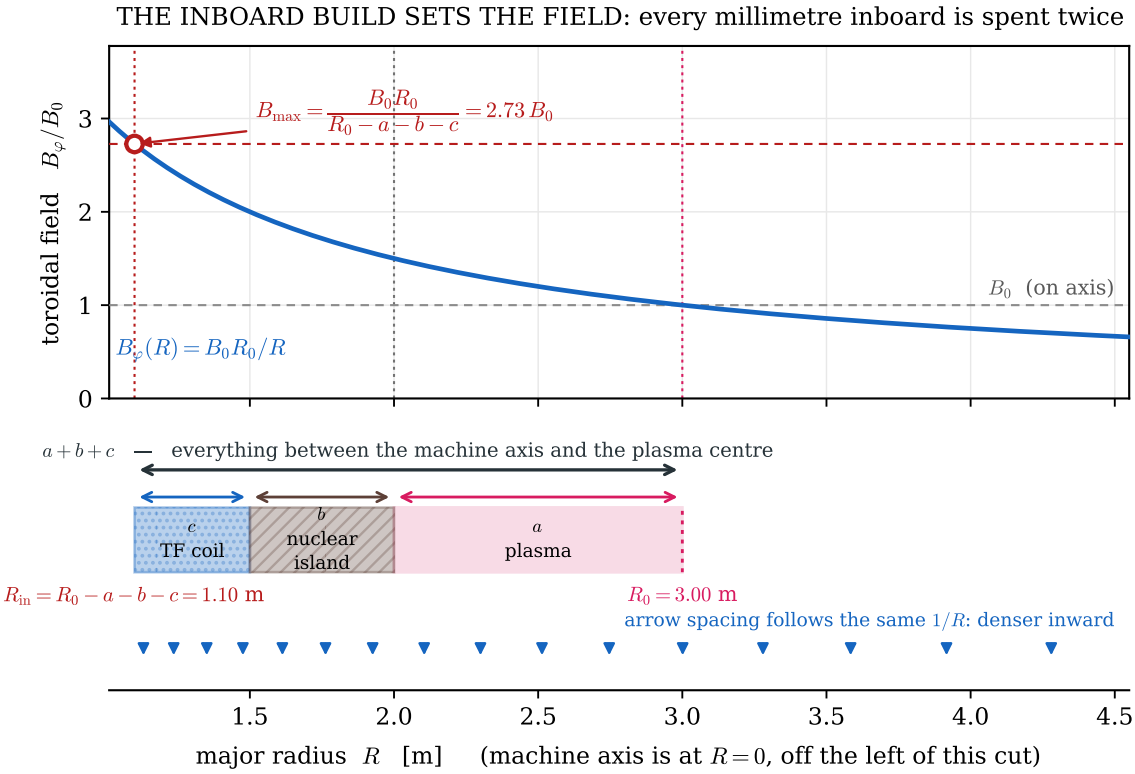
$$B_{\max} = \frac{B_0 R_0}{R_0 - a - b - c}. \tag{89}$$

Equivalently,

$$B_0 = B_{\max}\left(1 - \frac{a+b+c}{R_0}\right). \tag{90}$$

This is the first compact expression of the radial-build problem. A larger plasma minor radius, a thicker nuclear island, or a thicker TF coil leaves less inboard radius. At fixed material limit $B_{\max}$, each reduces the attainable central field B_0.

The equation also explains why a tokamak cannot be designed by choosing a desirable B_0 and adding the blanket and magnets afterwards. Those supposedly external components feed back directly into the field available to the plasma.



Rearranged, Eq. (90): $B_0 = B_{\max}(1 - (a+b+c)/R_0) = 2.73 \times (1 - 1.90/3.00) = 1.00\,B_0$ — the identity, evaluated. The consequence is the one the primer draws out: at a FIXED material limit $B_{\max}$, a larger plasma, a thicker nuclear island or a thicker coil each reduce the attainable B_0. The three compete for the same inboard radius, and none of them is free. a, b, c here are ILLUSTRATIVE: b is not derived until the nuclear-island section and c not until the coil-stress section.

Figure 21. The inboard radial ledger connects nuclear thickness b and magnet thickness c to the field available at the plasma. Dimensions shown are illustrative.

47. A field coil must carry current and survive its own force

A magnetic field stores energy. In vacuum its energy density is

$$u_B = \frac{B^2}{2\mu_0}. \tag{91}$$

One way to obtain this expression is to build a solenoid gradually. Its magnetic energy is $U = \frac{1}{2}LI^2$; dividing by volume and substituting the ideal-solenoid relations $L = \mu_0 N^2 A/\ell$ and $B = \mu_0 NI/\ell$ gives $U/(A\ell) = B^2/(2\mu_0)$.

The same scale appears as a magnetic pressure. If a field-filled region expands by a small volume dV at fixed flux, the released magnetic energy supplies mechanical work $p_B dV$. The resulting stress scale is

$$p_B = \frac{B^2}{2\mu_0}. \tag{92}$$

At high field this is enormous: the dependence is quadratic, so doubling field quadruples this stress scale. A TF coil is therefore not just superconducting cable. It needs a structural load path that prevents the winding pack from being crushed, pulled apart, or driven beyond allowable strain.

For a transparent thickness estimate, let a representative magnetic force be $F \sim p_B A_{\rm load}$ and let structural material of cross-sectional area $A_{\rm str}$ be limited to stress $\sigma_{\rm allow}$. Force balance requires

$$\sigma_{\rm allow} A_{\rm str} \gtrsim p_B A_{\rm load}.$$

If the geometry converts these areas into a representative support length ℓ_s, then the structural contribution obeys the scaling

$$c_{\rm str} \sim \frac{B_{\rm max}^2}{2\mu_0 \sigma_{\rm allow}} \ell_s \times (\text{geometry factor}). \tag{93}$$

The geometry factor incorporates tensile, centering, bending and wedging loads. It connects the field-pressure scale to the required support thickness.

Scope. The support-thickness estimate uses a representative load path. Detailed sizing requires the coil geometry and material allowables at the operating temperature and fluence.

48. The other part of coil thickness: enough current-carrying area

Stress is not the only reason a coil occupies space. Ampère's law requires a total enclosed current approximately

$$NI \simeq \frac{2\pi R_{\rm in} B_{\rm max}}{\mu_0}. \tag{94}$$

The product NI, called ampere-turns, cannot be carried through zero area. If $J_{\rm wp}$ is the allowed current density averaged over the complete winding pack (including conductor, stabilizer and internal pack allowances), the required current-carrying area scales as

$$A_{\rm wp} \gtrsim \frac{NI}{J_{\rm wp}}. \tag{95}$$

For a representative coil height $h_{\rm wp}$, this yields a winding-pack thickness scale

$$c_{\rm wp} \sim \frac{2\pi R_{\rm in} B_{\rm max}}{\mu_0 J_{\rm wp} h_{\rm wp}}. \tag{96}$$

The area-to-thickness conversion uses the representative height $h_{\rm wp}$. Its geometric factor depends on how the total ampere-turns are distributed among coils.

The full primary parameter is therefore a radial ledger:

$$c = c_{\rm str} + c_{\rm wp} + c_{\rm ins} + c_{\rm cool} + c_{\rm margin}. \tag{97}$$

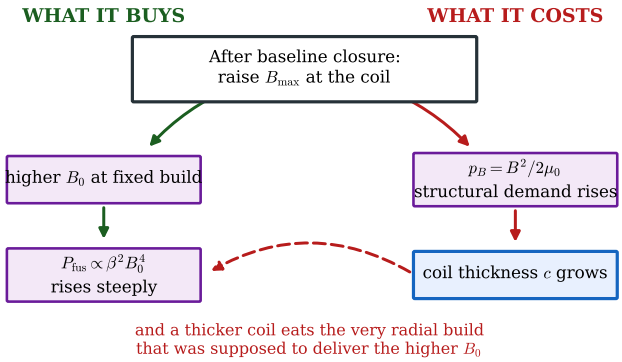
Here $c_{\rm str}$ denotes support outside the winding pack. The remaining allowances cover insulation, cooling and margins outside that pack; internal allowances already included in $J_{\rm wp}$ are counted only once. The first two terms, Eqs. (93) and (96), are the two irreducible physical demands: survive the field and produce the field.

49. Higher field changes both performance and magnet build

We can now say precisely what a higher-field magnet changes. At fixed radial geometry, Eq. (90) makes a higher allowable $B_{\rm max}$ available as a higher central field B_0. But Eqs. (93) and (96) also show the price: field increases magnetic stress quadratically and imposes a technology-dependent current-density requirement. Both effects enter the magnet-build calculation.

High-temperature superconductors (HTS) enter the argument after the baseline closure identifies a field requirement beyond the conventional magnet model. They enter this trade through the field and current density their conductors can sustain. Their value to a reactor depends on the resulting plasma performance, magnet build and engineering cost. The reference calculation will compare field choices within the same coupled constraints.

Design consequence. A higher allowable coil field can raise B_0, while stress and current-density requirements consume radial space. The benefit requires a coupled calculation.



Explanatory sketch. No magnet technology, vendor or numerical performance claim is implied; this is the shape of the trade-off only.

Figure 22. Higher coil field raises the available plasma field and the structural demand. The radial build couples those two effects.

50. The primary design vocabulary is complete

The primary parameter set is now complete, with $n = n_i$, $I = I_p$ and $f_B = f_{\mathrm{bs}}$:

$$a,\ R_0,\ \kappa,\ b,\ c,\ T,\ n,\ p,\ \tau_E,\ B_0,\ \beta,\ I,\ q_*,\ f_B.$$

The set combines geometry, plasma state, magnetic balance and the radial builds forced by neutron transport and magnet engineering. Its variables are coupled:

$$p = 2nT, \qquad q_* \propto \frac{a^2 B_0}{R_0 I}, \qquad B_0 = B_{\max}\left(1 - \frac{a+b+c}{R_0}\right).$$

These relations show how a component change propagates through the machine.

The next part applies Freidberg's 2015 reference calculation. It will select explicit engineering and nuclear assumptions, express the primary parameters as coupled functions rather than isolated knobs, and then compare the demanded plasma state with four operational tests: Greenwald density, Troyon beta, kink safety factor, and achievable bootstrap fraction. The core question will be whether all four inequalities can hold simultaneously for a conventional baseline.

Design consequence. The parameter set now supports a numerical closure. The reference calculation will test whether its operating constraints overlap.

Scope. The reference is Freidberg, Mangiarotti and Minervini (2015), Sections 3–6. Its reduced blanket and analytic TF-coil models supply the numerical design assumptions.

51. Entering the 2015 reference calculation

The first fifty sections established the machine's components and primary parameters. We now ask what plasma state a specified electric output demands. Freidberg, Mangiarotti and Minervini's 2015 calculation closes this problem with a chosen wall loading, thermal efficiency, coil-field ceiling, structural and winding-pack limits, and recirculating-power budget.

Model used here. The baseline combines mission choices, engineering limits, nuclear modelling and an empirical confinement fit. Its conclusion applies to that combination; changing an input defines a new case.

> **Design consequence.** The reference plant must satisfy density, beta, kink-safety and steady-state-current constraints at the same operating point. We first calculate the state demanded by its mission.

52. The reference design asks the machine to do a specific job

For the 2015 baseline, the primary imposed mission choices are

$$P_E = 1000\,\mathrm{MW_e}, \qquad W_n = 4\,\mathrm{MW\,m^{-2}}, \qquad \eta_T = 0.40.$$

P_E is desired electric output, W_n is the chosen upper bound on average neutron wall loading, and η_T converts fusion heat into electric power:

$$P_E = \eta_T P_F. \tag{98}$$

Here P_F is the thermal power credited to fusion and blanket reactions in the source's accounting model. These inputs set the plant mission and conversion assumption.

The reference magnet model also takes

$$B_{\max} = 13\,\mathrm{T}, \qquad \sigma_{\mathrm{allow}} = 600\,\mathrm{MPa}, \qquad J_{\mathrm{wp}} = 20\,\mathrm{MA\,m^{-2}}.$$

These values define the conventional-superconductor and structural baseline. They are reference inputs supporting the primary Table-1 parameters.

53. Power and wall loading turn two dimensions into one family

For the shaped boundary, the source uses the standard mean-square ellipse-perimeter approximation. Revolving it around the major radius gives the first-wall area

$$S_{\mathrm{wall}} \simeq 4\pi^2 R_0 a \left(\frac{1+\kappa^2}{2}\right)^{1/2}. \tag{99}$$

This agrees with the circular result $4\pi^2 R_0 a$ when $\kappa = 1$.

The D–T neutron carries $E_n = 14.1\,\mathrm{MeV}$ of the fusion energy $E_{\mathrm{DT}} = 17.6\,\mathrm{MeV}$. With E_F the credited thermal energy per fusion reaction in the reference accounting,

$$P_n = \frac{E_n}{E_F} P_F = \frac{E_n}{E_F}\frac{P_E}{\eta_T}.$$

Equating neutron power to the allowed wall loading times the wall area, $P_n = W_n S_{\text{wall}}$, gives

$$R_0 a = \frac{E_n P_E}{4\pi^2 E_F W_n \eta_T} \left(\frac{2}{1+\kappa^2} \right)^{1/2}. \tag{100}$$

At fixed mission, energy accounting and elongation, the right-hand side is constant:

$$R_0(a) = \frac{C_{\text{wall}}}{a}, \qquad C_{\text{wall}} \equiv \frac{E_n P_E}{4\pi^2 E_F W_n \eta_T} \left(\frac{2}{1+\kappa^2} \right)^{1/2}. \tag{101}$$

Model used here. The area estimate uses an ellipse-perimeter approximation and average wall loading. Local loading requires the actual first-wall surface and neutron transport.

> **Design consequence.** Each trial minor radius fixes a major radius. The inboard build must determine the field available to that plasma.

54. The radial build makes the field a coupled result

Section 53 makes R_0 a function of a. For its particular analytic TF-coil calculation, Freidberg defines the inboard field-build fraction

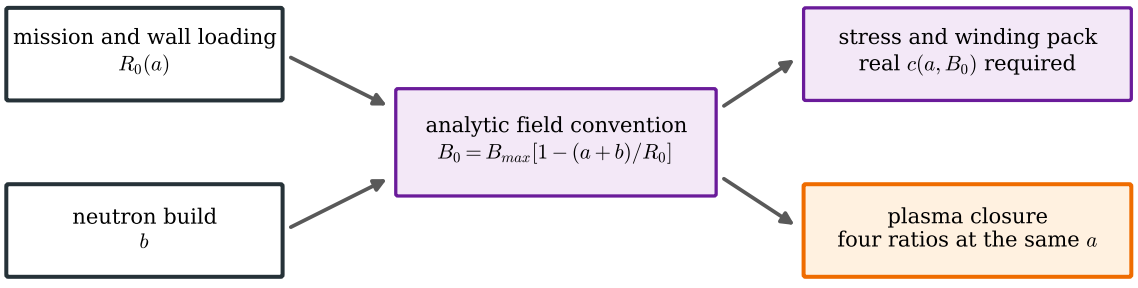
$$\epsilon_B = \frac{a+b}{R_0}$$

and writes

$$B_0 = B_{\max}(1-\epsilon_B).$$

Model used here. This is the source's analytic field convention: c is obtained separately from the stress and winding-pack equations. The fuller physical ledger in §§46–48 explains the coil thickness; reproducing this scan uses $(a+b)/R_0$ in ϵ_B.

The trial geometry and neutron build determine B_0, while the mechanical and ampere-turn requirements determine whether a real coil thickness c exists.



Trial radius a sets a candidate; coil feasibility and plasma allowances jointly test it.

Dependencies within the source's reduced radial-build convention.

Figure 23. The mission and radial build reduce the baseline to a family indexed by minor radius. Each member must meet the plasma constraints.

55. Fusion power fixes the required pressure

The D–T reaction rate in an equal fuel mixture is

$$\mathcal{R} = \frac{n^2}{4}\langle\sigma v\rangle(T),$$

because $n_D = n_T = n/2$. Multiplying by the energy released per reaction and integrating over plasma volume gives

$$P_{\mathrm{fus}} = \frac{\langle\sigma v\rangle(T) E_{\mathrm{DT}}}{4} \int n^2 \, dV. \tag{102}$$

For a first transparent estimate, use volume-averaged n and T:

$$P_{\mathrm{fus}} \simeq \frac{\langle\sigma v\rangle(T) E_{\mathrm{DT}}}{4} \, n^2 V.$$

The equal-temperature quasineutral model gave $p = 2nT$, hence

$$P_{\mathrm{fus}} \simeq \frac{\langle\sigma v\rangle(T) E_{\mathrm{DT}}}{16 T^2} \, p^2 V. \tag{103}$$

Power balance therefore determines a required pressure:

$$p_{\mathrm{req}} \simeq \left[\frac{16 T^2 P_{\mathrm{fus,req}}}{\langle\sigma v\rangle(T) E_{\mathrm{DT}} V}\right]^{1/2}. \tag{104}$$

The mission and available volume therefore demand a pressure. The reference model uses $T \simeq 14\,\mathrm{keV}$, near the broad favourable range of $\langle\sigma v\rangle/T^2$ for D–T.

Model used here. The estimate above uses uniform density and temperature. The numerical reference calculation uses the declared profiles introduced in §68.

> **Design consequence.** Fusion output fixes a pressure requirement. Alpha heating must then sustain the thermal energy stored at that pressure.

56. Ignition turns pressure into a required confinement time

The neutron carries energy out of the plasma; for D–T, the alpha fraction is

$$\frac{E_\alpha}{E_{\mathrm{DT}}} = \frac{3.5}{17.6} \simeq \frac{1}{5}.$$

For an ignited plasma, alpha heating must replace the energy conducted and transported out of the core. Let W be thermal energy stored in the plasma and define

$$\tau_E = \frac{W}{P_{\mathrm{loss}}}.$$

The ignition balance is $P_\alpha = P_{\mathrm{loss}} = W/\tau_E$, or

$$\tau_{E,\mathrm{req}} = \frac{W}{P_\alpha}. \tag{105}$$

In the equal-temperature model, the thermal energy density is $w = \frac{3}{2}(n_e + n_i)T = 3nT = \frac{3}{2}p$. Thus $W \simeq \frac{3}{2}pV$. With $P_\alpha \simeq P_{\mathrm{fus}}/5$,

$$\tau_{E,\mathrm{req}} \simeq \frac{15 pV}{2 P_{\mathrm{fus}}}. \tag{106}$$

Design consequence. Power and wall loading determine the required pressure and confinement time. The regression will translate that time into a current demand.

57. Before testing a limit: what kind of statement is it?

Three kinds of evidence enter the four tests:

Derived relation	follows from stated equations and stated approximations; for example, $p = 2nT$ and $q_* \propto a^2 B_0/(R_0 I)$.
Physics model	follows after adopting a specified equilibrium, profile, or kinetic model; its regime and profile assumptions accompany the result.
Empirical boundary	summarizes experimental operation or a regression; its coefficient and applicability range come from that evidence.

Greenwald density is empirical; the practical Troyon coefficient combines stability modelling and experimental evidence. The kink threshold depends on the adopted MHD model, profiles and wall. Bootstrap current requires a kinetic closure and compatible profiles.

For every constraint define a demand-to-allowance ratio,

$$\mathcal{R} \equiv \frac{\text{quantity demanded by the reference design}}{\text{quantity allowed by the stated constraint}}. \tag{107}$$

All relevant ratios must be at most one at the same design point.

58. The one empirical input that determines the required current

The 2015 reference case uses an ELMy H-mode confinement regression. H-mode has a strong edge transport barrier. ELMy identifies discharges with recurring edge-localized-mode relaxations, which expel edge energy and particles.[1] In the declared units, the regression is

$$\tau_E = 0.145 H \, \frac{I^{0.93} R_0^{1.39} a^{0.58} \kappa^{0.78} \bar{n}^{0.41} B_0^{0.15} A^{0.19}}{P_\alpha^{0.69}} \quad \text{s}. \tag{108}$$

Here I is in MA, R_0 and a in m, $\bar{n}$ in $10^{20}\,\mathrm{m}^{-3}$, B_0 in T, P_α in MW, $A = 2.5$ is the mean ion mass used by the fit, and H is the confinement multiplier. The coefficient belongs to these units and the selected regression. Here A denotes mean ion mass; the geometric aspect ratio is written explicitly as R_0/a.

Scope. This is the empirical closure in Freidberg, Mangiarotti and Minervini (2015), Eq. (28). Treat the regression as data with a stated applicability range. Its current, size and power exponents describe the fitted operating regime.

The conventional baseline uses $H = 1$; varying H later tests a specified improvement in confinement.

[1] A freely accessible ITER explanation describes H-mode's steep edge pressure region and the associated edge-localized modes: https://www.iter.org/node/20687/progress-elm-physics-and-elm-control.

59. Solving the confinement fit for the current

The ignition calculation has already supplied $\tau_{E,\mathrm{req}}$. Set it equal to the right side of Eq. (108) and isolate the only unknown in this step, I:

$$I^{0.93} = \frac{\tau_{E,\mathrm{req}} P_\alpha^{0.69}}{0.145 H R_0^{1.39} a^{0.58} \kappa^{0.78} \bar{n}^{0.41} B_0^{0.15} A^{0.19}}.$$

Raising both sides to the power $1/0.93$ gives

$$I_{\mathrm{req}} = \left[\frac{\tau_{E,\mathrm{req}} P_\alpha^{0.69}}{0.145 H R_0^{1.39} a^{0.58} \kappa^{0.78} \bar{n}^{0.41} B_0^{0.15} A^{0.19}} \right]^{1/0.93} \mathrm{MA}. \tag{109}$$

This current makes the fitted confinement time equal the ignition requirement.

After substituting the reference design's previously determined relations, Freidberg obtains the compact scaling

$$I_{\mathrm{req}} \simeq 12.1 \, \frac{a^{1.63}}{B_0^{0.16}} \ \mathrm{MA}, \tag{110}$$

The coefficient absorbs the fixed baseline inputs and stated units. Current grows strongly with a and depends weakly on B_0 in this reduced relation. At $a = 1.34\,\mathrm{m}$ and $B_0 = 6.83\,\mathrm{T}$, the source obtains $I_{\mathrm{req}} = 14.3\,\mathrm{MA}$.

> **Design consequence.** The confinement requirement fixes the current. Its poloidal field must leave sufficient field-line twist for the adopted stability criterion.

60. High required current immediately tests field-line twist

The safety factor was introduced as the number of toroidal circuits made by a field line per poloidal circuit. At the plasma edge, more plasma current produces more poloidal field, so it reduces this twist count. For the elongated reference geometry, the paper uses

$$q_* = \frac{2\pi a^2 B_0}{\mu_0 R_0 I} \left(\frac{1+\kappa^2}{2} \right). \tag{111}$$

The first factor is the Ampère-law estimate already derived from $B_\theta \sim \mu_0 I/(2\pi a)$; the shape factor records the paper's chosen elongation correction. Thus substituting I_{req}, not an independently chosen current, produces the required q_*.

A helical displacement can release magnetic energy associated with the plasma current. The ideal-MHD kink criterion depends on current and pressure profiles, wall response, rotation and mode number.

The reference model uses the practical external-kink/disruption condition

$$q_* \geq q_K, \qquad q_K \simeq 2. \tag{112}$$

Model used here. The adopted threshold represents a practical external-kink/disruption boundary. A disruption is an abrupt loss of controlled equilibrium and confinement, often followed by rapid current termination and large transient loads.

The ratio is $\mathcal{R}_K = q_K/q_*$: raising the demanded current lowers q_* and increases this ratio.

61. Four tests at one operating point

At the same reference design point, evaluate four distinct ratios:

$$\mathcal{R}_G = \frac{n}{n_G}, \qquad n_G = \frac{I}{\pi a^2} \times 10^{20}\,\mathrm{m}^{-3}, \tag{113}$$

$$\mathcal{R}_T = \frac{\beta}{\beta_T}, \qquad \beta_T = \beta_N \frac{I}{aB_0}, \tag{114}$$

$$\mathcal{R}_K = \frac{q_K}{q_*}, \qquad q_K \simeq 2, \tag{115}$$

$$\mathcal{R}_B = \frac{f_B}{f_{\mathrm{NC}}}. \tag{116}$$

The Greenwald expression uses I in MA and a in m; its density comparison follows the reference model's averaging convention. The Troyon expression here uses beta in percent and B_0 in T, so the adopted normalized coefficient is $\beta_N = 2.8$. $\mathcal{R}_B$ compares the required bootstrap fraction f_B with the calculated neoclassical fraction f_{NC}.

> **Design consequence.** Confinement-driven current enters all four comparisons. We test density, pressure and twist first, then calculate the available bootstrap share.

62. First audit: the density requirement

The fusion-power calculation demands the reference average density

$$n = 1.43 \times 10^{20}\,\mathrm{m}^{-3}.$$

The Greenwald reference density, using the confinement-demanded $I = 14.3\,\mathrm{MA}$ and $a = 1.34\,\mathrm{m}$, is

$$n_G = \frac{14.3}{\pi(1.34)^2} \times 10^{20}\,\mathrm{m}^{-3} = 2.54 \times 10^{20}\,\mathrm{m}^{-3}.$$

Therefore

$$\mathcal{R}_G = \frac{1.43}{2.54} = 0.56 < 1. \tag{117}$$

The reference point lies below the adopted density boundary.

Scope. This comparison reproduces the source's density convention and reactor-scale extrapolation of the Greenwald relation.

63. Second audit: pressure is barely allowed

The same reference design requires

$$\beta = 4.13\%.$$

Its adopted normalized-beta coefficient is $\beta_N = 2.8$. With $I = 14.3\,\mathrm{MA}$, $a = 1.34\,\mathrm{m}$, and $B_0 = 6.83\,\mathrm{T}$,

$$\beta_T = 2.8\,\frac{14.3}{(1.34)(6.83)}\% = 4.39\%.$$

Thus

$$\mathcal{R}_T = \frac{4.13}{4.39} = 0.94 < 1. \tag{118}$$

The reference point passes with little beta margin for profile changes, control reserve or modelling uncertainty.

Scope. The normalized coefficient 2.8 is the source's stability allowance; its applicability depends on equilibrium, wall, profile and control assumptions.

64. Third audit: the confinement-driven current fails the kink test

Insert the same demanded current into Eq. (111). The reference calculation gives

$$q_* = 1.56.$$

The adopted disruption-avoidance threshold is $q_K \simeq 2$, so

$$\mathcal{R}_K = \frac{2}{1.56} = 1.28 > 1. \tag{119}$$

The demanded current leaves insufficient edge twist for the adopted kink criterion. Reducing it would raise q_*, while lowering the confinement time predicted by the same baseline regression.

Design consequence. The reference current fails the kink test. Its steady-state supply must also be checked against external drive and bootstrap current.

65. Fourth audit: the required bootstrap share

A steady-state tokamak cannot rely on a transformer indefinitely. Its current ledger must be

$$I = I_{\mathrm{CD}} + I_{\mathrm{bs}},$$

where I_{CD} is externally driven current and I_{bs} is bootstrap current. Divide by the required total current:

$$1 = \frac{I_{\mathrm{CD}}}{I} + f_B, \qquad f_B = 1 - \frac{I_{\mathrm{CD}}}{I}. \tag{120}$$

This is a current accounting identity; the two sources must supply its terms.

For the reference recirculating-power and lower-hybrid current-drive model, the paper obtains $I_{\mathrm{CD}} = 2.25\,\mathrm{MA}$. With $I = 14.3\,\mathrm{MA}$,

$$f_B = 1 - \frac{2.25}{14.3} = 0.84. \tag{121}$$

A large total current and limited external-drive allowance demand a large bootstrap share. The attainable fraction f_{NC} follows from the kinetic closure, pressure profiles and magnetic geometry.

Design consequence. The ledger requires $f_B = 0.84$. Trapped-particle dynamics and radial gradients determine how much self-current the plasma can supply.

66. Where bootstrap current actually comes from

Sections 36–37 introduced trapped particles and an outward-decreasing pressure profile. Their coupling provides the bootstrap drive.

In a uniform straight field, particles moving parallel to **B** in the two directions can cancel one another's current. In a torus, trapped particles bounce between mirror points and do not complete a full parallel circuit. A pressure gradient means that the population arriving from the core side of a narrow radial shell is not identical to the population arriving from the edge side. Grad-B and curvature drifts then separate the two populations slightly. Collisions transfer part of that imbalance between trapped and passing particles. The cancellation of parallel electron motion is therefore incomplete: a toroidal current remains.

The pressure profile supplies the free energy. Flattening the density and temperature profiles removes this drive; steepening them increases the drive and the associated stability burden. Bootstrap current belongs in the same closure calculation as beta and kink safety.

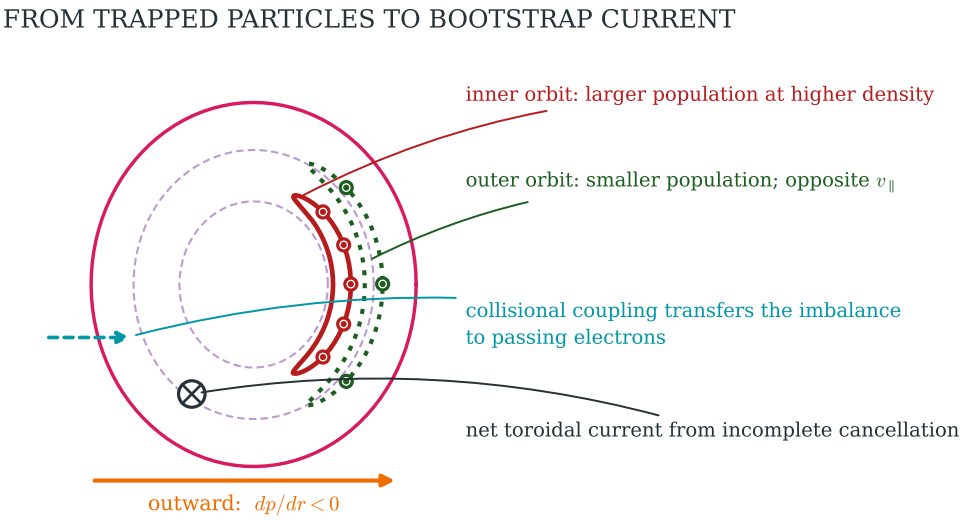


Figure 24. Trapped orbits, radial gradients and collisional coupling produce a parallel-current imbalance. The kinetic closure quantifies its magnitude.

> **Design consequence.** Pressure gradients can supply part of the required current. Quantifying that share requires a kinetic closure on the same profiles and geometry.

67. The kinetic closure for bootstrap current

The mathematical object behind the preceding paragraph is a drift-kinetic equation. In schematic form for a distribution function $f(\mathbf{x}, \mathbf{v})$ it is

$$\left(v_\parallel \mathbf{b} + \mathbf{v}_d\right)\cdot\nabla f + \frac{qE_\parallel}{m}\frac{\partial f}{\partial v_\parallel} = C[f]. \tag{122}$$

Here $\mathbf{b} = \mathbf{B}/B$, $\mathbf{v}_d$ represents the guiding-center drifts, and $C[f]$ is the Coulomb collision operator. Trapped orbits and collisional coupling enter through this kinetic description.

Freidberg, Mangiarotti, and Minervini therefore choose a particular large-aspect-ratio, low-collisionality neoclassical closure. They assume equal ion and electron temperatures, $T_i = T_e = T$, one effective ion charge, $Z = 1$, and a circular area-equivalent radius

$$\hat{a} = \kappa^{1/2} a, \qquad \rho = \frac{r}{\hat{a}}. \tag{123}$$

Their local bootstrap-current model is their Eq. (41):

$$J_B(\rho) = -2.44 \left(\frac{r}{R_0}\right)^{1/2} \frac{p}{B_\theta} \left(\frac{1}{n}\frac{dn}{dr} + 0.055\,\frac{1}{T}\frac{dT}{dr}\right). \tag{124}$$

Model used here. Equation (124) is the source's Eq. (41), a large-aspect-ratio, low-collisionality closure for $T_i = T_e$ and $Z = 1$. Its coefficients include the kinetic response. Gradients, poloidal field and the area integral are evaluated explicitly below.

68. The selected profiles make the pressure-gradient term calculable

The same paper does not leave $n(r)$ and $T(r)$ implicit. Its standard-profile choice (its Eq. (20)) is

$$T = 2\bar{T}(1-\rho^2), \qquad p = 2.5\bar{p}(1-\rho^2)^{3/2}, \qquad n = 1.5\bar{n}(1-\rho^2)^{1/2}. \tag{125}$$

The numerical factors make the bars cross-sectional averages. For example, $\int_0^1 2\rho\,[2\bar{T}(1-\rho^2)]\,d\rho = \bar{T}$. Because $d\rho/dr = 1/\hat{a}$, ordinary differentiation gives

$$\frac{1}{n}\frac{dn}{dr} = \frac{1}{1.5\bar{n}(1-\rho^2)^{1/2}} \left[1.5\bar{n}\,\frac{1}{2}(1-\rho^2)^{-1/2}\,(-2\rho)\,\frac{1}{\hat{a}}\right] = -\frac{\rho}{\hat{a}(1-\rho^2)}, \tag{126}$$

$$\frac{1}{T}\frac{dT}{dr} = \frac{1}{2\bar{T}(1-\rho^2)} \left[2\bar{T}\,(-2\rho)\,\frac{1}{\hat{a}}\right] = -\frac{2\rho}{\hat{a}(1-\rho^2)}. \tag{127}$$

Thus the bracket in Eq. (124) is

$$\frac{1}{n}\frac{dn}{dr} + 0.055\frac{1}{T}\frac{dT}{dr} = -\frac{(1+0.110)\rho}{\hat{a}(1-\rho^2)}. \tag{128}$$

Substitution of Eqs. (125) and (128) into Eq. (124) then gives, with $2.44(1.110)(2.5) = 6.77 \simeq 6.8$,

$$J_B(\rho) = 6.8\,\frac{\bar{p}}{\hat{a}^{1/2} R_0^{1/2}} \frac{\rho^{3/2}(1-\rho^2)^{1/2}}{B_\theta(\rho)}. \tag{129}$$

The direction follows the conventions for B_θ and current. The radial shape also depends on $B_\theta(\rho)$, which comes from the total-current profile.

69. The poloidal field is obtained from the assumed total-current profile

The total-current profile determines the remaining denominator $B_\theta(\rho)$. The paper selects

$$J_T(\rho) = \frac{I}{\pi\hat{a}^2}\,\frac{9\rho^{1/4}}{8}\,\frac{\alpha^2(1-x)e^{\alpha x}}{e^\alpha - 1 - \alpha}, \qquad x = \rho^{9/4}, \quad \alpha = 2.53. \tag{130}$$

This is another profile assumption, chosen to put appreciable current away from the axis. It is normalized correctly: using $\rho = x^{4/9}$ gives

$$\rho^{1/4}\rho\,d\rho = \frac{4}{9}\,dx, \qquad \frac{I_{\rm enc}}{I} = \int_0^x \frac{\alpha^2(1-u)e^{\alpha u}}{e^\alpha - 1 - \alpha}\,du. \tag{131}$$

The antiderivative is

$$\int \alpha^2(1-u)e^{\alpha u}\,du = (1+\alpha-\alpha u)\,e^{\alpha u}, \tag{132}$$

so evaluation between 0 and x yields

$$\frac{I_{\rm enc}}{I} = \frac{(1+\alpha-\alpha x)e^{\alpha x} - 1 - \alpha}{e^\alpha - 1 - \alpha}. \tag{133}$$

Ampere's law $2\pi r B_\theta = \mu_0 I_{\rm enc}$ now gives

$$b_\theta(\rho) \equiv \frac{B_\theta}{\mu_0 I/(2\pi\hat{a})} = \frac{1}{\rho}\left[\frac{(1+\alpha-\alpha x)e^{\alpha x} - 1 - \alpha}{e^\alpha - 1 - \alpha}\right]. \tag{134}$$

At $\rho = 1$, $I_{\rm enc} = I$ and $b_\theta = 1$, confirming the normalization. Near the axis the selected profile gives $b_\theta \propto \rho^{5/4}$, so the bootstrap current density scales as $\rho^{1/4}$ and remains regular.

Design consequence. The total-current profile fixes the poloidal field. Integrating the resulting bootstrap density gives the available current share.

70. Integrate the local current, then test the steady-state requirement

In the area-equivalent circular cross-section, $dA = 2\pi\hat{a}^2\rho\,d\rho$ after integrating over the polar angle. Thus

$$I_B = 2\pi\hat{a}^2 \int_0^1 J_B(\rho)\rho\,d\rho \tag{135}$$

$$= 2\pi\hat{a}^2 \int_0^1 \left[6.8\frac{\bar{p}}{\hat{a}^{1/2}R_0^{1/2}}\frac{\rho^{3/2}(1-\rho^2)^{1/2}}{B_\theta}\right]\rho\,d\rho. \tag{136}$$

Replace B_θ using Eq. (134), divide by I, and collect constants. Since $4\pi^2(6.8) = 268.5$, the result is the paper's Eq. (44):

$$f_{\rm NC} \equiv \frac{I_B}{I} = 268\frac{\hat{a}^{5/2}\bar{p}}{\mu_0 R_0^{1/2} I^2}\int_0^1 \frac{\rho^{5/2}(1-\rho^2)^{1/2}}{b_\theta(\rho)}\,d\rho. \tag{137}$$

Numerical quadrature at $\alpha = 2.53$ gives the dimensionless integral 0.318 (to three significant figures).

For the reference point,

$$\kappa = 1.70, \quad \hat{a} = \sqrt{1.70}(1.34\,\text{m}) = 1.747\,\text{m}, \quad \bar{p} = 7.57\,\text{atm}, \quad R_0 = 5.34\,\text{m}, \quad I = 14.3\,\text{MA}, \tag{138}$$

where $1\,\text{atm} = 101325\,\text{Pa}$ and $1\,\text{MA} = 10^6\,\text{A}$. Equation (137) then gives

$$f_{\rm NC} = 0.44. \tag{139}$$

The required value from Eq. (121) was $f_B = 0.84$. The audited comparison is therefore

$$\mathcal{R}_B = \frac{f_B}{f_{\rm NC}} = \frac{0.84}{0.44} = 1.91 > 1. \tag{140}$$

The selected standard-profile model supplies about half the required self-current.

71. The two coupled closure failures

The reference point gives:

Test	ratio	status	meaning
Greenwald density	$\mathcal{R}_G = 0.56$	passes	density is below the adopted empirical boundary
Troyon beta	$\mathcal{R}_T = 0.94$	marginal	pressure is near the adopted MHD boundary
Kink safety	$\mathcal{R}_K = 1.28$	fails	required current gives too little q_* margin
Bootstrap closure	$\mathcal{R}_B = 1.91$	fails	calculated self-current is below required current share

The mission demands pressure and volume; ignition and confinement demand current. That current lowers q_*, while the chosen pressure profile supplies only a limited bootstrap share under the kinetic closure.

> **Design consequence.** The reference point fails kink safety and steady-state current supply. We must test whether another size in the same family closes both.

72. Extending the calculation across machine size

The reference point $a = 1.34\,\mathrm{m}$ was chosen in the source by imposing $R_0/a = 4$. It is a useful check, but it cannot establish that the conventional baseline fails at every size. The wall-loading calculation fixes instead

$$R_0(a) = \frac{7.16}{a}\ \mathrm{m} \qquad (\kappa = 1.70,\ 2015\ \text{baseline inputs}), \tag{141}$$

so a is the remaining geometric trial variable.

For every trial a, the source's analytic radial-build convention supplies

$$\epsilon_B(a) = \frac{a+b}{R_0(a)}, \qquad B_0(a) = B_{\max}[1-\epsilon_B(a)], \quad b = 1.20\,\mathrm{m}, \quad B_{\max} = 13\,\mathrm{T}. \tag{142}$$

The TF-coil stress and winding-pack model determines $c(a)$ and bounds the geometrically admissible range. Equation (142) retains the analytic field convention stated in §54.

The already-derived power and ignition relations become compact functions of a:

$$\bar{p}(a) = \frac{8.76}{a^{1/2}}\ \mathrm{atm}, \qquad \bar{n}(a) = \frac{1.66}{a^{1/2}} \times 10^{20}\,\mathrm{m}^{-3}, \qquad \tau_E(a) = 0.81 a^{1/2}\ \mathrm{s}. \tag{143}$$

These functions retain the baseline mission, profile and wall-loading assumptions; a is entered in metres.

73. The current-drive allowance must move with the trial design

The external-drive allowance changes with density, field and size. At each trial radius, calculate it from the same recirculating-power budget.

The power allocated to waves is

$$P_{\mathrm{CD}} = \eta_{\mathrm{RF}}\eta_{\mathrm{kly}} f_{\mathrm{RP}} P_E = (0.80)(0.50)(0.10)P_E = 40\,\mathrm{MW}. \tag{144}$$

The source's lower-hybrid model writes the local efficiency in its conventional units as

$$\eta_{\mathrm{CD}} \equiv \frac{R_0 \bar{n} I_{\mathrm{CD}}}{P_{\mathrm{CD}}} \simeq \frac{1.2}{n_{\parallel}^2}. \tag{145}$$

Here $\bar{n}$ is measured in $10^{20}\,\mathrm{m}^{-3}$, I_{CD} in MA, and P_{CD} in MW. The refractive index is calculated at the intended deposition surface, $\rho_m \simeq 0.8$, from

$$n_{\parallel} \simeq \frac{\omega_{pe}}{\Omega_e} + \left(1 + \frac{\omega_{pe}^2}{\Omega_e^2}\right)^{1/2} \left(1 - \frac{\omega_{LH}^2}{\omega^2}\right)^{1/2}, \qquad \omega \simeq 2\omega_{LH}(\rho_m). \tag{146}$$

The frequencies follow from the local density and field:

$$\omega_{pe}^2 = \frac{ne^2}{\epsilon_0 m_e}, \qquad \Omega_e = \frac{eB}{m_e}, \qquad \omega_{LH}^2 = \frac{\omega_{pi}^2}{1 + \omega_{pe}^2/\Omega_e^2}. \tag{147}$$

Thus each trial design gives $n_{\parallel}(a)$, then $\eta_{\mathrm{CD}}(a)$, then $I_{\mathrm{CD}}(a) = P_{\mathrm{CD}}\eta_{\mathrm{CD}}/[R_0(a)\bar{n}(a)]$, and finally

$$f_B(a) = 1 - \frac{I_{\mathrm{CD}}(a)}{I(a)}. \tag{148}$$

Model used here. The lower-hybrid efficiency and accessibility relations specify the source's current-drive closure at $\rho_m \simeq 0.8$. Holding wave power fixed still allows I_{CD} to vary with the trial design.

74. Four ratio functions, one admissibility question

For each permitted a, use the confinement regression and the relations above to obtain $I(a)$, then calculate

$$\mathcal{R}_G(a) = \frac{\bar{n}(a)}{I(a)/[\pi a^2]}, \tag{149}$$

$$\mathcal{R}_T(a) = \frac{\beta(a)}{2.8\, I(a)/[aB_0(a)]}, \tag{150}$$

$$\mathcal{R}_K(a) = \frac{2}{q_*(a)}, \tag{151}$$

$$\mathcal{R}_B(a) = \frac{f_B(a)}{f_{\mathrm{NC}}(a)}. \tag{152}$$

The denominator in the first line is in the usual Greenwald units. The 2.8 in the second line is the stated Troyon-normalized-beta model input, in percent convention. The 2 in the third line is the adopted kink-safety threshold. The final denominator is the profile-integrated neoclassical result derived in §§66–70.

The calculation has a clean pass rule:

$$\mathcal{R}_{\max}(a) = \max\{\mathcal{R}_G, \mathcal{R}_T, \mathcal{R}_K, \mathcal{R}_B\}, \qquad \text{admissible only if} \qquad \mathcal{R}_{\max}(a) \le 1. \tag{153}$$

The scan used here is implemented in the accompanying transparent arithmetic file:

`freidberg_baseline_scan.py`

It evaluates the analytic coil, confinement, lower-hybrid and bootstrap closures directly.

Design consequence. An operating window requires the intersection of all four allowances. The scan tests whether changing size can produce that intersection.

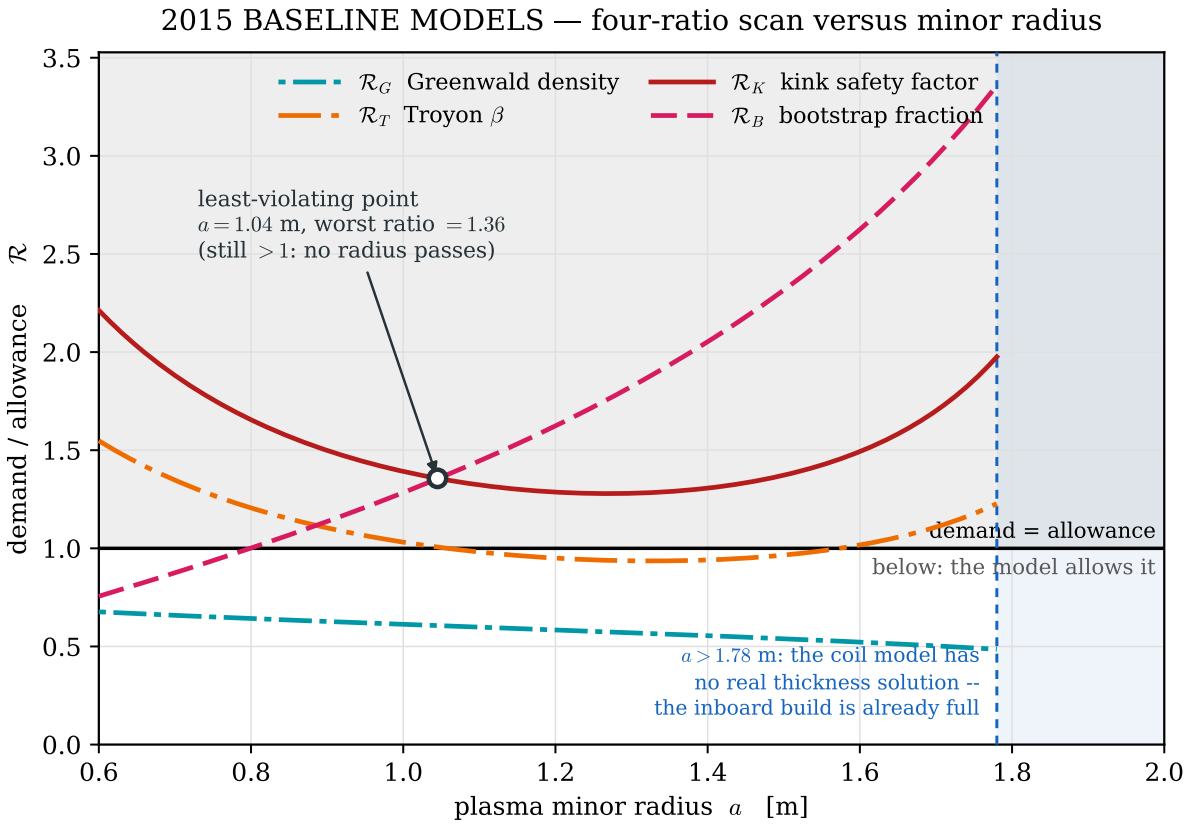
75. The conventional baseline has no simultaneous window in the scan

The table below gives representative outputs of the fully disclosed calculation. The valid high-a end is limited by the source's TF-coil stress/current-density model: with the stated inputs, its last real radial-build solution is $a \simeq 1.782\,\mathrm{m}$. Beyond that point the model has no real coil thickness, so the inboard build has failed before a plasma ratio can be evaluated. The small-a end is already excluded by the rapidly worsening kink and beta ratios.

a (m)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$	$\mathcal{R}_{\max}$
0.40	0.72	2.31	3.56	0.53	3.56
0.80	0.64	1.21	1.66	1.00	1.66
1.00	0.61	1.03	1.39	1.28	1.39
1.05	0.61	1.00	1.35	1.36	1.36
1.20	0.58	0.95	1.29	1.62	1.62
1.34	0.56	0.94	1.29	1.91	1.91
1.60	0.52	1.02	1.49	2.62	2.62

The table is rounded to two decimals; a displayed ratio of 1.00 can lie slightly above unity. The scan uses rounded analytic coefficients, so its ratios can differ slightly from the source reference-point values quoted in §§62–70.

The least-violating sampled point is near $a = 1.05\,\mathrm{m}$, with both $\mathcal{R}_K \simeq 1.35$ and $\mathcal{R}_B \simeq 1.36$. Across the scan, decreasing size worsens kink and beta, while increasing size raises the bootstrap requirement relative to the available share. Greenwald density remains below its boundary. This reproduces the trade shown in the source's Fig. 5.



Source: freidberg_baseline_scan.py; fixed 2015 mission, coil, confinement and current-supply models.

Figure 25. The four demand-to-allowance ratios for the 2015 baseline. Simultaneous admissibility requires every curve at or below unity.

76. The first conventional conclusion is conditional and specific

Within the chosen mission, wall-loading, blanket, TF-coil, confinement, lower-hybrid, standard-profile and neoclassical-bootstrap models, changing size alone does not close both kink safety and steady-state current supply.

A remedy must reduce the confinement-driven current or improve the available current supply and twist margin while retaining the other allowances. The next tests vary confinement (H), then the engineering field ceiling ($B_{\max}$), before changing the plant mission and geometry.

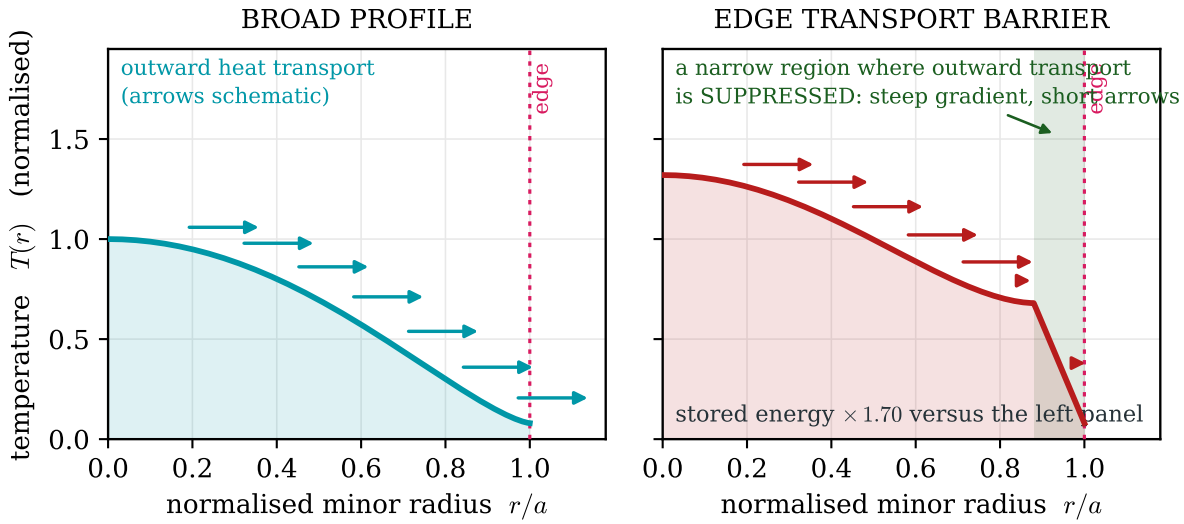
Design consequence. The baseline identifies the current-closure problem that improved confinement and higher field must solve. HTS enters when the required field exceeds the conventional magnet model.

77. Remedy one: improved confinement

The first remedy asks whether the plasma can retain energy longer than the reference H-mode regression predicts:

$$\tau_E = H\tau_{E,\mathrm{ref}}, \qquad H > 1. \tag{154}$$

Here H measures confinement relative to the regression in §58. Pressure and current-profile control can change the magnetic shear and flow environment of turbulent eddies; transport barriers can reduce outward heat loss. Maintaining such a state requires control compatible with alpha heating in an ignited plasma.



Illustrative profiles at uniform density: stored energy is proportional to $\int T(r)\, r\, dr$.

$H = \tau_E/\tau_{E,ref}$ additionally requires the loss power and a specified reference regression.

Figure 26. Illustrative temperature profiles and transport barriers. The confinement multiplier H is evaluated against a specified regression; these profiles illustrate transport control rather than a predicted reactor operating point.

Scope. An assumed $H > 1$ specifies the confinement performance required of the scenario. Its sustained value must be demonstrated in the relevant operating regime.

78. Better confinement lowers current—and moves every coupled test

Equation (109) contains H only in the denominator inside a power $1/0.93$. Holding the other inputs of that equation fixed gives

$$I_{\mathrm{req}}(H) = I_{\mathrm{req}}(1)\, H^{-1/0.93}. \tag{155}$$

Thus a larger H lowers the current that the empirical fit demands. Since $q_* \propto 1/I$ at fixed geometry,

$$q_*(H) = q_*(1)\, H^{1/0.93}. \tag{156}$$

This is the intended benefit: less current produces more field-line twist and hence more margin in the adopted kink test.

The same reduced current tightens two allowance ratios. The Greenwald allowance is proportional to I, and the adopted Troyon pressure allowance is also proportional to I. At fixed geometry and fusion pressure,

$$\mathcal{R}_G(H) = \mathcal{R}_G(1)H^{1/0.93}, \qquad \mathcal{R}_T(H) = \mathcal{R}_T(1)H^{1/0.93}. \tag{157}$$

In words: lowering current helps kink safety but makes a fixed density and pressure look larger relative to current-based operating boundaries.

Bootstrap closure shifts in the helpful direction at fixed geometry. From Eq. (137), $f_{\mathrm{NC}} \propto p/I^2$, while $f_B = 1 - I_{\mathrm{CD}}/I$. Lower I both raises the calculated bootstrap fraction and reduces the required bootstrap share if the current-drive allowance is held fixed.

> **Design consequence.** Lower current helps kink safety and bootstrap closure while reducing density and beta allowances. The remedy requires a simultaneous recalculation.

79. The H scan: close bootstrap first, then inspect the remaining margins

For each trial H, the 2015 procedure chooses the radius that closes the steady-state current ledger:

$$f_B\big(a, H\big) = f_{\mathrm{NC}}\big(a, H\big). \tag{158}$$

The fixed-geometry scalings above explain the direction of each effect. In the scan, a is readjusted at each H, so the tabulated current and geometry change together. The radius is selected by this equality, then the other ratios are tested using the changed current in the safety-factor relation. The companion script uses the nearest bootstrap match on its 0.001 m radius grid. Tables below round ratios to two decimals; 1.00 can represent a value slightly above or below unity.

H	a (m)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
1.00	0.799	0.64	1.21	1.66	1.00
1.10	0.990	0.68	1.15	1.26	1.00
1.20	1.167	0.72	1.17	1.06	1.00
1.26	1.263	0.74	1.20	1.00	1.00
1.30	1.323	0.75	1.24	0.97	1.00
1.40	1.459	0.78	1.37	0.93	1.00

The table exhibits the trade directly. Near $H = 1.26$, bootstrap and kink closure are both obtained and the Greenwald ratio remains below one, but the Troyon ratio is about 1.20: the selected 2.8% normalized-beta allowance is exceeded. Larger H only increases that beta demand in this family.

Scope. The paper reports a required normalized beta of 3.6% at its threshold case; the printed analytic relations give about 3.37% on the stated radius grid. Both exceed the 2.8% allowance. This source–reconstruction discrepancy is retained; the paper's complete numerical implementation is unavailable.

80. What the advanced-confinement remedy has established

Improved confinement can remove the baseline kink and bootstrap failures, but this family then exceeds the adopted beta allowance. The route therefore requires reproducible confinement enhancement, profile control and an MHD-stability basis for the higher normalized beta.

Design consequence. The remaining requirement is pressure-stability margin. We next hold $H = 1$ and test whether a higher coil-field ceiling can provide it.

81. Freidberg's second remedy: raise the field ceiling

The second remedy keeps $H = 1$ and raises the maximum inboard TF-coil field $B_{\max}$. Using the source's analytic radial-build convention from §54,

$$B_0 = B_{\max}(1 - \epsilon_B). \tag{159}$$

At fixed fractional build, the central field rises in proportion. The mission still requires the same pressure, and

$$\beta = \frac{2\mu_0 p}{B_0^2}. \tag{160}$$

Thus at fixed pressure and geometry, doubling B_0 quarters β. This is the first reason high field can relieve a pressure-stability problem: it increases the magnetic pressure available to balance the same plasma pressure.

The reference compact expression for beta had 13 T absorbed into its numerical coefficient. Restore that dependence when varying $B_{\max}$:

$$\beta(\%) = \frac{1.31}{a^{1/2}(1 - \epsilon_B)^2} \left(\frac{13\,\mathrm{T}}{B_{\max}} \right)^2. \tag{161}$$

This is the beta definition with the source's power-and-size relations substituted. The explicit field factor updates the coefficient when moving away from 13 T. $B_{\max}$ remains a supporting engineering input: it changes B_0, coil thickness c and the coupled plasma requirements.

82. Higher field changes plasma and mechanical requirements

First hold geometry fixed as a thought experiment. The selected confinement regression gave the compact current requirement

$$I_{\mathrm{req}} \propto B_0^{-0.16}.$$

This direct reduction in demanded current is weak. The full benefit requires recalculating safety factor, drive efficiency, bootstrap and radial build together.

The safety-factor relation makes one helpful direction visible without any empirical claim:

$$q_* \propto \frac{a^2 B_0}{R_0 I}. \tag{162}$$

Higher toroidal field and lower current both raise q_* at fixed geometry. The same field imposes a mechanical load:

$$u_B = p_B = \frac{B_{\max}^2}{2\mu_0}. \tag{163}$$

For a fixed allowable stress, the higher Lorentz load requires more structure. The analytic coil model calculates the corresponding thickness c.

Design consequence. Higher field lowers beta and raises twist, while magnetic pressure grows as $B_{\max}^2$. The scan must retain the accompanying structural demand.

83. Why field also changes the selected current-drive calculation

The reference current ledger limits externally driven current because wave power is limited. Its lower-hybrid model does not treat current-drive efficiency as a constant:

$$\eta_{\mathrm{CD}} \simeq \frac{1.2}{n_{\parallel}^2}, \qquad I_{\mathrm{CD}} = \frac{P_{\mathrm{CD}}\eta_{\mathrm{CD}}}{R_0\bar{n}}. \tag{164}$$

Here $n_{\parallel} \equiv ck_{\parallel}/\omega$ is the parallel refractive index of the launched wave, with c here denoting the speed of light. In the selected outside-launch approximation it contains the ratio

$$\frac{\omega_{pe}}{\Omega_e}, \qquad \omega_{pe}^2 = \frac{ne^2}{\epsilon_0 m_e}, \qquad \Omega_e = \frac{eB}{m_e}. \tag{165}$$

Here ω_{pe} is the electron plasma frequency and Ω_e is the electron cyclotron frequency. At fixed density, increasing local B increases Ω_e and decreases ω_{pe}/Ω_e. In this particular lower-hybrid launch model, the resulting change in $n_{\parallel}$ improves the available current-drive efficiency. That improvement reduces the bootstrap share required by the ledger

$$f_B = 1 - \frac{I_{\mathrm{CD}}}{I}.$$

Model used here. The scan retains the source's 40 MW wave-power allocation, lower-hybrid accessibility model, profiles and $H = 1$. It varies $B_{\max}$ and repeats the coupled calculation. The current-drive benefit depends on this launch and damping model; other wave systems require their own calculation.

84. The high-field scan: one closure condition, then four independent tests

For every trial $B_{\max}$, retain the mission and wall-loading relations, compute $R_0(a)$, $B_0(a, B_{\max})$, the analytic TF build c, pressure, density, confinement-demanded current, lower-hybrid drive, and bootstrap fraction. As in the H scan, choose the remaining radius by the steady-state closure condition

$$f_B(a, B_{\max}) = f_{\mathrm{NC}}(a, B_{\max}). \tag{166}$$

Then inspect all four ratios. The implementation is `code/calculations/freidberg_baseline_scan.py`, using the same 0.001 m radius grid.

$B_{\max}$ (T)	a (m)	B_0 (T)	c (m)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
13.0	0.799	10.10	1.02	0.64	1.21	1.66	1.00
15.0	0.875	11.20	1.27	0.65	1.00	1.30	1.00
17.0	0.944	12.19	1.55	0.65	0.85	1.06	1.00
17.6	0.964	12.47	1.64	0.65	0.82	1.00	1.00
18.0	0.976	12.66	1.70	0.65	0.80	0.97	1.00
20.0	1.036	13.53	2.05	0.65	0.70	0.83	1.00

Table note. The reconstruction's first all-pass point is 17.65 T on 0.01 T field and 0.001 m radius grids.

The table has a simple reading. Greenwald changes little along this bootstrap-matched family. Troyon becomes comfortable before the threshold. The kink ratio is the last ratio to fall to one. The four ratios close near $B_{\max} = 17.6\,\mathrm{T}$, consistent with the threshold reported in the 2015 paper.

Scope. The threshold describes overlap of the selected confinement, current-drive, bootstrap and stability models. Its numerical value inherits those models and the grid resolution.

85. The price is measurable: more engineered volume and more exhaust burden

The increasing c column is already a direct mechanical cost of high field. The source also records two simple engineering figures of merit. Its engineered-volume proxy is

$$\frac{V_I}{P_E} = \frac{V_B + V_{\mathrm{TF}}}{P_E}, \tag{167}$$

Here V_B and V_{TF} are blanket and TF-coil volumes in the specified geometry. The ratio measures engineered material per electric megawatt.

The source's parallel-heat-flux proxy follows the alpha power along the field:

$$Q_{\parallel} = \frac{P_\alpha B_0}{R_0}. \tag{168}$$

Scope. $Q_{\parallel}$ has units $\mathrm{MW\,T\,m^{-1}}$ and measures the parallel-exhaust burden. Converting it to a material surface heat flux requires scrape-off-layer width, magnetic expansion, radiation and target geometry.

For the source's compared cases, moving from its advanced-confinement case at 13 T to the high-field all-constraint case near 17.6 T increases V_I/P_E from 1.07 to $1.87\,\mathrm{m^3\,MW^{-1}}$ (about a factor 1.75) and reports $Q_{\parallel} = 657\,\mathrm{MW\,T\,m^{-1}}$, about 1.30 times its advanced-confinement comparison case. These engineering proxies are source-reported outputs.

> **Design consequence.** The high-field route closes the four plasma ratios with greater coil thickness, engineered volume and exhaust burden. A conductor must support that field while the magnet system carries its loads.

86. HTS is earned here: an enabler with a mechanical bill

A superconductor carries current with negligible DC resistance below its operating critical surface. Its usable current depends on field, temperature, angle, strain and conductor architecture:

$$I_c = I_c(B, T, \theta, \varepsilon, \text{conductor architecture}).$$

The conductor must remain below that surface throughout normal operation and credible transients, while its stabilizer, insulation, joints where present, cryogenic system, and protection system handle disturbances.

The scan gives a concrete requirement: this $H = 1$ route needs a coil-field ceiling near 17.6 T, above the conventional 13 T baseline.

Rare-earth barium copper oxide (REBCO), commonly called an HTS conductor, is a serious candidate because it can retain useful current capability in fields and temperatures where a conventional low-temperature-superconductor design is much more constrained. This is no longer purely aspirational: a representative-scale REBCO TF model coil was built and tested to 20.1 T peak field on conductor at 20 K. [1]

HTS provides access to higher field while the magnetic-pressure load in Eq. (163) remains. Magnet design must close stress and strain, quench detection and energy extraction, radiation tolerance, joint resistance, manufacturing, cryogenic power and replacement.

Design consequence. The high-field calculation gives HTS a specific role and operating requirement. The next remedy keeps the conventional field ceiling and changes plant output.

87. Freidberg's third remedy: change the plant scale

The third remedy raises electric output P_E from 1000 $\mathrm{MW_e}$. At fixed neutron wall loading, additional fusion power requires more wall area:

$$R_0 a = 7.16 \left(\frac{P_E}{1000\,\mathrm{MW}} \right) \mathrm{m}^2, \tag{169}$$

where the coefficient retains the 2015 blanket, shape, wall-loading, and thermal conversion assumptions.

At fixed a, R_0 and plasma volume rise with output. The plant gains toroidal length while retaining the assigned neutron power per square metre.

For this test Freidberg holds the original plasma and magnet assumptions:

$$H = 1, \qquad B_{\max} = 13\,\mathrm{T}, \qquad W_n = 4\,\mathrm{MW\,m^{-2}},$$

and again uses $f_B = f_{\mathrm{NC}}$ to select the compatible a for each trial output. The test asks whether scale alone produces overlap of the four constraints.

88. Output and volume grow together at fixed wall loading

At the fixed temperature and profiles of the reference model, fusion power has the form

$$P_{\mathrm{fus}} \propto p^2 V. \tag{170}$$

For a torus of fixed elongation, $V \propto R_0 a^2$. Insert Eq. (169):

$$V \propto (R_0 a) a \propto P_E a.$$

The thermal fusion power is proportional to P_E, so

$$P_E \propto p^2 (P_E a) \quad \Longrightarrow \quad p \propto a^{-1/2}.$$

[1] Z. S. Hartwig *et al.*, The SPARC Toroidal Field Model Coil Program, open preprint, 2023: https://arxiv.org/abs/2308.12301. The report also records 40.5 kA terminal current, large Lorentz loading, and nearly 1 GPa structural-case stress.

The factor P_E cancels. At fixed a, the mission asks for a proportionally larger volume at proportionally larger power, leaving the required average pressure unchanged. With the fixed $T \simeq 14\,\mathrm{keV}$ model and $p = 2nT$, the same is true of the average density:

$$n \propto a^{-1/2}.$$

The confinement time makes the cancellation visible a second way. The stored thermal energy scales as $W \sim pV \propto P_E a^{1/2}$, while alpha heating scales as $P_\alpha \propto P_E$. Thus

$$\tau_{E,\mathrm{req}} = \frac{W}{P_\alpha} \propto a^{1/2}, \tag{171}$$

again with no direct P_E dependence at fixed a.

Design consequence. At fixed a, increased output requires proportionally more volume at the same pressure. Any current benefit must come from the larger major radius in the confinement and drive relations.

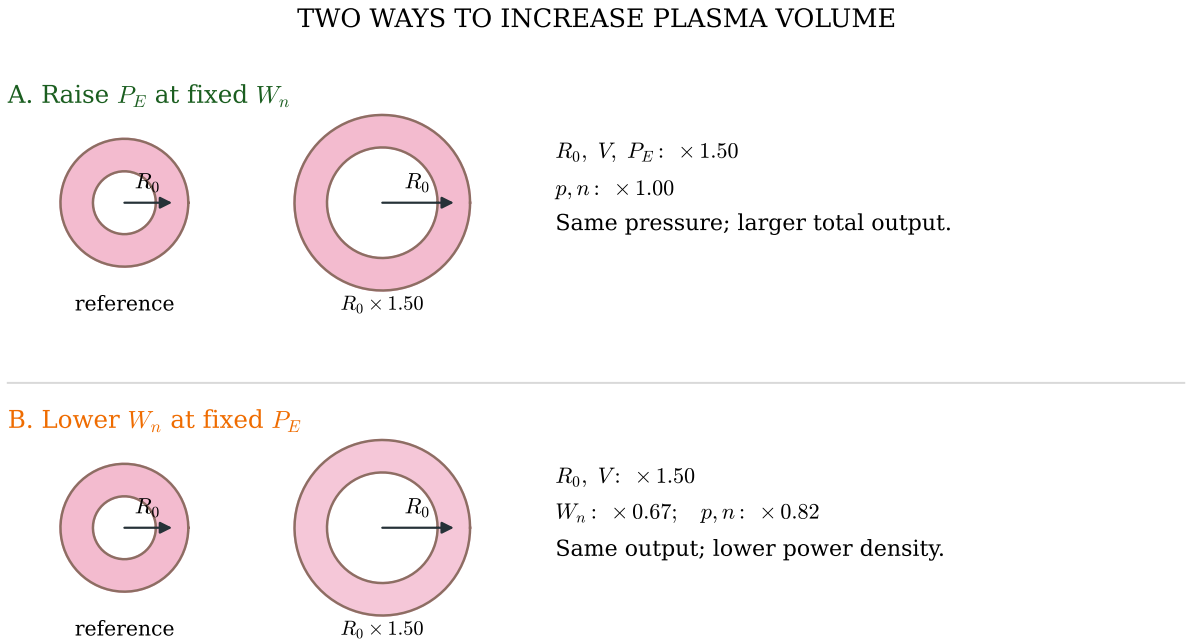


Figure 27. Two ways to increase volume at fixed minor radius and elongation. Raising output at fixed wall loading preserves the required pressure; lowering wall loading at fixed output reduces it.

89. The enlarged machine changes the current requirement

The selected confinement relation, solved for current in §59, has the dependence

$$I^{0.93} \propto \frac{\tau_{E,\mathrm{req}} P_\alpha^{0.69}}{R_0^{1.39} a^{0.58} n^{0.41} B_0^{0.15}}.$$

At fixed a, section 88 showed that $\tau_{E,\mathrm{req}}$ and n do not carry a direct power-output factor. In contrast,

$$P_\alpha^{0.69} \propto P_E^{0.69}, \qquad R_0^{1.39} \propto P_E^{1.39}.$$

Ignoring for one line the smaller change in B_0 caused by the changed radial geometry, their ratio is $P_E^{-0.70}$. Raising both sides to $1/0.93$ gives

$$I_{\mathrm{req}} \propto P_E^{-0.70/0.93} \simeq P_E^{-0.75}. \tag{172}$$

This is the central mechanism of the power-output remedy. The larger major radius helps the empirical confinement scaling more strongly than the increased alpha power hurts it, so the regression asks for less current at a fixed cross-section.

The current-drive budget must be updated at the same time. If the selected recirculating-power fraction remains fixed, $P_{\mathrm{CD}} \propto P_E$. But

$$I_{\mathrm{CD}} = \frac{P_{\mathrm{CD}}\eta_{\mathrm{CD}}}{R_0 n}$$

contains the same factor $R_0 \propto P_E$ in its denominator. At fixed a its direct power dependence approximately cancels, leaving field and wave-accessibility changes to determine the remainder. The required total current, however, falls according to Eq. (172). Therefore the bootstrap fraction required by the ledger falls:

$$f_B = 1 - \frac{I_{\mathrm{CD}}}{I}.$$

Scope. The exponent -0.75 follows from the selected regression, fixed wall loading, fixed temperature and profiles, and the stated power budget. The full scan also updates B_0 and selects a through bootstrap matching.

90. The power-output scan: the model closes near $1.55\,\mathrm{GW_e}$

For each P_E, the calculation holds $H = 1$ and $B_{\max} = 13\,\mathrm{T}$, imposes $f_B = f_{\mathrm{NC}}$, and then tests all four ratios.

P_E (MW)	a (m)	R_0 (m)	B_0 (T)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
1000	0.799	8.96	10.10	0.64	1.21	1.66	1.00
1200	1.030	8.34	9.52	0.71	1.09	1.31	1.00
1400	1.263	7.94	8.97	0.77	1.02	1.10	1.00
1550	1.436	7.73	8.57	0.81	0.99	1.00	1.00
1600	1.493	7.67	8.44	0.82	0.99	0.97	1.00
1800	1.717	7.51	7.95	0.87	0.97	0.89	1.00

Table note. Reconstruction outputs on 1 MW output and 0.001 m radius grids. Rounded source values are quoted separately in §98.

The kink and Troyon ratios are the last two to close. The scan closes near $1.55\,\mathrm{GW_e}$; Greenwald remains below one with appreciable margin.

Scope. This is four-ratio closure within the stated $H = 1$, 13 T model. The plant-scale consequences enter the engineering comparison below.

91. A larger plant trades plasma closure for absolute scale and heat exhaust

The source's engineered-volume proxy near $1.55\,\mathrm{GW_e}$ is

$$\frac{V_I}{P_E} = 1.17\,\mathrm{m^3\,MW^{-1}},$$

close to $1.07\,\mathrm{m^3\,MW^{-1}}$ in the advanced-confinement comparison. The volume per megawatt rises modestly, while total engineered volume increases with the larger plant output.

The same source reports its parallel-heat-flux proxy rising to

$$Q_{\parallel} = 673\,\mathrm{MW\,T\,m^{-1}}$$

at the high-output point, compared with $507\,\mathrm{MW\,T\,m^{-1}}$ in its advanced-confinement comparison case. The higher value identifies a greater parallel-exhaust burden within the source model.

> **Design consequence.** Larger output closes the selected constraints at conventional field and confinement, with greater absolute plant scale and exhaust demand.

92. Freidberg's fourth remedy: lower the permitted wall loading

The last single-knob remedy changes the allowed average neutron wall loading W_n. The reference value was $4\,\mathrm{MW\,m^{-2}}$. Lowering it means demanding a larger first-wall area for the same neutron power:

$$R_0 a = 7.16 \left(\frac{4\,\mathrm{MW\,m^{-2}}}{W_n}\right)\mathrm{m^2} \qquad (P_E = 1000\,\mathrm{MW_e}). \tag{173}$$

Lower neutron power per unit area requires more first-wall surface at the same output.

The immediate trade is visible before any plasma fit is used. At fixed a, $R_0 \propto W_n^{-1}$ and $V \propto R_0 a^2 \propto W_n^{-1}$. The same fusion output is produced in a larger volume. That lowers the required power density, pressure, and density, but it makes the reactor physically larger. Freidberg holds

$$P_E = 1000\,\mathrm{MW_e}, \qquad H = 1, \qquad B_{\max} = 13\,\mathrm{T},$$

and again imposes bootstrap closure to select $a(W_n)$ before testing the four ratios.

> **Design consequence.** Lower wall loading reduces required power density by increasing wall area and volume. We next calculate how that change reaches the current requirement.

93. Lower wall loading reduces the required pressure

At fixed output and temperature, use again

$$P_{\mathrm{fus}} \propto p^2 V.$$

From Eq. (173), $V \propto W_n^{-1} a$ at fixed elongation. Holding total fusion power fixed therefore gives

$$p \propto \left(\frac{W_n}{a}\right)^{1/2}, \qquad n \propto \left(\frac{W_n}{a}\right)^{1/2}. \tag{174}$$

Here reducing W_n reduces the average pressure and density required to make the same total power. The stored energy scales as

$$W \sim pV \propto \left(\frac{W_n}{a}\right)^{1/2}\left(\frac{a}{W_n}\right) = \left(\frac{a}{W_n}\right)^{1/2}.$$

The alpha power is unchanged because the total fusion output is unchanged. Hence

$$\tau_{E,\mathrm{req}} \propto \left(\frac{a}{W_n}\right)^{1/2}. \tag{175}$$

The required confinement time rises. Its effect on current must be combined with the larger R_0, lower density and changed field in the regression. Figure 27 compares this route with increased output.

94. The net current direction follows from all terms together

Use the same selected confinement dependence:

$$I^{0.93} \propto \frac{\tau_{E,\mathrm{req}} P_\alpha^{0.69}}{R_0^{1.39} a^{0.58} n^{0.41} B_0^{0.15}}.$$

At fixed a, Eqs. (174) and (175) give

$$\tau_{E,\mathrm{req}} \propto W_n^{-1/2}, \qquad R_0^{-1.39} \propto W_n^{1.39}, \qquad n^{-0.41} \propto W_n^{-0.205}.$$

Multiplying the three explicit wall-loading powers gives

$$W_n^{-0.500+1.390-0.205} = W_n^{0.685}.$$

Consequently,

$$I_{\mathrm{req}} \propto W_n^{0.685/0.93} \simeq W_n^{0.74}, \tag{176}$$

apart from the smaller self-consistent change in B_0. Lowering wall loading lowers the current demanded by the selected regression. It also lowers beta directly through Eq. (174), while the bootstrap-matched scan also changes a. Its combined geometry, field and current changes improve the kink ratio; increasing R_0 alone would lower q_* at fixed a, B_0 and I.

With fixed output, P_{CD} is fixed. As W_n falls, $R_0 n \propto W_n^{-1/2}$ grows, which tends to reduce the externally driven current in the selected efficiency model. Yet I_{req} falls more rapidly in this family; the required bootstrap share still decreases after all quantities are recalculated. The scan therefore recalculates external drive at each trial point.

95. The wall-loading scan closes near $2.1\,\mathrm{MW\,m^{-2}}$

The calculation now scans W_n while retaining $P_E = 1000\,\mathrm{MW_e}$, $H = 1$, and $B_{\max} = 13\,\mathrm{T}$. At each value it first sets $f_B = f_{\mathrm{NC}}$ and then tests the same four ratios.

W_n (MW m^{-2})	a (m)	R_0 (m)	B_0 (T)	$\mathcal{R}_G$	$\mathcal{R}_T$	$\mathcal{R}_K$	$\mathcal{R}_B$
4.00	0.799	8.96	10.10	0.64	1.21	1.66	1.00
3.50	0.890	9.19	10.04	0.65	1.11	1.49	1.00
3.00	1.009	9.46	9.96	0.67	1.00	1.32	1.00
2.50	1.171	9.78	9.85	0.68	0.89	1.14	1.00
2.10	1.349	10.11	9.72	0.69	0.80	1.00	1.00
2.00	1.403	10.21	9.68	0.70	0.78	0.97	1.00

Table note. The reconstruction's highest all-pass wall loading is 2.09 MW m^{-2} on 0.01 MW m^{-2} loading and 0.001 m radius grids.

The kink criterion is again the final one to close. The four ratios overlap when wall loading falls to about 2.1 MW m^{-2}.

Scope. The plasma constraints close by selecting a larger, lower-power-density machine under the original confinement and coil-field assumptions.

96. The reward is lower exhaust proxy; the price is a much larger plant

The source reports the low-wall-loading case's engineered-volume proxy as

$$\frac{V_I}{P_E} = 2.59\,\mathrm{m^3\,MW^{-1}}$$

near $W_n = 2.1\,\mathrm{MW\,m^{-2}}$, compared with $1.07\,\mathrm{m^3\,MW^{-1}}$ in its advanced-confinement comparison: about a factor 2.42 increase. This is qualitatively expected from the enlarged radial and toroidal dimensions, though the exact number belongs to its specified blanket and TF-coil geometry.

The reported parallel-heat-flux proxy moves in the favorable direction, from 507 to 374 MW T m^{-1} in those compared cases. The lower value indicates a reduced parallel-exhaust burden. A target design still requires power sharing, scrape-off-layer transport, radiation, detachment and material response.

Design consequence. Lower wall loading improves the exhaust proxy while raising engineered volume per megawatt. The four remedies can now be compared on a common basis.

97. One comparison ledger: which changes actually close the four tests?

Each route changes one input and recalculates the same coupled requirements:

Option	Changed demand	Result in the stated model
Improved confinement	$H : 1 \to 1.26$	Kink and bootstrap close, but the Troyon ratio is about $1.20 > 1$; not a full closure.
High field	$B_{\max} : 13 \to 17.6\,\mathrm{T}$	All four ratios close; magnet stress, coil build, and engineered-volume proxy rise.
Higher output	$P_E : 1000 \to 1550\,\mathrm{MW_e}$	All four ratios close; absolute plant size and heat-exhaust proxy rise.
Lower wall loading	$W_n : 4 \to 2.1\,\mathrm{MW\,m^{-2}}$	All four ratios close; engineered volume per megawatt rises strongly.

The confinement route closes kink and bootstrap but exceeds the beta allowance. The other three pass the four-ratio screen through distinct changes in field, plant scale or wall area. Bootstrap equality selects a state; all remaining inequalities must still be satisfied there.

98. The source's final cases

The 2015 paper reports the following rounded states. These are source outputs; the preceding scan tables report the independent reconstruction.

	reference	$H = 1.26$	high field	high output	low wall load
$B_{\max}$ (T)	13	13	17.6	13	13
P_E (MW)	1000	1000	1000	1554	1000
W_n ($\mathrm{MW\,m^{-2}}$)	4.0	4.0	4.0	4.0	2.1
a (m)	1.34	1.26	0.97	1.44	1.35
R_0 (m)	5.34	5.57	7.39	7.70	10.2
I (MA)	14.3	10.0	7.64	11.2	8.52
q_K/q_*	1.28	1.00	1.00	1.00	1.00
f_B/f_{NC}	1.90	1.00	1.00	1.00	1.00
β/β_T	0.94	1.21	0.82	1.00	0.80

Scope. The source values are rounded and retain its profile, coil and empirical-boundary assumptions. The companion calculation reproduces the three closure thresholds at the resolutions stated above; source and reconstructed values remain separately identified.

> **Design consequence.** Reducing current helps only when the accompanying beta and supply requirements also close. The successful routes make different demands on the plant.

99. What the conventional calculation establishes

The reference state fails kink safety and bootstrap closure under the selected mission, geometry, coil, profile, confinement and current-drive assumptions. The size scan finds no common operating window within that baseline. The remedy scans identify changes that restore overlap and the engineering burden accompanying each.

Scope. The Greenwald and normalized-beta allowances are operating boundaries; confinement and current drive use empirical or model closures. Bootstrap depends on the profiles and kinetic model, and the analytic coil calculation supplies a reduced engineering constraint.

Design consequence. The reconstruction identifies which changed assumptions remove the current-closure failures. Reactor development must then satisfy the additional physical and engineering requirements outside this screen.

100. Requirements beyond the four-ratio screen

The four-ratio calculation provides a core-plasma screening test. Power-exhaust design adds scrape-off-layer transport, divertor geometry, impurity radiation, detachment, transient loads and material lifetime. Disruption assessment adds vertical stability, control authority, error fields, energetic particles, resistive walls and nonlinear evolution to the edge-safety-factor criterion.

A maintainable nuclear island requires neutron-damage allowances, tritium inventory control, remote handling and component replacement. The magnet and plant studies add cryogenic power, joint reliability, fatigue, availability, construction, licensing, fuel cycle and grid integration. These requirements define the work needed to develop a state that passes the four ratios into an integrated plant.

101. From conventional closure to the aspect-ratio question

The machine's requirements have established confinement, current supply, the nuclear island and a finite magnet build. The primary set

$$a,\ R_0,\ \kappa,\ b,\ c,\ T,\ n,\ p,\ \tau_E,\ B_0,\ \beta,\ I,\ q_*,\ f_B$$

now connects physical mechanisms to the numerical design. The conventional reference needs more current than its kink margin and bootstrap supply permit; high field is one route to closure, with corresponding structural and protection demands.

The next question is what changing aspect ratio does to these same requirements. Spherical-tokamak geometry must be tested for its effect on current and field while retaining the nuclear and magnet obligations in the reduced central space. The radial-build ledger provides the starting point.

102. A spherical tokamak changes one geometric ratio, not the laws of plasma

Define the plasma aspect ratio by

$$A \equiv \frac{R_0}{a}. \tag{177}$$

A conventional tokamak has A of order 3–4 or larger. A low-aspect-ratio tokamak, commonly called a spherical tokamak (ST), has A of order 1–2. The word spherical describes the compact toroidal shape; the plasma remains a torus, not a filled sphere. It still needs TF field, poloidal field, equilibrium control, heating, exhaust, a vacuum boundary, and a nuclear island.

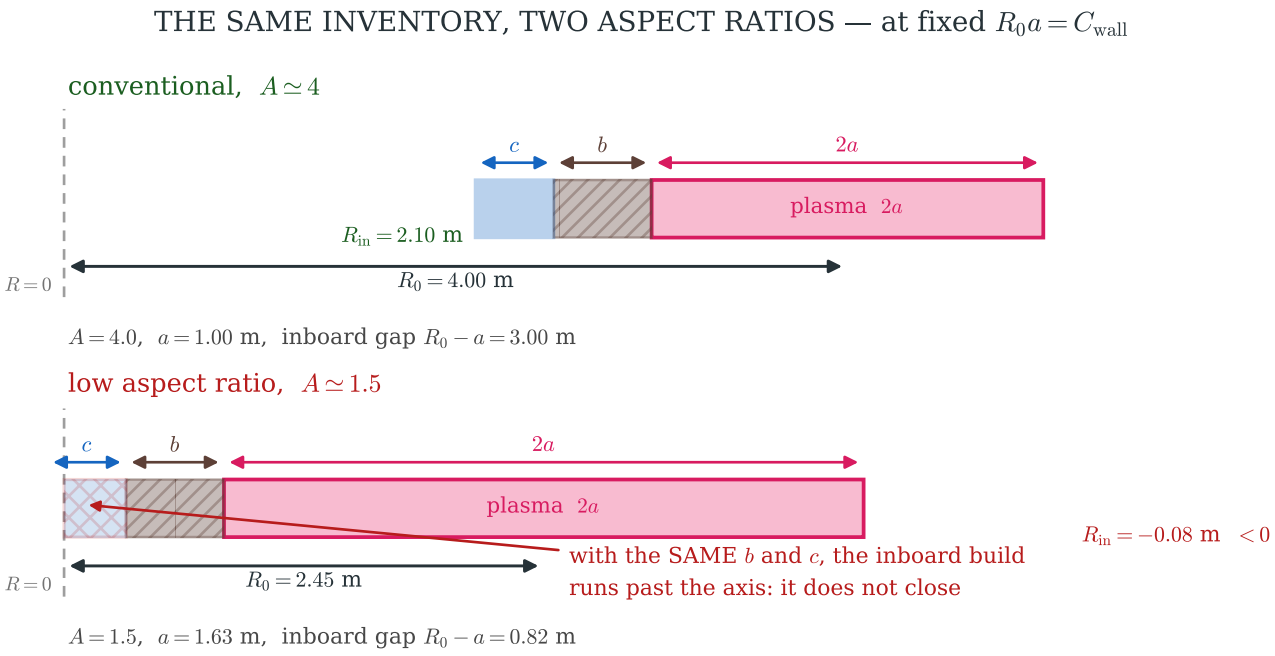
At a fixed fusion mission and wall-loading product $R_0 a = C_{\mathrm{wall}}$, the two geometric variables can instead be written in terms of A:

$$R_0 = (AC_{\mathrm{wall}})^{1/2}, \qquad a = \left(\frac{C_{\mathrm{wall}}}{A}\right)^{1/2}. \tag{178}$$

Thus lowering A makes the cross-section larger while bringing the plasma centre closer to the symmetry axis. It is not simply a uniformly shrunken conventional machine.

Design consequence. Aspect ratio is a causal geometric variable. Lowering it can alter field-line twist, trapped-particle physics, volume, and radial space at once. We must derive each direction separately before calling low aspect ratio an advantage.

PLACE IMAGE HERE — Reset geometry sketch: aspect ratio changes the torus, not the physics inventory. Show two midplane cuts with the same labelled $R_0 a = C_{\mathrm{wall}}$: a conventional $A \simeq 4$ plasma and a low-$A \simeq 1.5$ plasma. Mark R_0, a, and the symmetry axis. Retain only the already-earned plasma, nuclear-island bracket, and TF-system location as simplified outlines. Make the low-A inboard gap visibly smaller. Do not label either design as automatically superior and do not yet add an HTS brand or an ST-specific divertor.



Both rows hold $R_0 a = 4.00\,\mathrm{m}^2$, so the first-wall area is the same. $b = 0.50$ m and $c = 0.40$ m are fixed IN METRES: they are set by neutron attenuation and by conductor and structure, not by how round the plasma is — scaling them with a would assume the answer. Everything above is then arithmetic: $a = \sqrt{C_{\mathrm{wall}}/A}$, $R_0 = Aa$.

NEITHER design is superior here, and the figure does not say one is. It says something narrower: low A cannot be adopted while holding b and c fixed — the inboard allocations have to be renegotiated. That is exactly what the following sections do, and why they quote their own shielding allocation and centre-column envelope instead of inheriting the conventional ones. The physics inventory has not changed; the room it fits into has.

Explanatory sketch; midplane cuts at a common radial scale. No HTS brand and no ST divertor: neither is derived yet.

Figure 40: Reset geometry sketch: aspect ratio changes the torus, not the physics inventory. *Idealized schematic; not a fabrication drawing.*

103. Low aspect ratio strengthens toroidal field variation and trapping

The toroidal-field relation remains

$$B_\varphi(R)R = B_0R_0.$$

At the inboard and outboard plasma edges, $R = R_0 - a$ and $R = R_0 + a$. Hence

$$\frac{B_{\text{out}}}{B_{\text{in}}} = \frac{R_0 - a}{R_0 + a} = \frac{A - 1}{A + 1}. \tag{179}$$

When A falls, this ratio falls: the inboard field becomes much stronger than the outboard field. The mirror argument from page 36 then says that a larger fraction of particles can be trapped. In the simple low-collisionality estimate used earlier,

$$f_t \simeq \left(1 - \frac{B_{\text{min}}}{B_{\text{max}}}\right)^{1/2} \simeq \left(\frac{2}{A + 1}\right)^{1/2}. \tag{180}$$

For orientation, this expression gives $f_t \simeq 0.63$ at $A = 4$ and $f_t \simeq 0.89$ at $A = 1.5$.

More trapped particles make a larger bootstrap response physically plausible because bootstrap current begins with trapped-particle/passing-particle collisional asymmetry. But Eq. (180) is not a bootstrap fraction. The large-aspect-ratio local closure used on pages 67–70 is no longer controlled at $A \sim 1$–2. A finite-aspect-ratio kinetic calculation must replace it before claiming a numerical f_{NC} for an ST.

Model and limits. The direction of the trapping effect follows from the toroidal-field geometry and magnetic-moment argument. The resulting bootstrap fraction, profile, and collisionality dependence are finite-aspect-ratio neoclassical physics, not a number that can be extrapolated from Freidberg's large-aspect-ratio coefficient.

104. Aspect ratio also changes the kink–beta coupling

The shaped edge safety-factor estimate was

$$q_* = \frac{2\pi a^2 B_0}{\mu_0 R_0 I}\left(\frac{1+\kappa^2}{2}\right) = \frac{2\pi}{\mu_0}\left(\frac{1+\kappa^2}{2}\right)\frac{aB_0}{AI}. \tag{181}$$

Normalized beta contains the same dimensional group aB_0/I:

$$\beta_N \equiv \beta(\%)\frac{aB_0}{I_{\mathrm{MA}}}. \tag{182}$$

Here $I_{\mathrm{MA}} = I/(10^6\,\mathrm{A})$ and $\beta(\%) = 100\beta$. Substitute

$$\frac{aB_0}{I} = \frac{\beta_N}{10^8\beta}$$

into Eq. (181) to obtain

$$q_* = \frac{0.05}{A}\left(\frac{1+\kappa^2}{2}\right)\frac{\beta_N}{\beta}. \tag{183}$$

The numerical 0.05 comes only from $\mu_0 = 4\pi \times 10^{-7}$ and the conventional percent–MA units in β_N.

Equation (183) is a geometric insight, not a stability limit. At fixed β, β_N, and shape, lower A raises q_*. Equivalently, at a required q_* it permits a larger normalized beta within this relation. This is the first-principles reason low aspect ratio is a plausible response to the Freidberg kink–beta coupling: it changes the geometry of the current that produces the stabilizing poloidal field.

Audit note. The equation does *not* prove that an ST is stable at a chosen β_N. The allowable β_N remains an equilibrium, wall, profile, rotation, and kinetic-MHD question. Low aspect ratio changes the relevant stability physics, so the conventional Troyon coefficient 2.8 cannot simply be reused as an ST theorem.

105. What an ST can plausibly relieve in the Freidberg closure

The conventional reference failed because its required current simultaneously made q_* too small and made steady-state replacement too demanding. Low aspect ratio attacks both mechanisms in direction, though not yet in certified magnitude:

- Equation (183) shows that smaller A increases q_* for the same normalized-beta and shape inputs. This can relax the kink-versus-pressure coupling that was restrictive in the conventional scan.
- Equation (180) shows stronger toroidal trapping. Combined with the pressure gradients already required by fusion, this creates a physical route to larger bootstrap current and a smaller externally driven current share.
- At fixed $R_0 a$, Eq. (178) gives a larger minor radius. That changes volume, current profile, wall area distribution, shaping, and power-exhaust geometry; none may be held fixed without calculation.

This is enough to formulate a responsible test, not to announce its result. The question immediately next is narrow and mathematical: does a finite-aspect-ratio ST calculation satisfy the two ratios that failed in Freidberg's conventional baseline, namely the kink ratio and the non-inductive current ratio? We will use a published low-A equilibrium and current-drive model, state every replacement assumption, and evaluate those two ratios. The centre-stack problem is *not* a premise of that test; it is the separate engineering problem revealed only after the two Freidberg constraints have been answered.

Integrated ST studies make the same methodological point. Menard *et al.* report low-aspect-ratio pilot-plant configurations only after free-boundary equilibrium, detailed device layout, and three-dimensional neutronics are coupled; their configurations must also meet tritium breeding, divertor, and maintenance requirements.[4]

[4] J. E. Menard *et al.*, Fusion nuclear science facilities and pilot plants based on the spherical tokamak, *Nuclear Fusion* 56 (2016) 106023, freely accessible at https://iopscience.iop.org/article/10.1088/0029-5515/56/10/106023/pdf.

106. The ST question is exactly Freidberg's two failed ratios

Freidberg's conventional reference failed two current-centred tests:

$$\mathcal{R}_K = \frac{q_K}{q_*} > 1, \qquad \mathcal{R}_B = \frac{1 - I_{\mathrm{CD}}/I}{f_{\mathrm{NC}}} > 1.$$

The first says that the confinement-demanded current leaves too little safety factor; the second says that the calculated self-current is smaller than the self-current share required after permitted external current drive is subtracted. For the conventional reference, pages 64 and 70 obtained 1.28 and 1.91, respectively.

The ST test uses the same accounting, but it must replace the two large-aspect-ratio inputs rather than reuse them carelessly:

$$\mathcal{R}_{K,\mathrm{ST}} = \frac{q_{K,\mathrm{ST}}}{q_{*,\mathrm{ST}}}, \qquad \mathcal{R}_{B,\mathrm{ST}} = \frac{1 - I_{\mathrm{CD,allow}}/I_{\mathrm{ST}}}{f_{\mathrm{BS,ST}}}.$$

Here $q_{K,\mathrm{ST}}$ is an ST-specific kink boundary, $f_{\mathrm{BS,ST}}$ is obtained from an ST finite-aspect-ratio bootstrap model, and $I_{\mathrm{CD,allow}}$ is the external current that the plant can afford to supply under an independently stated power and efficiency constraint. The logic has not changed. A candidate closes Freidberg's bootstrap obstruction only when those two inputs are independently obtained and the ratio is at most one.

Design consequence. The centre-stack question is not included in Eq. (). It must not be used to weaken or postpone the answer. The task first is to calculate the ST kink ratio and to compare its finite-A bootstrap prediction with an independently justified current-drive allowance.

107. The finite-aspect-ratio bootstrap model is declared

Menard *et al.* use a low-aspect-ratio equilibrium-informed bootstrap closure of the form

$$f_{\mathrm{BS}} = C_{\mathrm{BS}}\sqrt{\epsilon}\,\beta_{\mathrm{pol,th}}, \qquad \epsilon \equiv \frac{a}{R_0} = \frac{1}{A}, \tag{184}$$

where $\beta_{\mathrm{pol,th}}$ is the thermal poloidal beta and C_{BS} is a profile/safety-factor coefficient. Their stated parameterization is

$$C_{\mathrm{BS}} = \max\left(1.2 - 0.2\frac{q_*}{q_{\min}},\, 0.6\right). \tag{185}$$

Here $q_{\min}$ is the minimum safety factor of the assumed equilibrium/current profile. It is an input to this finite-A closure, not a new universal constant. [5]

The definition of poloidal beta makes the physical content visible:

$$\beta_{\mathrm{pol,th}} \equiv \frac{2\mu_0\langle p_{\mathrm{th}}\rangle}{B_{\mathrm{pol}}^2}.$$

It increases when thermal pressure rises relative to the poloidal magnetic field. Equation (184) then combines the pressure-gradient source of bootstrap current with the low-A orbit geometry through $\sqrt{\epsilon}$. This is the finite-A replacement for Freidberg's large-aspect-ratio local integral on pages 67–70.

Model and limits. The new coefficient and the equilibrium-derived poloidal beta make Eq. (184) a model closure. That is acceptable and necessary: Freidberg's own numerical bootstrap result also required a declared neoclassical closure. What would be unacceptable is to use the conventional large-A coefficient at $A \simeq 1.7$ and call the extrapolation a proof.

[5] J. E. Menard *et al.*, cited on page 105, Sec. 2.1.5, Eqs. and surrounding text on bootstrap/current-drive modeling. They state that this form is consistent with low-aspect-ratio equilibrium calculations; it is not deduced from Maxwell's equations alone.

108. A published finite-A candidate supplies every quantity the two tests need

The following is not a generic ST slogan. It is a specified low-aspect-ratio systems calculation from Menard *et al.*: $A = 1.7$, $\kappa = 2.75$, $B_{T0} = 3\,\mathrm{T}$, ITER-$H_{98} = 1.25$, and 80 MW of neutral-beam heating/current-drive power. Their density scan at $R_0 = 1.6\,\mathrm{m}$ reports

$$\beta_N = 4.0\text{–}4.7, \qquad q_* \geq 3, \qquad I = 10.8\text{–}12\,\mathrm{MA}.$$

At normalized density near 0.8, their current model gives a bootstrap fraction of about $f_{\mathrm{BS}} = 0.76$ and a beam-driven share of about 0.24. [6]

These inputs give an ST safety factor and an ST bootstrap/current-drive distribution at the same operating point. They demonstrate that the authors can construct a fully non-inductive scenario in their model. They do *not*, by themselves, state the independently derived current-drive allowance required in the numerator of Eq. (). That distinction is the difference between constructing a current balance and testing Freidberg's economic current-drive constraint.

Audit note. The displayed $\beta_N = 4.0$–4.7 range characterizes this candidate, but it is not silently used as a passed stability test in the present two-constraint comparison. Menard *et al.* provide a separate normalized-beta/stability assessment for their ST equilibria. That analysis must be evaluated explicitly before this thesis makes any broader high-β_N stability claim.

PLACE IMAGE HERE — Freidberg two-constraint closure: published low-A ST operating point. Use a two-column, deliberately non-engineering graphic. Left: the conventional Freidberg reference with $q_* = 1.56 < 2$ and required/calculated bootstrap 0.84/0.44. Right: the specified $A = 1.7$ ST model with $q_* \geq 3$ against $q_{K,\mathrm{ST}} = 2.8$, and current fractions $f_{\mathrm{BS}} \simeq 0.76$ plus $f_{\mathrm{NBI}} \simeq 0.24$. Label the right-hand current sum as an identity imposed by a fully non-inductive scenario, not as an independently passed Freidberg ratio. State the resulting required external current explicitly. Do not draw magnets, centre stacks, HTS, or reactor hardware; this image separates the resolved kink result from the still-conditional bootstrap question.

[6] Menard *et al.*, cited on page 105, Secs. 2.1.3–2.1.6 and Figs. 8–9. Values read from the published scan are shown to two significant figures. The paper calculates total non-inductive current as the sum of bootstrap and neutral-beam current.

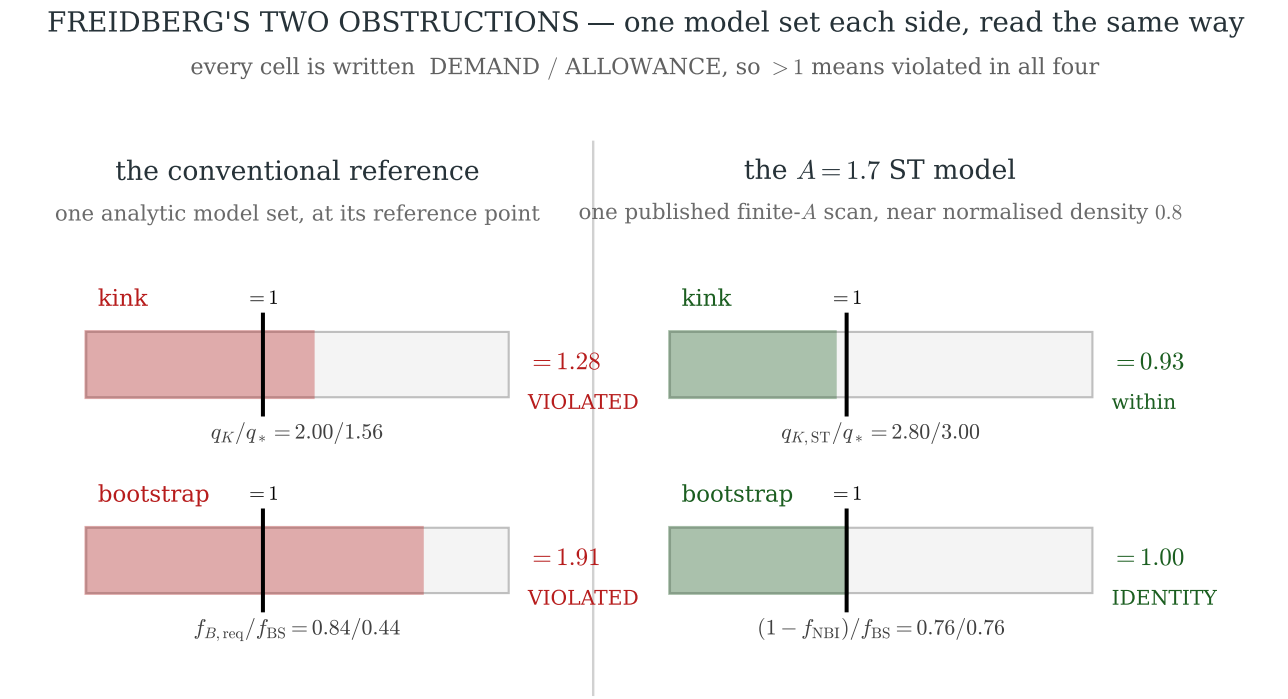


Figure 41: Freidberg's two obstructions, read with their required logical inputs visible. The ST kink comparison is an actual threshold test. The ST 0.76/0.24 current split is an identity of Menard's fully non-inductive scenario, not an independent bootstrap-ratio pass: it identifies the external current the scenario must provide. Both columns are declared model results, not properties of tokamaks or of STs.

109. First answer: the ST candidate closes the kink obstruction

The relevant ST disruption evidence used by Menard *et al.* is more demanding than Freidberg's conventional reference threshold: NSTX disruption data show a marked increase in disruptivity below $q_* \simeq 2.8$. Their stated ST design requirement is therefore

$$q_{K,\mathrm{ST}} = 2.8.$$

[7] The same candidate scan gives $q_* \geq 3$. Substitute these published values:

$$\mathcal{R}_{K,\mathrm{ST}} = \frac{2.8}{q_*} \leq \frac{2.8}{3.0} = 0.93 < 1. \tag{186}$$

This is an affirmative answer to Freidberg's kink obstruction within the declared finite-A model. It does not claim that every ST is kink stable; it demonstrates that the low-A candidate can satisfy a stricter ST-specific safety-factor criterion than the $q_K = 2$ criterion that the conventional reference failed. The geometrical reason anticipated on page 104 is now joined to an explicit operating-point test.

[7] Menard *et al.*, cited on page 105, Sec. 2.1.3. The numerical boundary comes from an NSTX data-based stability/disruption criterion and remains profile, wall, and operating-regime dependent.

110. The bootstrap result identifies the price; it does not yet test Freidberg's constraint

The current ledger does not change with aspect ratio:

$$I_{\mathrm{ST}} = I_{\mathrm{BS}} + I_{\mathrm{CD}}.$$

Divide by I_{ST}. The Menard candidate gives, to the precision supported by the published scan,

$$\frac{I_{\mathrm{BS}}}{I_{\mathrm{ST}}} \simeq 0.76, \qquad \frac{I_{\mathrm{CD}}}{I_{\mathrm{ST}}} \simeq 0.24,$$

and the two fractions sum to one. This equality is not an independent test. The Menard scenario is constructed to be fully non-inductive, so

$$\frac{I_{\mathrm{BS}}}{I} + \frac{I_{\mathrm{CD}}}{I} = 1$$

by its current-accounting definition. Dividing its own complement by its own bootstrap fraction produces 1 identically; it could not reveal a failure.

What the published point does give us is the physical and engineering burden that an independent Freidberg test must meet:

$$I_{\mathrm{CD,needed}} = (1 - f_{\mathrm{BS}})I \simeq 0.24I = 2.6\text{–}2.9\,\mathrm{MA}. \tag{187}$$

The scenario assumes 80 MW of 0.5 MeV neutral-beam heating/current-drive power to obtain this fully non-inductive balance. It therefore demonstrates a substantial finite-A bootstrap fraction and states the associated external-drive burden; it does not independently establish that a plant can afford that burden.

The proper Freidberg test remains conditional until an economic/recirculating-power analysis supplies $I_{\mathrm{CD,allow}}$ independently:

$$\mathcal{R}_{B,\mathrm{ST}} = \frac{1 - I_{\mathrm{CD,allow}}/I}{f_{\mathrm{BS}}} \leq 1 \quad \Longleftrightarrow \quad I_{\mathrm{CD,allow}} \geq I_{\mathrm{CD,needed}}. \tag{188}$$

This is the question the next research pass must answer. It is more rigorous than calling the identity $0.76/0.76 = 1$ a measured closure.

The condition can be priced with published scalars alone. Freidberg's allowance was 100 MW of wall electricity for current drive, of which 40 MW reaches the plasma (a 50% wall-to-klystron and 80% absorption chain); Menard's pilot-plant analysis likewise assumes a neutral-beam wall-plug efficiency of 0.4.[8] Keeping the published 80 MW beam state would therefore draw $80/0.4 = 200\,\mathrm{MW_e}$, twice that allowance; fitting under it would need a wall-plug efficiency of 0.8, or, if only the allowance's 40 MW were injected, a drive efficacy of about $2.6\,\mathrm{MA}/40\,\mathrm{MW} \simeq 65\,\mathrm{kA/MW}$ at that state. These are bounds on the published state, not on a redesigned ST: lowering the beam power changes heating, fast-ion pressure, fusion power, current, and bootstrap fraction together, and the FNSF point is a $Q_{\mathrm{DT}} \simeq 2$ neutron-source mission rather than a $1\,\mathrm{GW_e}$ plant. The affordability question can only be settled on a power-plant-mission ST scan carrying its own recirculating-power budget.

[8] Freidberg *et al.*, Sec. 4.6; Menard *et al.*, Sec. 2.1.6, where $\eta_{\mathrm{aux}} = 0.4$ is described as higher than presently achievable, and Sec. 5.3. The arithmetic is reproduced in `thesis/code/calculations/st_drive_affordability.py`.

111. Verdict: ST closes the kink obstruction; bootstrap closure has a stated condition

The published finite-aspect-ratio ST model answers one of Freidberg's two current-centred obstructions affirmatively:

$$\mathcal{R}_{K,\mathrm{ST}} \leq 0.93 < 1.$$

It also demonstrates a large bootstrap fraction, $f_{\mathrm{BS}} \simeq 0.76$, and makes the missing condition quantitative: the operating point needs approximately 2.6–2.9 MA of independently affordable current drive.

We must not call that second result closure until its current-drive allowance has been obtained independently. Freidberg's conventional comparison was meaningful because the permitted drive came from a separate recirculating-power calculation while the bootstrap fraction came from a neoclassical calculation. Reusing the complement of a fully non-inductive ST current balance would erase that distinction.

Design consequence. Low aspect ratio has now been shown to close the kink obstruction in a published finite-A model and to raise the bootstrap fraction to a quantified, promising level. Whether it closes Freidberg's bootstrap obstruction depends on the separate question $I_{\mathrm{CD,allow}} \geq 2.6$–$2.9\,\mathrm{MA}$.

112. Only after that answer does the separate centre-stack problem begin

The conclusion on page 111 does not say that the ST has no further problems. It says the kink question is answered affirmatively and the bootstrap question has been reduced to an explicit independent current-drive requirement. Lowering A simultaneously brings the plasma closer to the symmetry axis and strengthens the inboard toroidal field. That creates a new engineering question: can a D–T reactor fit shielding, breeding, TF conductor, support, cooling, protection, and perhaps current-initiation function inside the remaining central radius?

That question is logically separate. It cannot retroactively turn the resolved kink ratio into a failure, any more than it can replace the independent current-drive analysis still required for bootstrap closure. The remainder of this chapter changes its subject explicitly: from *what does low-A plasma physics change?* to *what engineering debt does the low-A solution create, and can HTS help pay it?*

113. The same geometry takes radial space away from the centre stack

The inboard radial ledger is the unavoidable counterargument. Let c_{cs} denote the radial build occupied by the centre-stack TF conductor, its structural case, insulation, cooling, and any allocated central-solenoid function. The innermost radius available to the toroidal field is

$$R_{\mathrm{cs}} = R_0 - a - b - c_{\mathrm{cs}} = R_0\left(1 - \frac{1}{A}\right) - b - c_{\mathrm{cs}}. \tag{189}$$

The same $BR =$ constant relation now gives

$$B_{\mathrm{cs}} = \frac{B_0 R_0}{R_{\mathrm{cs}}}, \qquad B_0 = B_{\mathrm{cs}}\left[1 - \frac{1}{A} - \frac{b + c_{\mathrm{cs}}}{R_0}\right]. \tag{190}$$

Every term in the square bracket has a physical meaning. Lowering A subtracts geometric room; b is demanded by neutrons; c_{cs} is demanded by current, structure, insulation, cooling, and possibly start-up flux. The bracket must be positive even before a usable central field can exist.

As $A \to 1$, the purely geometric factor $1 - 1/A$ tends to zero. No improved plasma stability can reverse that subtraction. A low-A reactor must instead choose among a larger R_0, thinner-but-credible inboard nuclear and magnet systems, a higher permissible B_{cs}, different start-up/current-drive architecture, or some combination. This is the centre-stack problem in its most elementary form.

PLACE IMAGE HERE — Cumulative construction state: low aspect ratio reveals the centre stack. Start with the complete conventional radial-build state, then reduce A while preserving the plasma, first wall/nuclear island b, and TF function. On the inboard side show the shrinking interval $R_{\mathrm{cs}} = R_0 - a - b - c_{\mathrm{cs}}$. Split c_{cs} into conductor, structure, insulation/cooling, and optional central- solenoid allocation. Label B_0 at the plasma and B_{cs} at the central column. The image must show that low A improves some plasma geometry while making this physical interval scarce; do not depict a solved reactor.

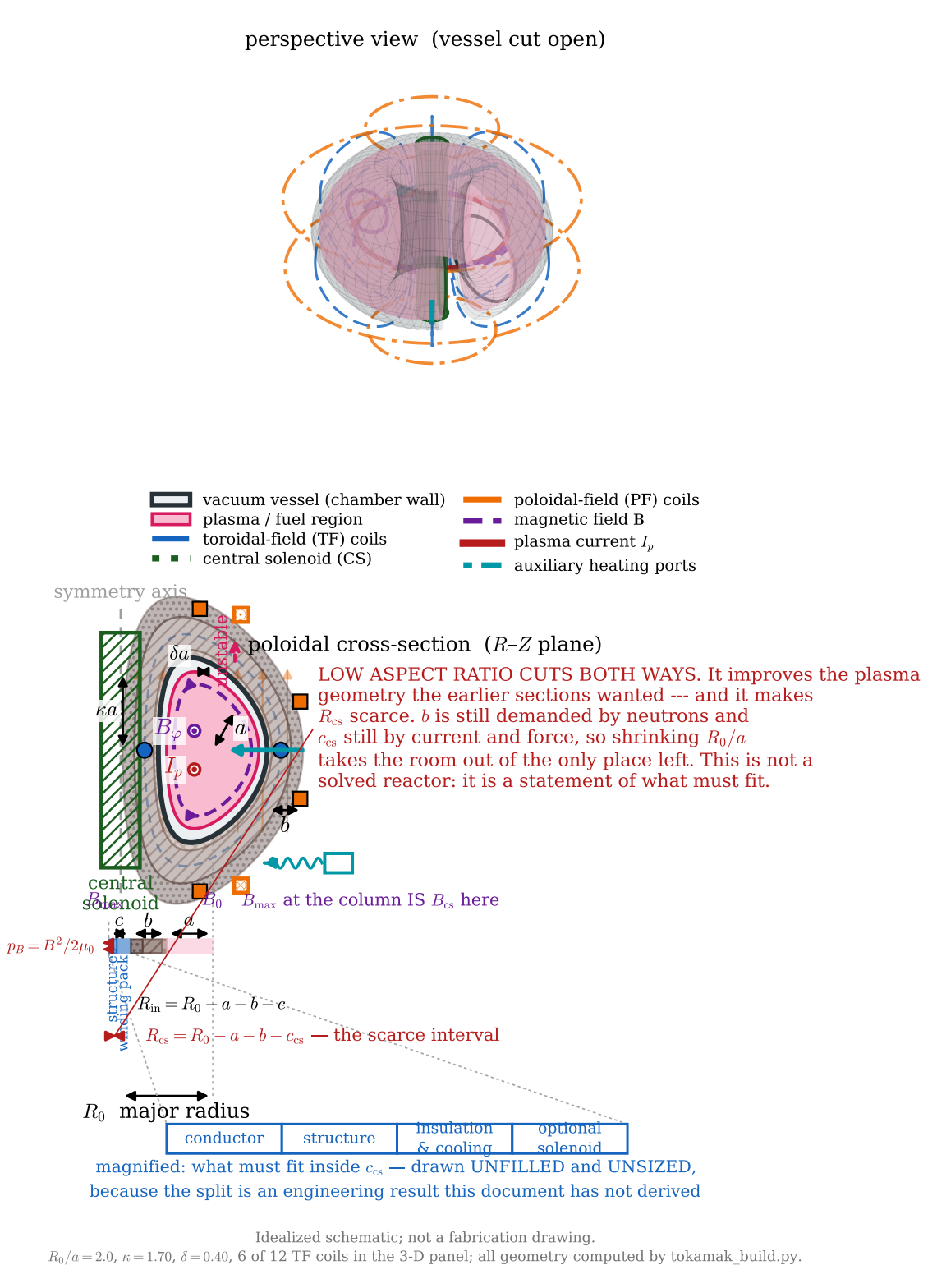


Figure 42: The same inventory as the complete radial build, at $A = 2$: the plasma, the nuclear island b and the TF function are all preserved — only the aspect ratio changes. What changes with it is the room left for R_{cs}. The magnified strip is drawn unfilled and unsized on purpose: that split is an engineering result this document has not derived.

114. HTS is a centre-stack lever, not a centre-stack eraser

Equation (190) explains why HTS becomes especially valuable in a low-aspect-ratio branch. For fixed desired B_0, reducing A raises the required central-column field B_{cs}. A conductor with credible operating current margin at higher field can make a smaller or more capable centre stack possible. It may therefore preserve some of the low-A gain that an ordinary conductor field ceiling would otherwise erase.

But the same equation returns us to page 47:

$$p_B = \frac{B_{cs}^2}{2\mu_0}.$$

The central column must carry that magnetic loading while remaining manufacturable, cooled, electrically insulated, protected, and maintainable in a neutron environment. HTS does not turn c_{cs} into zero; its conductor and support still occupy space, and higher field can increase the structural allocation. The central column also competes with shielding and tritium-breeding volume, which cannot be dismissed for a D–T plant.

> **Design consequence.** HTS can raise the credible field ceiling in the precise radial interval that low aspect ratio makes scarce. It is a powerful enabling lever for an ST, but it cannot by itself close the centre-stack equation.

115. What remains after the ST answers Freidberg's two questions

The narrowly posed Freidberg result is now established on page 111: the kink obstruction is closed within the declared finite-A model, and the bootstrap obstruction has been reduced to an explicit current-drive condition that still awaits an independent affordability calculation. That result does not settle every physical or engineering demand placed on a reactor. The following questions are separate and still require an integrated calculation:

1. a finite-aspect-ratio equilibrium and stability calculation, rather than the large-aspect-ratio kink and Troyon closures;
2. a finite-aspect-ratio neoclassical bootstrap and current-drive calculation, with explicit profiles and collisionality;
3. a transport and confinement basis applicable to the intended ST regime;
4. a radial centre-stack design that makes the bracket in Eq. (190) positive with credible stress, current density, shielding, cooling, protection, and maintenance margins;
5. a divertor, first-wall, blanket, and tritium-breeding calculation. This includes the exhaust problem Freidberg's four ratios intentionally did not close.

The centre-stack calculation begins next. Its task is not to reopen the two Freidberg ratios; one has been passed and the other has been reduced to a stated condition. Its task is to measure the exact geometric and mechanical debt created by the low-A solution. Only then can HTS, reduced shielding, larger radius, alternative current initiation, or modular maintenance be discussed as quantified responses rather than wishful labels.

116. The central column cannot evade Ampere's law

The toroidal field outside an axis-enclosing current path obeys

$$B_\varphi(R) = \frac{\mu_0 N I}{2\pi R}.$$

Evaluate it at the plasma centre:

$$NI = \frac{2\pi R_0 B_0}{\mu_0}. \tag{191}$$

This is a striking centre-stack fact. The required TF ampere-turns are set by the desired plasma product $R_0 B_0$, not by the small radius of the central column. Moving the current path inward raises the local field so that RB_φ remains constant; it does not make the required enclosed current disappear.

Let $J_{\mathrm{wp,cs}}$ be the allowed winding-pack current density after conductor, stabilizer, insulation, cooling, joints where relevant, and allocated structural fraction are counted honestly. The current-carrying cross-sectional area has the necessary lower bound

$$A_{\mathrm{wp,cs}} \geq \frac{NI}{J_{\mathrm{wp,cs}}} = \frac{2\pi R_0 B_0}{\mu_0 J_{\mathrm{wp,cs}}}. \tag{192}$$

It is a lower bound because a real column also needs support and non-conductor functions. The equation makes the danger of vague compactness claims visible: low aspect ratio shrinks the available column, while the TF current-carrying area is controlled by the plasma field target.

117. A first necessary radius inequality for the centre stack

Even in the unrealistically generous case that the entire circular column were available to the TF winding pack,

$$\pi R_{\mathrm{cs}}^2 \geq A_{\mathrm{wp,cs}}.$$

Substitute Eq. (192):

$$R_{\mathrm{cs}} \geq \left(\frac{2R_0 B_0}{\mu_0 J_{\mathrm{wp,cs}}} \right)^{1/2}. \tag{193}$$

This is not a design formula. It is a necessary condition with every real allocation set to zero except current-carrying area. A credible design needs a stronger inequality,

$$\pi R_{\mathrm{cs}}^2 \geq A_{\mathrm{wp,cs}} + A_{\mathrm{str}} + A_{\mathrm{ins/cool}} + A_{\mathrm{sol}} + A_{\mathrm{margin}}, \tag{194}$$

where A_{str} carries magnetic load, $A_{\mathrm{ins/cool}}$ provides electrical and thermal function, A_{sol} is any central-solenoid allocation, and the final term represents manufacturing, protection, and operational margin.

Combine Eq. (189) with Eq. (194). Lowering A reduces the left-side geometric radius, while the right side is set by field, materials, and operating functions. The centre-stack problem is therefore a geometric inequality test. It can fail before we ask whether the plasma is stable.

Design consequence. A proposed ST must publish a centre-stack area ledger, not merely an aspect ratio and a peak field. Without the conductor, structure, shielding, and start-up allocations, there is no check that the proposed central column exists.

118. Higher central field helps the plasma only by increasing mechanical demand

At the centre-stack radius, the local magnetic-pressure scale is

$$p_{B,\mathrm{cs}} = \frac{B_{\mathrm{cs}}^2}{2\mu_0}.$$

If a representative structural load path of area A_{load} is carried by material limited to σ_{allow}, force balance requires

$$A_{\mathrm{str}} \gtrsim \frac{p_{B,\mathrm{cs}}}{\sigma_{\mathrm{allow}}} A_{\mathrm{load}}. \tag{195}$$

The detailed geometry determines A_{load} and the actual stress distribution; the quadratic field dependence is already inescapable. As low A reduces R_{cs}, Eq. (190) pushes B_{cs} upward for a fixed B_0. The winding pack and its supporting structure then compete for the same shrinking area.

HTS can improve the conductor part of the ledger by providing useful current margin at higher field. It does not make the structural fraction vanish. In fact, if a design uses the higher field that HTS enables, Eq. (195) can require more support. That is why a high-field ST must report both conductor performance and a credible mechanical case.

Model and limits. Equation (195) is a lower-bound scaling, not a finite-element stress certification. Real centre columns need three-dimensional load paths, strain limits for the superconductor, cyclic-fatigue analysis, joints and terminations where used, electrical insulation, quench or fault protection, and neutron-degraded material properties.

119. The transformer also loses flux swing as the column shrinks

The transformer argument from page 20 gives a second, independent centre-stack constraint. A solenoid of effective radius R_{sol} changes linked magnetic flux by the scale

$$\Delta\Psi_{\mathrm{sol}} \sim \pi R_{\mathrm{sol}}^2 \Delta B_{\mathrm{sol}}. \tag{196}$$

Faraday's law converts this flux swing into the volt-seconds available to initiate and ramp plasma current. Since R_{sol} must fit inside the same central column, $R_{\mathrm{sol}} < R_{\mathrm{cs}}$, the available inductive flux falls at least quadratically as the column becomes narrow.

This fact connects directly to the Freidberg current ledger. A pulsed ST cannot claim steady operation simply because it can make an initial plasma current; it needs enough flux swing for its chosen pulse or a non-inductive start-up and sustainment route. Conversely, a steady-state ST with high bootstrap fraction has a second incentive to develop non-inductive current initiation: it can release central-column area otherwise reserved for a large solenoid.

The choice is an engineering-plasma co-design decision. Removing solenoid allocation from Eq. (194) may help the radial ledger, but transfers an obligation to radio-frequency, neutral-beam, helicity-injection, or other start-up/current-drive systems. It cannot be deleted from the thesis by a diagram.

120. The central-stack test is now ready to be made quantitative

The equations above turn the phrase centre-stack problem into five checkable questions:

1. What B_0, R_0, and A are demanded by the plasma mission?
2. Does Eq. (190) leave positive inboard radius after the required nuclear build b?
3. Does the available πR_{cs}^2 exceed the full area ledger in Eq. (194)?
4. Can the structural allocation carry the field-dependent load without exceeding stress or superconducting-strain limits?
5. If solenoid area is reduced, what explicit non-inductive start-up and current maintenance system replaces its flux swing?

The next step is not to assign convenient answers. We will define a transparent reference ST radial ledger and vary its field, radius, shielding, conductor current density, and solenoid allocation. That will identify whether an HTS-enabled central stack can close geometrically, where it fails, and which missing technologies would have to carry the remaining burden.

121. A published ST study shows why the ledger must be literal

It is useful to test the preceding geometry against a published allocation, provided we do not mistake one study's choices for universal constants. Menard *et al.* examined an illustrative FNSF configuration with $A = 2$, $R_0 = 1.87\,\mathrm{m}$, and a 30 cm centre-column radius. Their stated inboard nuclear protection included approximately 60 cm of water-cooled tungsten-carbide shielding; the corresponding radial picture also includes a 2 cm first wall. Their magnet assessment used a $70\,\mathrm{MA\,m^{-2}}$ winding-pack current-density assumption and imposed both support-stress and superconducting-strain constraints. [9]

The first calculation needs no plasma model. At $A = 2$,

$$a = \frac{R_0}{A} = 0.935\,\mathrm{m}, \qquad R_0 - a = 0.935\,\mathrm{m}.$$

Temporarily subtract only the stated shield and first-wall thickness:

$$0.935 - 0.600 - 0.020 = 0.315\,\mathrm{m}. \tag{197}$$

This rough geometry check is close to the study's 0.30 m centre-column radius. It is not an argument that 15 mm is spare room: it says that even before every cable, support, coolant channel, tolerance, and current-initiation function is drawn, the low-A inboard budget is already measured in centimetres.

> **Design consequence.** The centre stack is not an abstract penalty applied after plasma optimization. A realistic neutron shield can consume most of the inboard radius in an $A \simeq 2$ machine. A claimed compact ST must display its own radial ledger with the same honesty.

[9] Menard *et al.*, cited on page 105, Secs. 2.1 and 5.1, especially their Figs. 46–47. The numbers in this paragraph are inputs and results of that particular FNSF study, not universal requirements.

122. Why a D–T centre stack begins with shielding, not conductor

The inboard build b in Eq. (189) is not a decorative engineering margin. D–T fusion emits 14.1 MeV neutrons, which are uncharged. Magnetic confinement does not bend them. They stream into the first wall, blanket, shield, structural material, and magnets; a reactor must absorb their heat, breed tritium from lithium, and keep radiation damage to components within their allowed lifetime.

That is why the 60 cm shield in the illustrative calculation on page 121 cannot simply be traded for a better magnet. It was a neutron-protection choice for the study's mission and material assumptions. Changing it demands a new neutronics, heating, lifetime, and tritium-breeding calculation. The same study found that at low aspect ratio the inboard side can also be important to a tritium-breeding solution.[10]

This gives the radial order of operations:

plasma mission	⟶	neutron, heat, and tritium functions	⟶	space left for TF conductor, support, and st

Reversing that order—choosing a desired tiny centre column first and assigning the nuclear functions whatever space survives—would be the same category error as assuming a finished tokamak at page 1. The nuclear build is a design demand that helps reveal the permitted machine shape.

PLACE IMAGE HERE — Cumulative ST radial ledger: a neutron first, magnet second construction. Draw one clean outboard-midplane to inboard-midplane cross-section for the illustrative $A = 2$, $R_0 = 1.87$ m state. On the inboard side add components in causal order: plasma edge, first wall, approximately 0.60 m shielding as a study-specific allocation, then the approximately 0.30 m central-column envelope. Inside that envelope use unfilled subdivisions for TF winding, support, insulation/cooling, and optional solenoid–do not assert their final sizes. Include the arithmetic of Eq. (197) and a clear illustrative published-allocation, not-universal-reactor-template note.

[10] Menard *et al.*, cited on page 105, discuss the inboard shielding and breeding allocation in their Sec. 5.3. Their conclusion is configuration-specific; the general point here is only that breeding and shielding claim radial space before the magnet is sized.

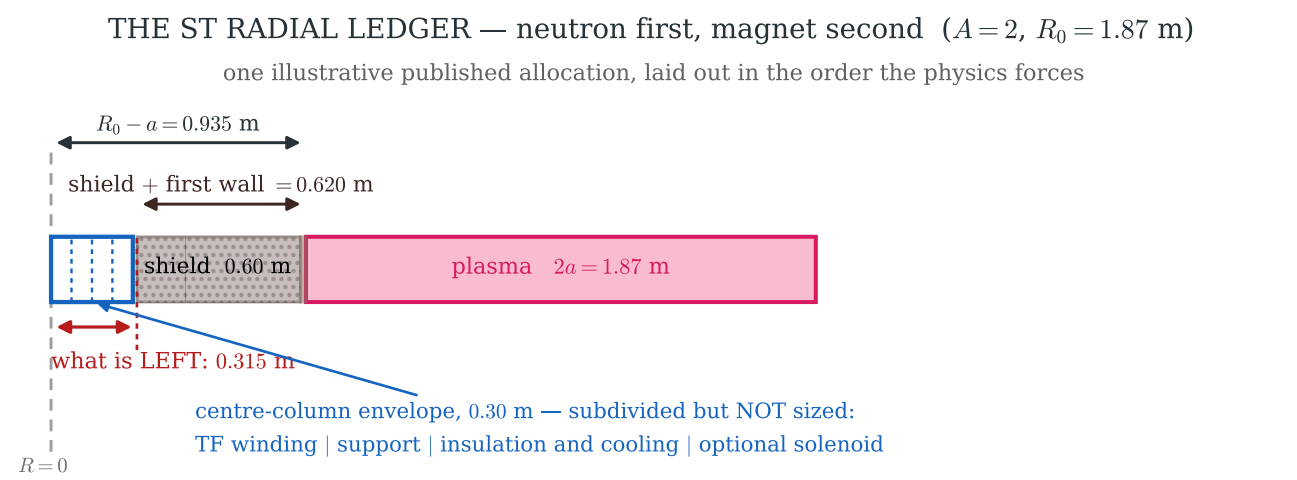


Figure 43: The published illustrative allocation, laid out in the order the physics forces: shield first, magnet into what is left. The arithmetic is Eq. (197) evaluated. The 15 mm difference from the study's stated column radius is not spare room — it is the residue of a check that has not yet paid for one cable, support, coolant channel or tolerance.

123. What HTS changes in this ledger

HTS changes a specific term: the credible current density and operating field margin of the TF winding pack. In the necessary bound,

$$A_{\mathrm{wp,cs}} \geq \frac{2\pi R_0 B_0}{\mu_0 J_{\mathrm{wp,cs}}},$$

raising a credibly usable $J_{\mathrm{wp,cs}}$ reduces the *minimum* current-carrying area for a prescribed $R_0 B_0$. It can therefore create room that a conventional conductor cannot. In the field relation

$$B_{\mathrm{cs}} = \frac{B_0 R_0}{R_{\mathrm{cs}}},$$

HTS can also make a larger central field technically plausible without immediately crossing the conductor's critical surface.

Those are real levers, but neither equation says that the whole right side of Eq. (194) scales down as $1/J_{\mathrm{wp,cs}}$. The support, insulation, cooling, protection, radiation, and manufacturing terms remain. Moreover, raising B_{cs} raises $p_{B,\mathrm{cs}} \propto B_{\mathrm{cs}}^2$, which can enlarge the structural term. A good HTS design may close the equation; HTS alone is not the missing calculation.

The published low-A work gives a concrete architectural direction rather than a shortcut. A later $A = 2$, $R_0 = 3\,\mathrm{m}$ HTS pilot-plant concept places all superconducting poloidal-field coils outside the TF system, uses continuous HTS TF coils, retains only a small central solenoid, and keeps inboard blanket/shield functions in the radial build.[11] Each choice answers one item in the ledger: outside PF coils avoid consuming central volume; a small solenoid makes non-inductive start-up more important; HTS improves the TF field option; blanket and shield remain non-negotiable.

Design consequence. HTS is the enabling material in a co-design, not an exemption from the co-design. Its success must be reported as a closed area, stress, radiation, and start-up ledger at the chosen R_0, A, and B_0.

[11] Menard *et al.*, cited on page 105, Sec. 5.4 and Fig. 53. It is a conditional systems study, not a demonstration that every one of its performance targets has been experimentally achieved.

124. What remains after the Freidberg answer

The causal argument has reached a useful, deliberately narrow conclusion. The finite-A calculation on pages 106–111 answered Freidberg's two current questions. The earlier first-principles geometry explains why that answer was possible:

1. Eq. (183) changes the current–safety-factor geometry in the favorable direction for the kink/current coupling.
2. Eq. (180) strengthens the trapping physics that underlies a route to bootstrap current.

HTS then supplies a plausible material lever for the adverse inboard field and winding-area problem created by the same geometry. Published integrated ST studies show that these ingredients can be put into a serious systems calculation.

That establishes the result relevant here: a specified finite-A ST model closes the kink constraint that failed in Freidberg's conventional baseline and reduces the bootstrap constraint to an explicit, quantified current-drive condition ($I_{\mathrm{CD,allow}} \geq 2.6$–$2.9\,\mathrm{MA}$, page 111), whereas merely raising H in that baseline did neither. It does not establish a theorem that every ST or every other reactor constraint is closed; exhaust, blanket, tritium, radiation, maintenance, and cost remain their own questions.

The next calculation must therefore have two ledgers, evaluated together:

finite-A plasma ledger	centre-stack reactor ledger
q, β_N, equilibrium/stability	b, R_{cs}, A_{wp}, A_{str}
bootstrap and driven current	shield, breeding, stress, start-up
transport and exhaust	HTS margin, protection, maintenance

The next section makes the centre-stack ledger into an explicit, auditable parameter scan. It does not relabel that separate engineering question as a failed Freidberg plasma constraint.

125. A dimensionless necessary test for a central TF winding

For a first scan, reserve the inboard nuclear build b before allocating the central column. The outer radius of the column envelope is then

$$R_{\mathrm{env}} = R_0 \left(1 - \frac{1}{A}\right) - b. \tag{198}$$

This is deliberately more generous than a finished design: it gives every remaining square millimetre to the central column before deciding how much is conductor, structure, cooling, or solenoid.

Let f_{wp} be the fraction of the envelope's circular cross-sectional area that the designer has explicitly allocated to the TF winding pack. It is an accounting input, not a material constant. Equation (192) then requires

$$\frac{2\pi R_0 B_0}{\mu_0 J_{\mathrm{wp,cs}}} \leq f_{\mathrm{wp}} \pi R_{\mathrm{env}}^2.$$

After cancelling π, define the dimensionless winding occupancy

$$\eta_{\mathrm{wp}} \equiv \frac{2R_0 B_0}{\mu_0 J_{\mathrm{wp,cs}} \left[R_0(1 - 1/A) - b\right]^2}. \tag{199}$$

The necessary test is simply

$$\eta_{\mathrm{wp}} \leq f_{\mathrm{wp}} < 1. \tag{200}$$

If $\eta_{\mathrm{wp}} > 1$, even a column made entirely of winding pack cannot carry the required TF current. If $f_{\mathrm{wp}} < \eta_{\mathrm{wp}} < 1$, a central column exists only on paper because the current pack has consumed space explicitly promised to the other functions.

Model and limits. This is a necessary electromagnetic-area test. It treats the winding pack as uniformly current carrying and does not certify field quality, stress, strain, quench protection, neutron lifetime, joints, penetrations, or manufacturability. Those omissions make passing the test easier, never harder.

126. Solving the test exposes a lower-aspect-ratio boundary

The test can be solved for the smallest aspect ratio permitted by the chosen area allocation. Take the positive square root of Eq. (200):

$$R_0\left(1-\frac{1}{A}\right)-b\geq\left(\frac{2R_0B_0}{\mu_0J_{\mathrm{wp,cs}}f_{\mathrm{wp}}}\right)^{1/2}.$$

Divide by R_0 and rearrange:

$$A\geq A_{\mathrm{min,wp}}\equiv\left[1-\frac{b}{R_0}-\left(\frac{2B_0}{\mu_0J_{\mathrm{wp,cs}}f_{\mathrm{wp}}R_0}\right)^{1/2}\right]^{-1}. \tag{201}$$

The bracket must be positive. If it is zero or negative, the stated R_0, B_0, nuclear build, winding-pack current density, and allocation cannot coexist at *any* aspect ratio in this simplified central-column arrangement.

Every trend in this equation has a physical reading:

- Increasing B_0 increases the required ampere-turns and raises the minimum allowed A.
- Increasing b/R_0 claims the inboard radius for nuclear function and raises the minimum allowed A.
- Increasing credible $J_{\mathrm{wp,cs}}$ or giving the winding pack a larger honest fraction f_{wp} lowers the electromagnetic area demand.
- Increasing R_0 helps twice: the nuclear build occupies a smaller fraction of the radius and the required winding area is less crowded into the column.

The last two points identify exactly what HTS and a less compact plant can improve. They also show why neither word automatically guarantees a viable ST.

127. A transparent teaching scan: field makes the low-A limit move

Use the deliberately declared teaching inputs

$$R_0 = 3.0\,\mathrm{m}, \quad b = 0.65\,\mathrm{m}, \quad J_{\mathrm{wp,cs}} = 70\,\mathrm{MA\,m^{-2}}, \quad f_{\mathrm{wp}} = 0.40.$$

The current-density scale is comparable to an assumption used in the illustrative Menard study on page 121; the radius, nuclear build, and 40% allocation here are *not* a proposed reactor. They merely make the algebra visible. Substitution in Eq. (201) gives:

B_0 (T)	$A_{\mathrm{min,wp}}$	η_{wp} at $A = 2$
4	1.97	0.378
6	2.24	0.567
8	2.54	0.756

At $A = 2$, the available envelope radius is $R_{\mathrm{env}} = 3(1 - 1/2) - 0.65 = 0.85\,\mathrm{m}$. The first row barely clears the assigned 40% winding allocation. The latter two do not: they require more winding area than has been allocated even before support, insulation, cooling, or a solenoid are tested. The table is reproducible from `centre_stack_ledger.py`; every input is displayed at the top of that file.

This is the exact sense in which HTS is a boost mechanism. If it raises the *credible system-level* $J_{\mathrm{wp,cs}}$ at the relevant field, temperature, strain, radiation, and protection margin, it can lower $A_{\mathrm{min,wp}}$. If the design instead spends the HTS benefit to raise B_0, the square-root field penalty pushes $A_{\mathrm{min,wp}}$ upward. The net result is a co-design decision, not a one-directional law.

Audit note. Passing the 4 T row does not make it a reactor. The table omits the structure term that grows with central field, the non-winding allocations, the neutron and tritium calculation behind b, and all plasma/stability/exhaust tests. It is intentionally a failure detector, not a feasibility certificate.